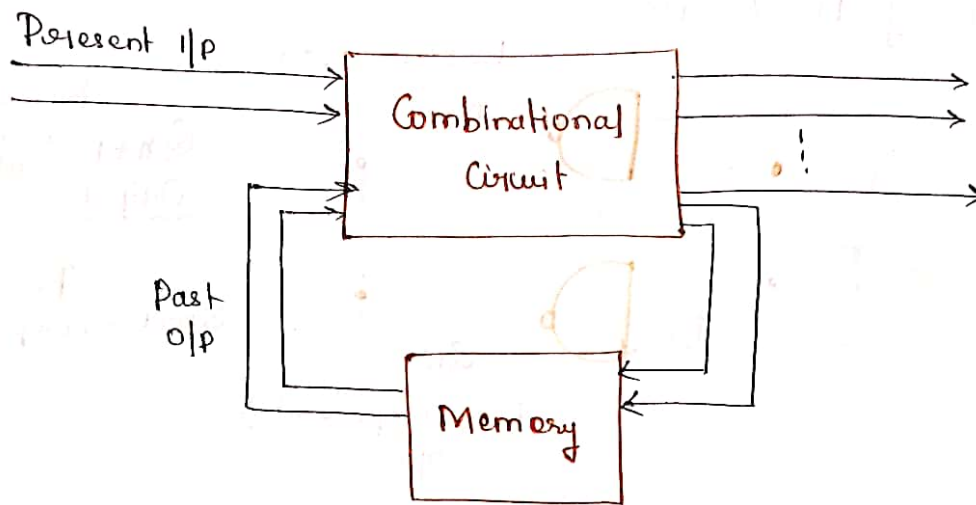


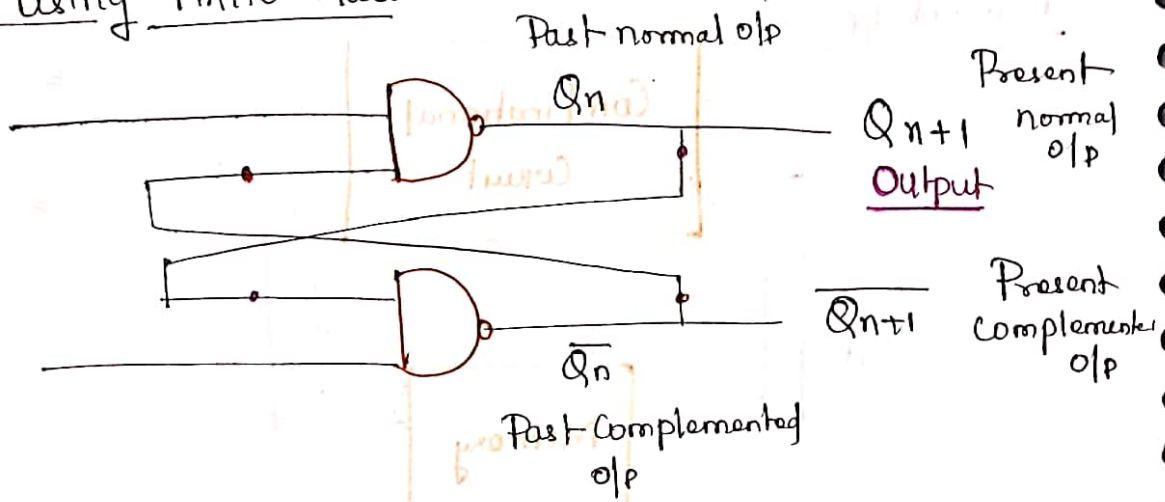
Unit-3 Sequential Circuits



1. In Combinational ckt, o/p at any time depends only on the present inputs. whereas in Sequential Circuit the o/p at any time depends on present input as well as past output.
2. As the o/p of Sequential ckt also depend on past o/p, there is a requirement of Memory element which is connected in the feedback path. Where as in Combinational ckt there is no requirement of Memory element or feedback path.
3. Due to the feedback path, Sequential ckt may be referred as closed system whereas Combinational ckt is referred as open system.
4. The Memory element used in digital Sequential ckt are -
 - i) latch
 - ii) Flip flop

* Latch (Cross Coupled Gates)

* Latch using NAND Gate



Output states of a latch / flipflop

① SET state

- latch/ff is said to be in SET state if present normal o/p or main output is at logic HIGH

$$\text{i.e. } Q_{n+1} = '1'$$

$$\& \overline{Q}_{n+1} = '0'$$

② RESET state

- latch/ff is said to be in RESET state if present normal o/p is at logic LOW

$$\text{i.e. } Q_{n+1} = '0'$$

$$\overline{Q}_{n+1} = '1'$$

III. HOLD state

latch/ff is said to be in hold state if the present o/p is same as the past o/p. i.e.

$$Q_{n+1} = Q_n$$

$$\& \quad \overline{Q}_{n+1} = \overline{Q}_n$$

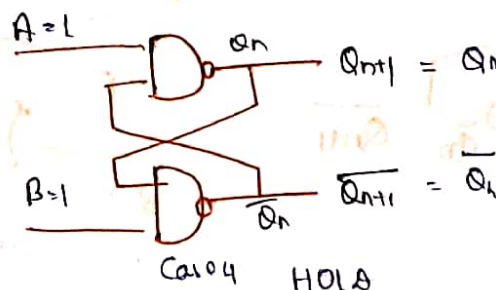
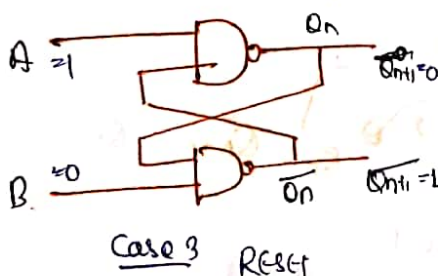
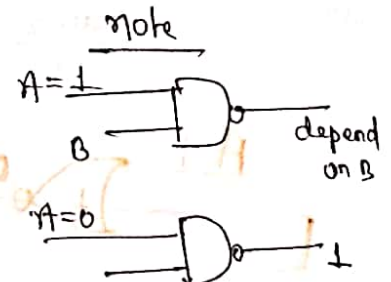
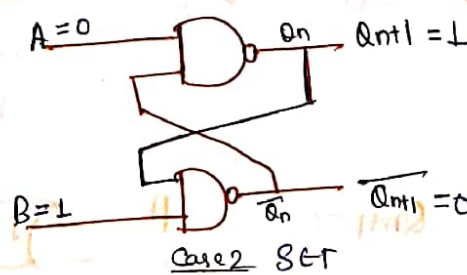
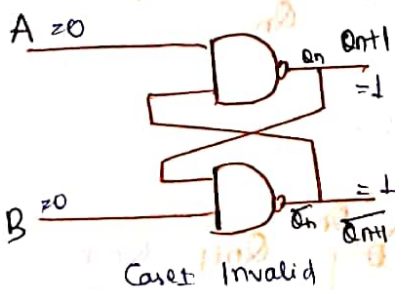
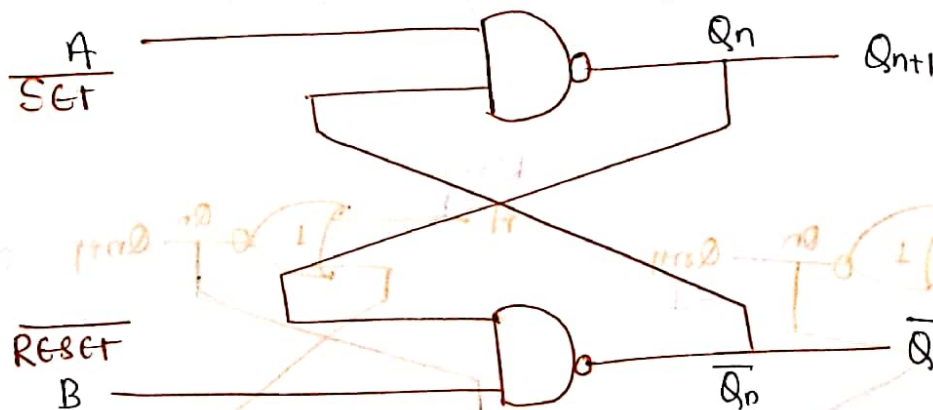
IV. Invalid state

latch/ff is said to be in invalid state if normal o/p is equal to complemented o/p.

$$\text{i.e. } Q_{n+1} = \overline{Q}_{n+1}$$

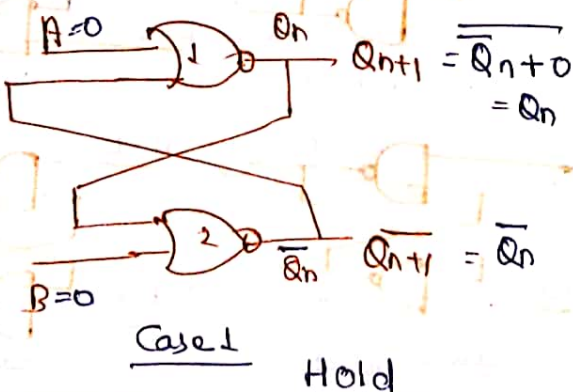
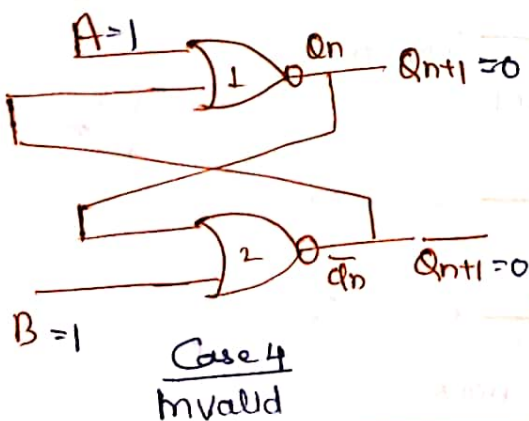
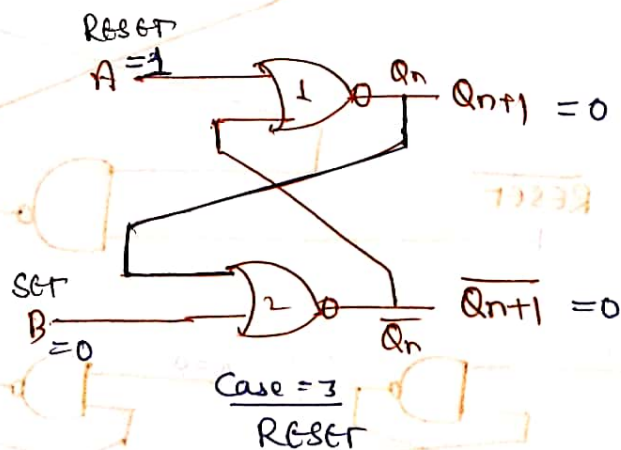
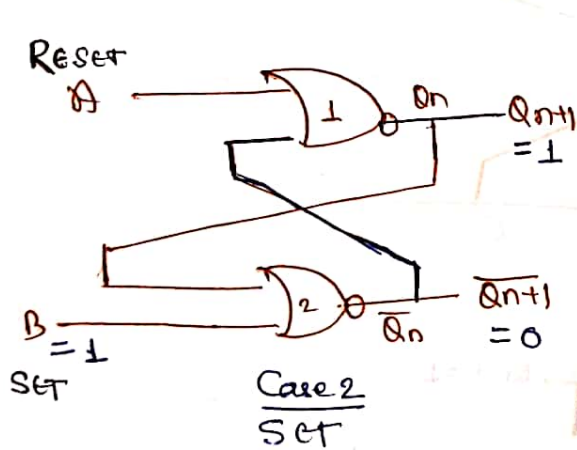
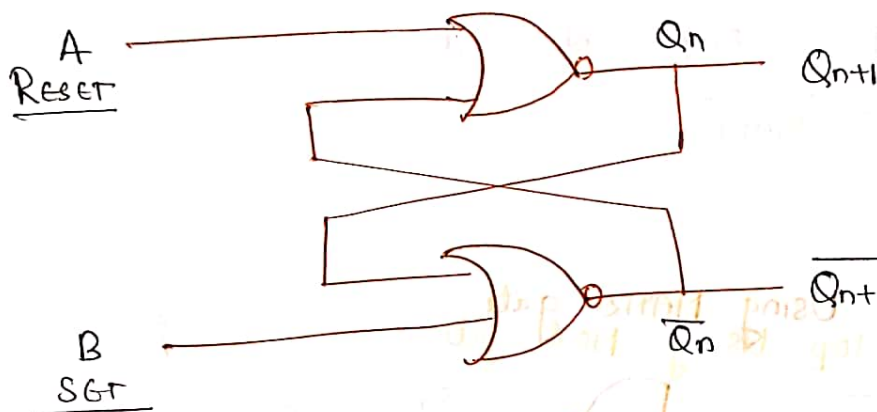
& Ⓚ

* SR Latch Using NAND gate:

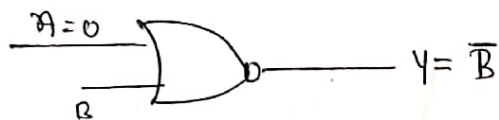
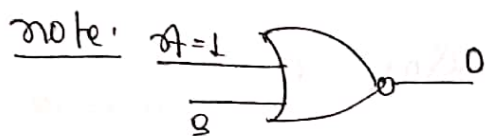


A	B	Q_{n+1}	State Q_{n+1}	State
0	0	1	1	Invalid
0	1	1	0	Set
1	0	0	1	Reset
1	1	Q_n	$\overline{Q_n}$	Hold

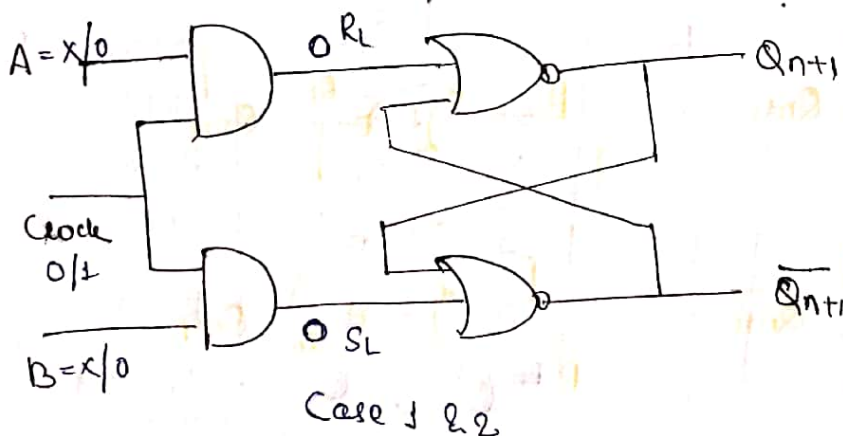
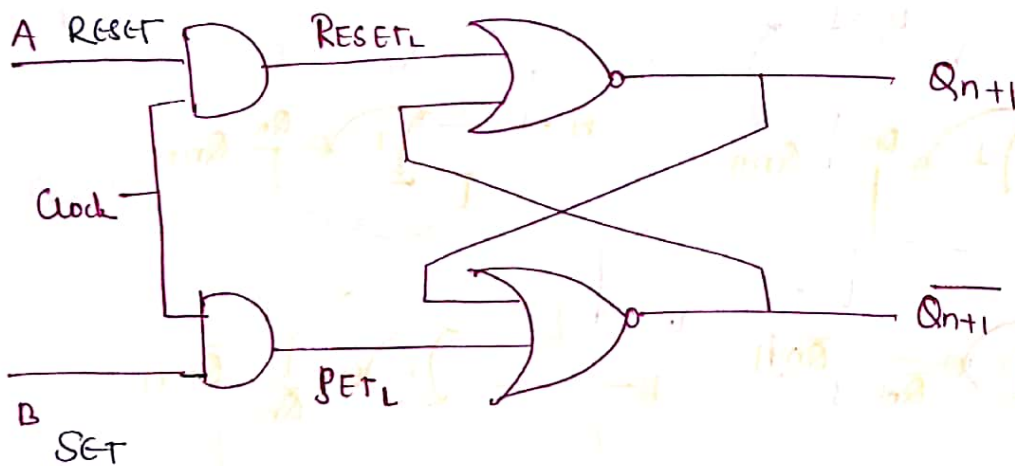
* SR Latch using NOR gate

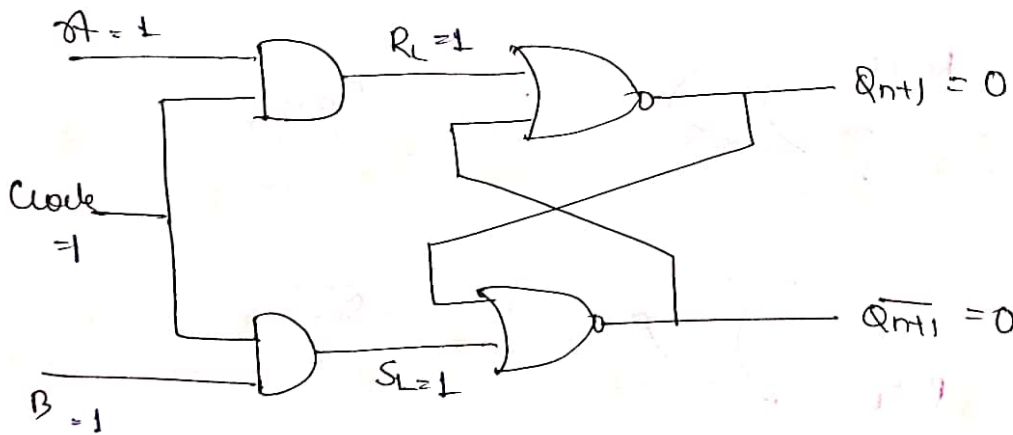
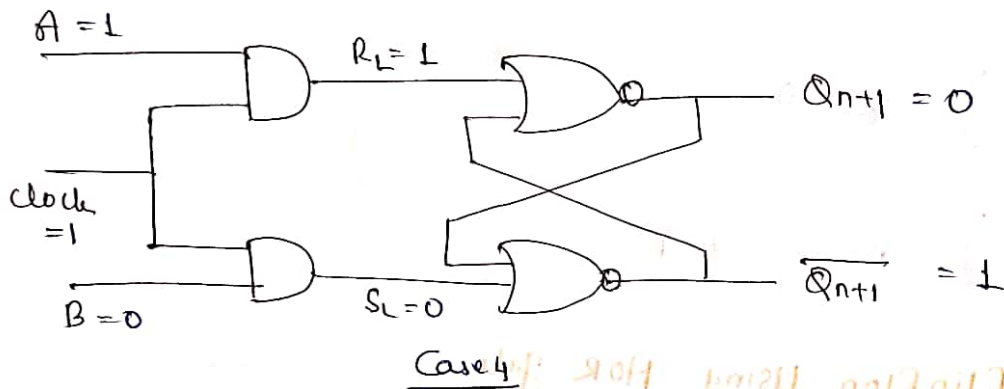
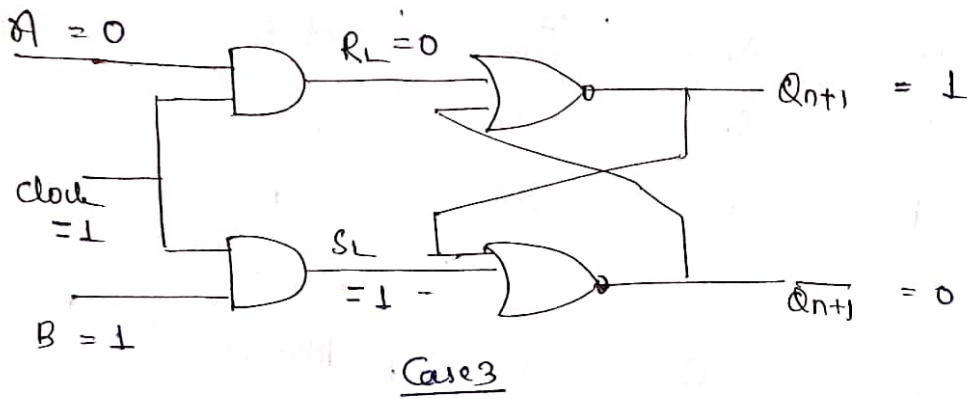


A	B	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	0	Q_n	$\overline{Q}_n$	HOLD
0	1	1	0	SET
1	0	0	1	RESET
1	1	0	0	INVALID



* SR FlipFlop Using NOR gate



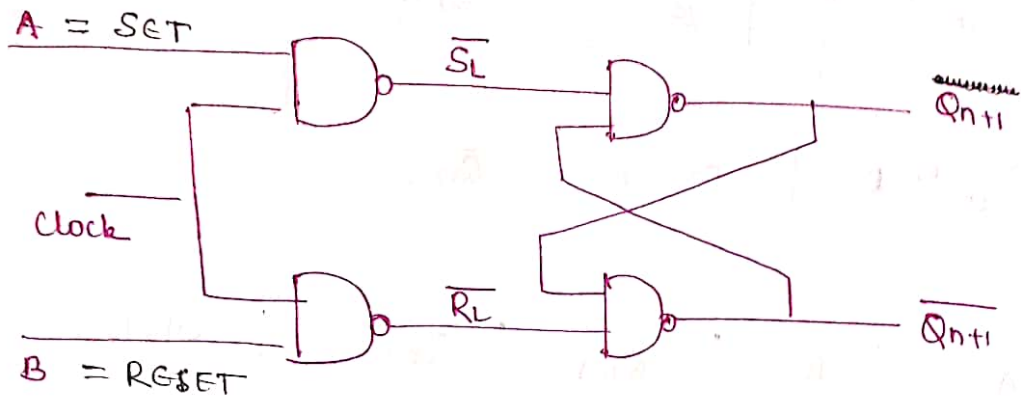


Table

Clock	A	B	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	X	X	Q_n	$\overline{Q}_n$	Hold
1	0	0	Q_n	$\overline{Q}_n$	Hold
1	0	1	1	0	SET
1	1	0	0	1	RESET
1	1	1	Invalid	Invalid	Invalid

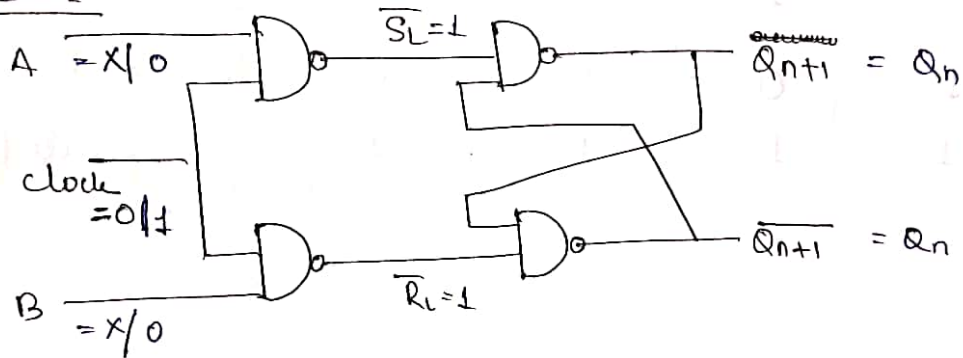
$\therefore B = \text{SET}$ $A = \text{RESET}$

* JK Flipflop Using NAND gate

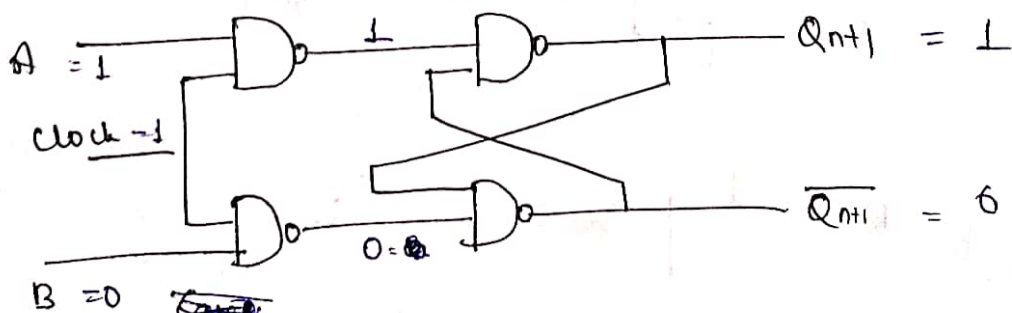
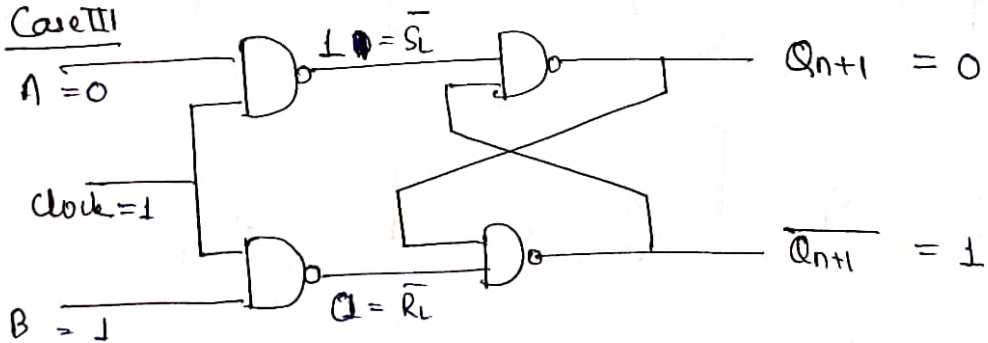


Working

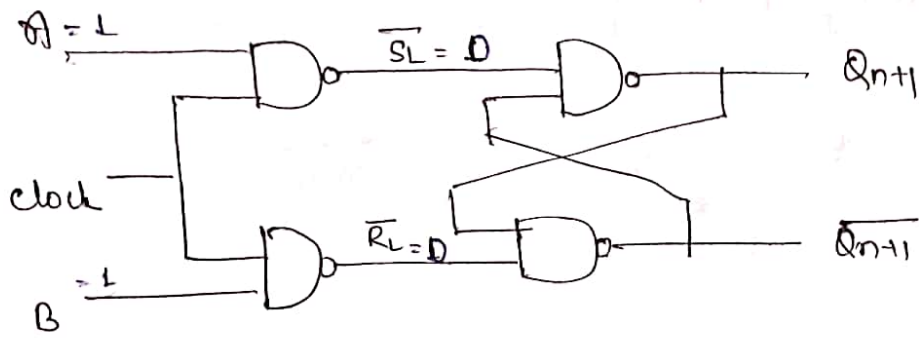
Case I



Case III

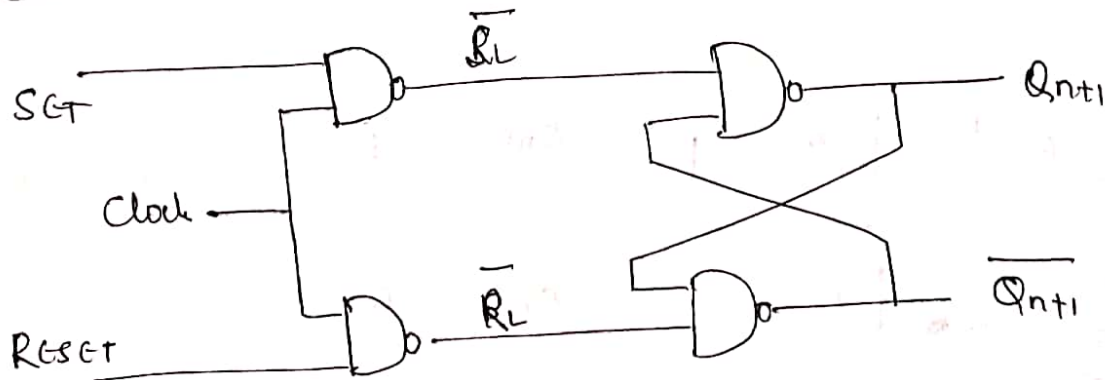
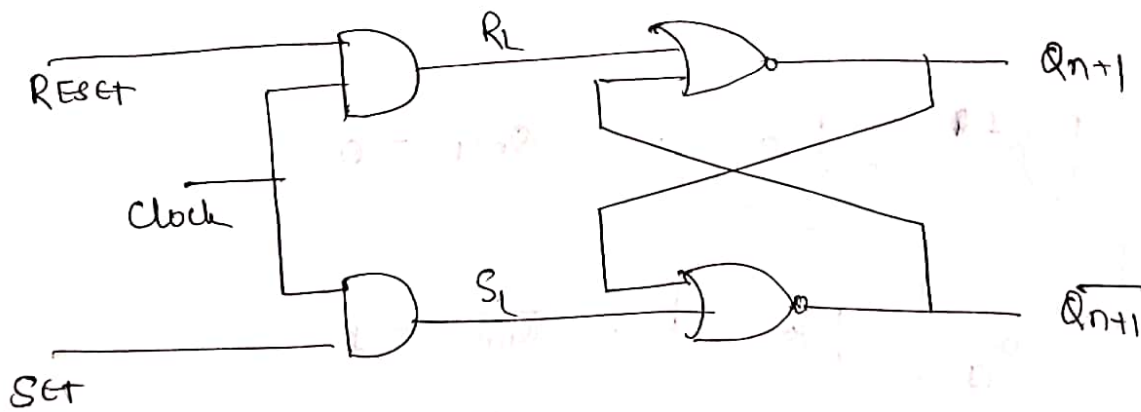


Case IV



Clock	A	B	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	X	X	Q_n	$\overline{Q_n}$	Hold
1	0	0	Q_n	$\overline{Q_n}$	Hold
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	1	1	Invalid

Conclusions



Truth table for SR Flipflop (NOR & NAND)

Clock	SET	RESET	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	X	X	Q_n	$\overline{Q}_n$	Hold
1	0	0	Q_n	$\overline{Q}_n$	Hold
1	0	1	0	1	Reset
1	1	0	1	0	Set
1	1	1	X	X	Invalid

(either 11 or 00)

* Characteristic table of SR Flipflop

Q_n	S	R	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	X

Q_n	$\overline{S}R$	$\overline{S}\overline{R}$	SR	$S\overline{R}$
$\overline{Q}_n$	0	0	X	1
Q_n	1	0	X	1

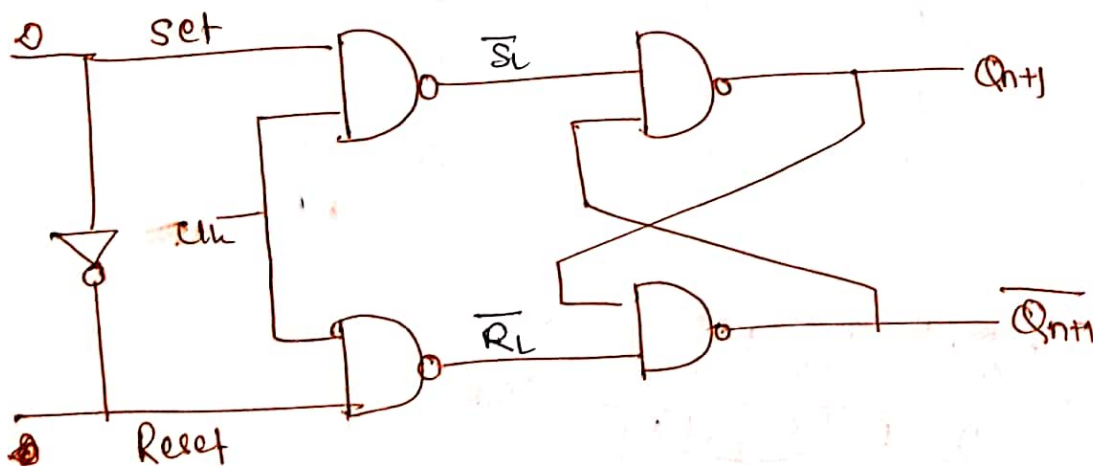
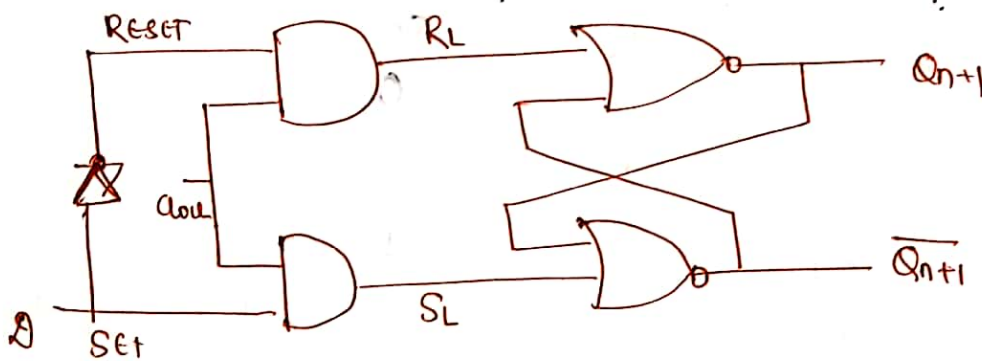
$$Q_{n+1} = S + Q_n \overline{R}$$

characteristic eqn
or Next state eqn

* Excitation Table of SR flip flop-

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

* SR Flip flop



note: Reset is $\overline{\text{Set}}$
(not gate used in Set Input)

$S \leftrightarrow \overline{S}$
$R \leftrightarrow \overline{R}$

Truth table for D flip flop (NAND & NOR)

Clock	D	S	R	Q_{n+1}	$\overline{Q_{n+1}}$	State
0	X	X	X	Q_n	$\overline{Q_n}$	Hold
1	0	0	1	0	1	
1	1	1	0	1	0	

Truth table for D FF (NAND NOR)

Clock	D	Q_{n+1}	$\overline{Q_{n+1}}$	State
0	X	Q_n	$\overline{Q_n}$	Hold
1	0	0	1	Reset
1	1	1	0	Set

D = Delay

Characteristic table for D FF

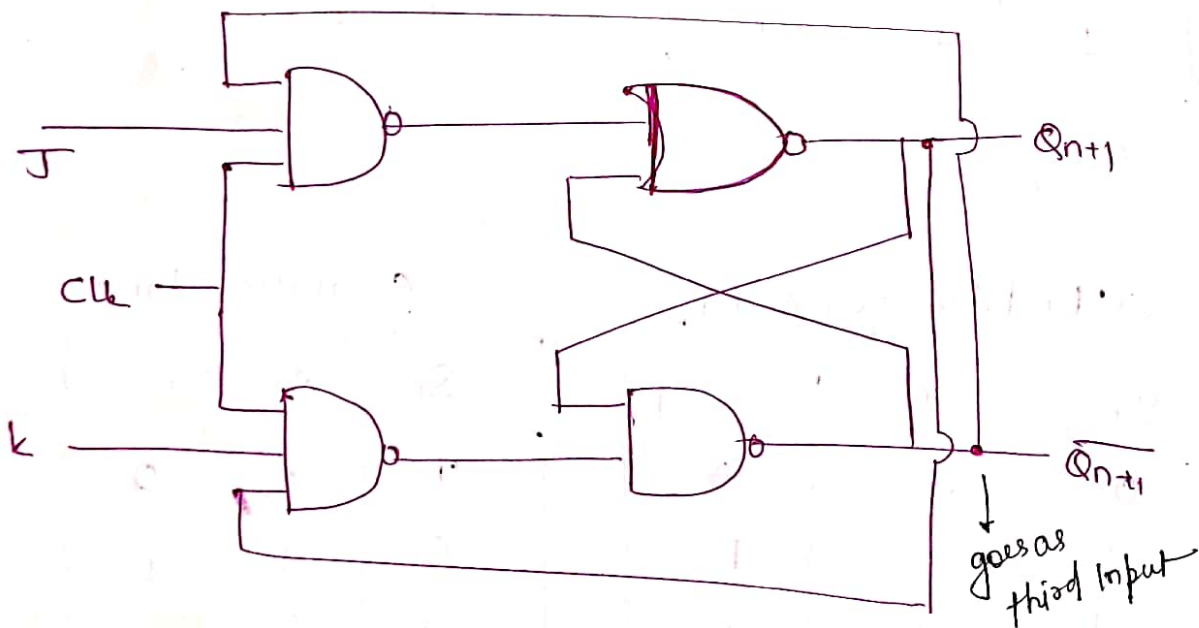
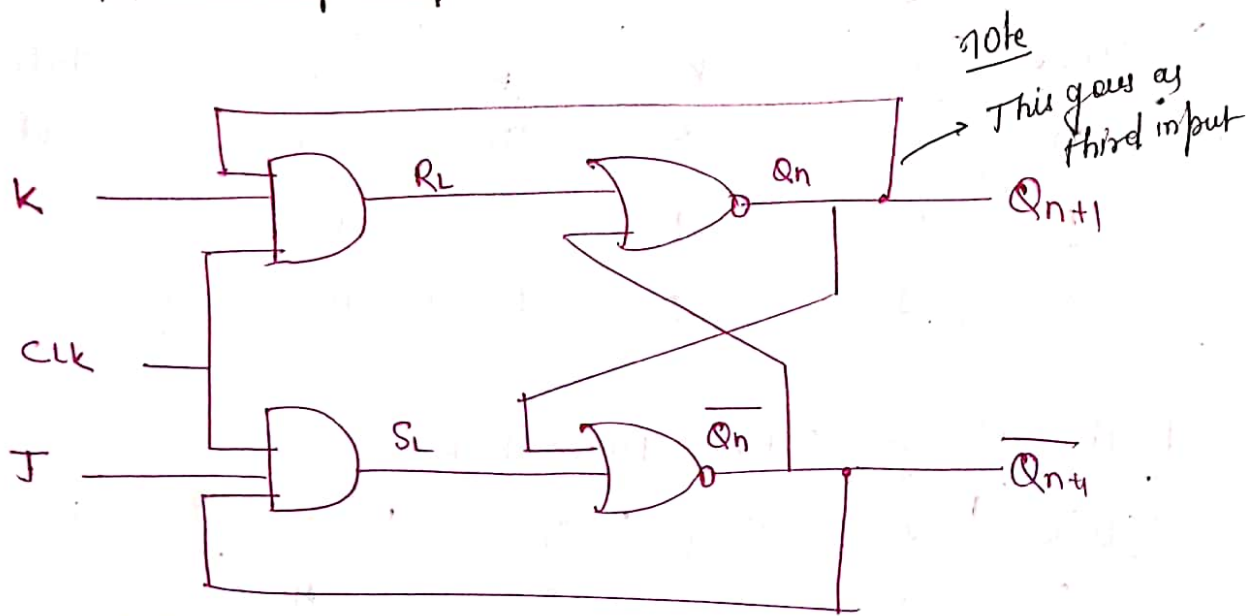
Q_n	D	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Excitation table

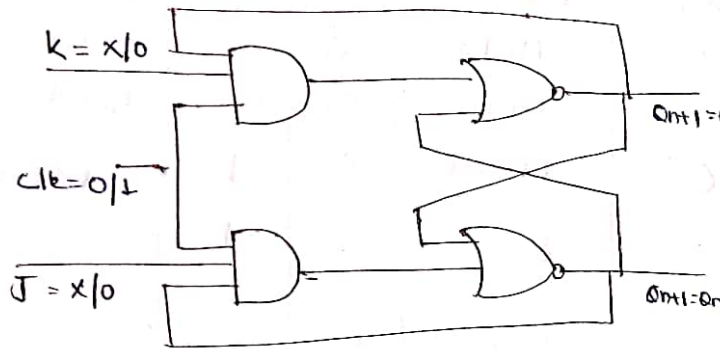
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

note:- To avoid Invalid state D flip flop is used.

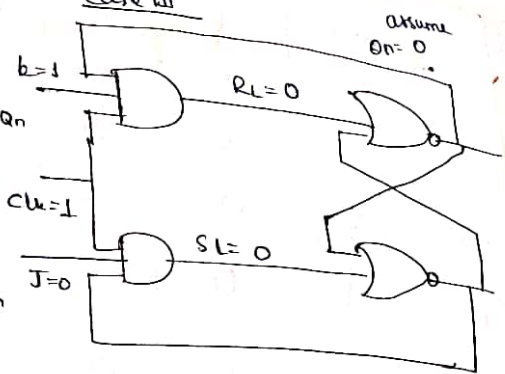
* Jk Flip flop



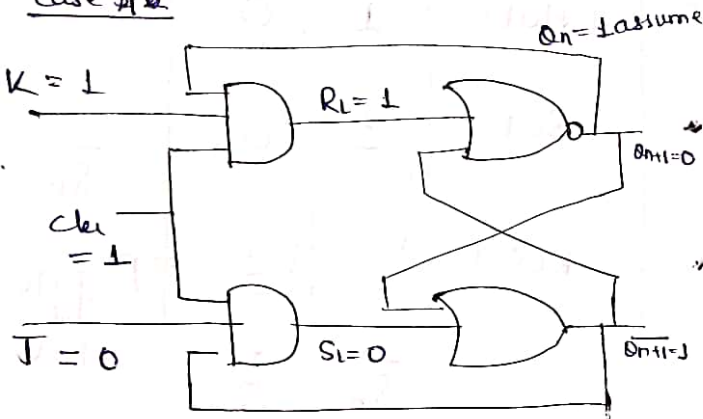
Case I & II



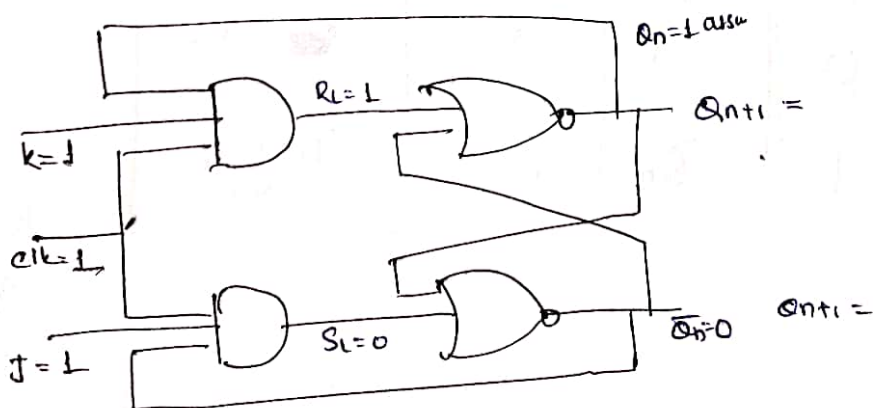
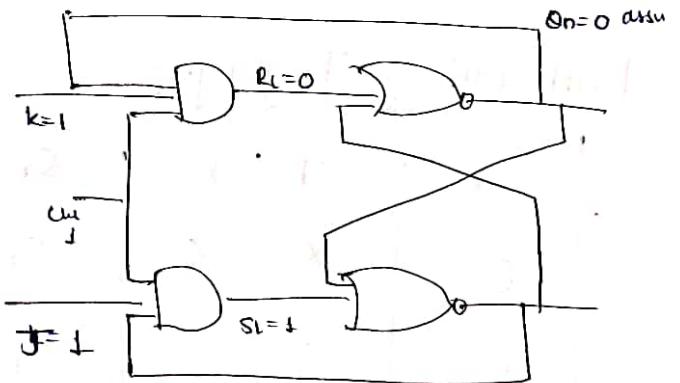
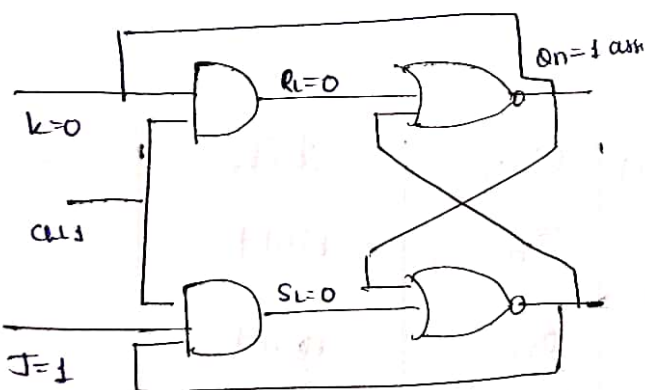
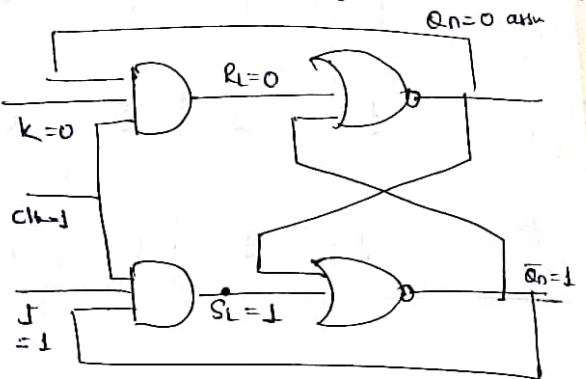
Case III



Case 4



Case 5



	Clock	J	K	Q_n	Latch state	Q_{n+1}	$\overline{Q}_{n+1}$	FF state
①	0	X	X	X	Hold	Q_n	$\overline{Q}_n$	Hold
②	1	0	0	X	Hold	Q_n	$\overline{Q}_n$	Hold
③	1	0	1	0^*	Hold	0	1	} RESET
④	1	0	1	1^*	RESET	0	1	
⑤	1	1	0	0^*	SET	1	0	} SET
⑥	1	1	0	1^*	HOLD	1	0	
⑦	1	1	1	0^*	SET	1	0	} $\overline{Q}_n$ Q_n
⑧	1	1	1	1^*	RESET	0	1	
						Q_n	Q_n	Toggle State

*Toggle state is exact complement of Hold state

Truth table of JK flip flop

clk	J	K	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	X	X	Q_n	$\overline{Q}_n$	Hold
1	0	0	Q_n	$\overline{Q}_n$	Hold
1	0	1	0	1	RESET
1	1	0	1	0	SET
1	1	1	$\overline{Q}_n$	Q_n	Toggle

Characteristic Table of JK

Q_n	J	k	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Toggle 1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0 Toggle

Q_n	Jk	$\bar{J}k$	$J\bar{k}$	$J\bar{k}$
$\bar{Q}_n$	0	0	1	1
Q_n	1	0	0	1

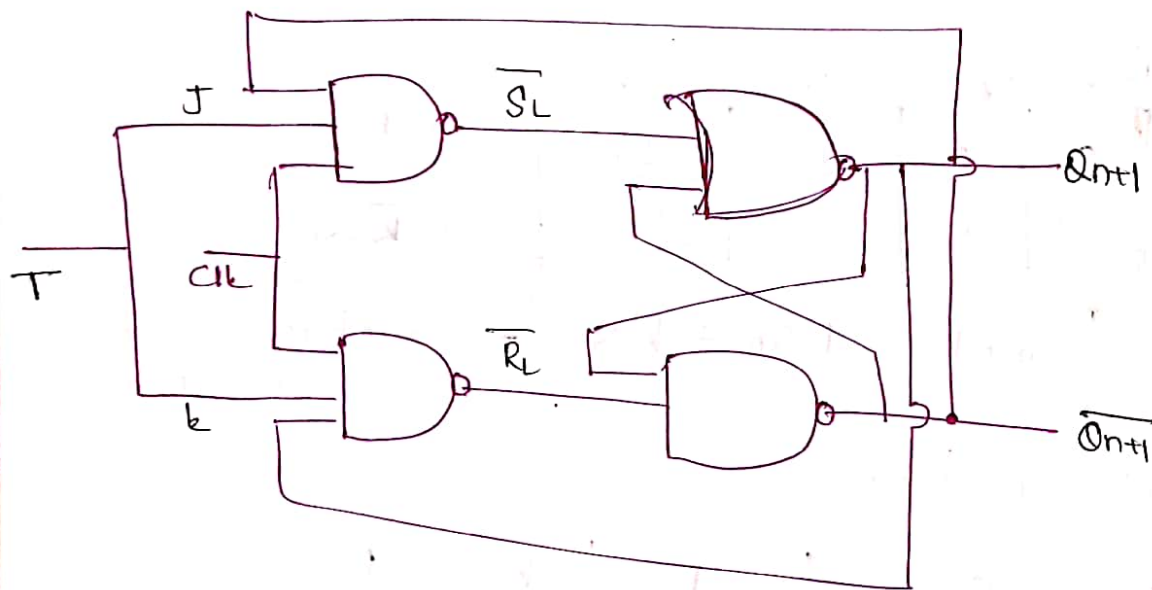
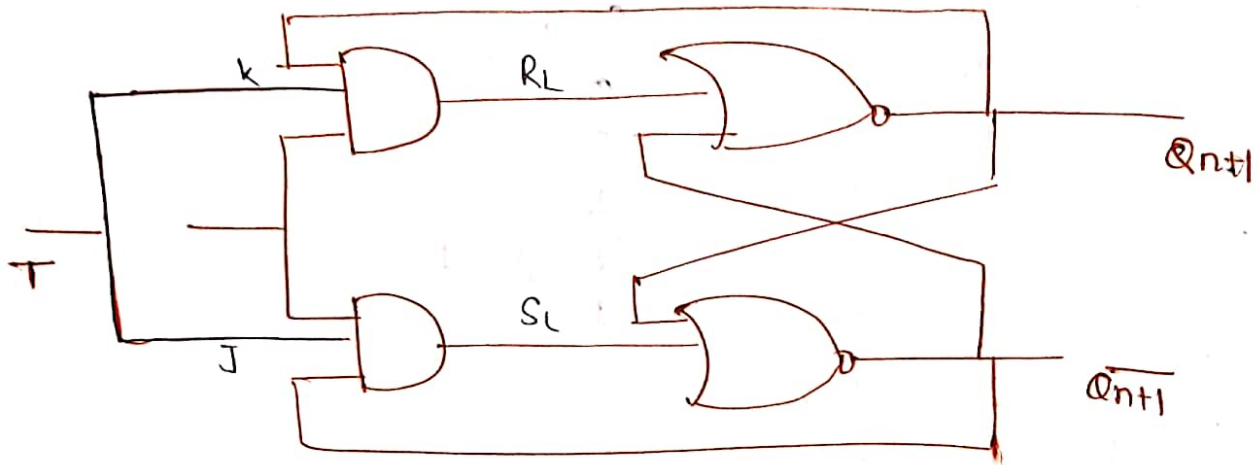
$J\bar{Q}_n$ (points to the 1 in the $J\bar{k}$ cell for $\bar{Q}_n$)
 $\bar{K}Q_n$ (points to the 1 in the $\bar{J}k$ cell for Q_n)

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n \rightarrow \text{characteristic equation}$$

* Excitation table

Q	Q_{n+1}	J	k
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

* T flipflop (Toggle flip-flop)



Truth table

clk	T	J	k	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	X	X	X	Q_n	$\overline{Q}_n$	Hold
1	0	0	0	Q_n	$\overline{Q}_n$	Hold
1	1	1	1	$\overline{Q}_n$	Q_n	Toggle

Clock	T	Q_{n+1}	$\overline{Q_{n+1}}$	State
0	X	Q_n	$\overline{Q_n}$	Hold
1	0	Q_n	$\overline{Q_n}$	Hold
1	1	$\overline{Q_n}$	Q_n	Toggle

Characteristic table of T FF

Q_n	T	Q_{n+1}
0	0	0
0	1	1
1	0	0
1	1	1

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$$Q_{n+1} = Q_n \oplus T$$

Conclusion

Q	Q_{n+1}	S	R	D	J	K	T
0	0	0	X	0	0	X	0
0	1	1	0	1	1	X	1
1	0	0	1	0	X	1	1
1	1	X	0	1	X	0	0

* Flip Flop Conversion

1. SR flipflop to JK flipflop

X $\longrightarrow$ Y
Excitation table Characteristic Table

Excⁿ table of SR

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

char. table of JK

J	K	Q_n	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

} Toggle

Conversion table

Q_n	Q_{n+1}	S	R
0	0	0	X
1	1	X	0
0	0	0	X
1	0	0	1
0	1	1	0
1	1	X	0
0	1	1	0
1	0	0	1

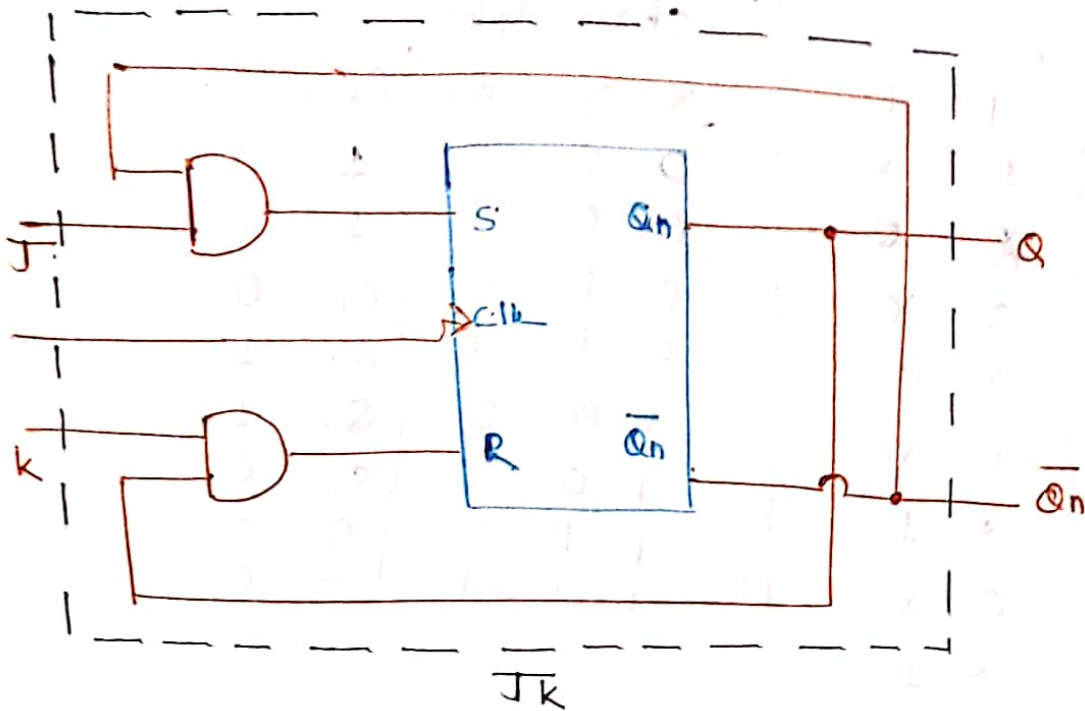
$$S(J, k, Q_n) =$$

	kQ_n	$\bar{k}Q_n$	$k\bar{Q}_n$	$\bar{k}\bar{Q}_n$
J	0	X	0	0
J	1	X	0	1

$$S(J, k, Q_n) = \bar{Q}_n = J\bar{Q}_n$$

$$R(J, k, Q_n) = kQ_n$$

	kQ_n	$\bar{k}Q_n$	$k\bar{Q}_n$	$\bar{k}\bar{Q}_n$
J	X	0	1	X
J	0	0	1	0



Que 1 An X, Y flipflop whose characteristic table is given is to be implemented using JK flipflop. This can be done, by making

- a) $J = X, K = Y$ b) $J = \bar{X}, K = Y$ c) $J = Y, K = \bar{X}$
 d) $J = \bar{Y}, K = X$

X	Y	Q_{n+1}
0	0	1
0	1	Q_n
1	0	$\bar{Q}_n$
1	1	0

Sol

$X, Y \longleftrightarrow$ JK
 excitation

excitation of JK

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Conversion table

Q_n	Q_{n+1}	J	k
0	1	1	X
1	1	X	0
0	Q_n 0	0	X
1	Q_n 1	X	0
0	$\bar{Q}_n$ 1	1	X
1	$\bar{Q}_n$ 0	X	1
0	0	0	X
1	0	X	1

char. table

X	Y	Q_n	Q_{n+1}
0	0	0	1
0	0	1	1
0	1	0	Q_n 0
0	1	1	Q_n 1
1	0	0	$\bar{Q}_n$ 1
1	0	1	$\bar{Q}_n$ 0
1	1	0	0
1	1	1	0

$J(Q_n, Q_{n+1})$
 $J(X, Y, Q_n)$

	$\bar{Y} \bar{Q}_n$	$\bar{Y} Q_n$	
$\bar{X}$	1	X	X
X	1	X	0

$$J(X, Y, Q_n) = \bar{Y}$$

$k(X, Y, Q_n)$

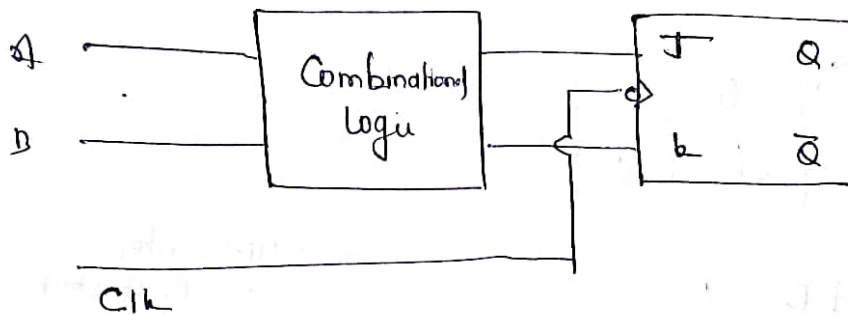
	$Y \bar{Q}_n$	$Y Q_n$	
X	X	0	0
X	X	1	X

$$k(X, Y, Q_n) = X$$

$$J = \bar{Y}, \quad k = X \quad // \quad \text{opt } 0$$

Q2. The following table has to be implemented with circuit shown in the figure. What is the o/p of the Combinational logic circuit to the J input

- a) $\overline{A \cdot B}$ b) $\overline{A}$ c) $\overline{B}$ d) $A \cdot B$



A	B	Q_{n+1}
0	0	Q_n'
0	1	1
1	0	Q_n
1	1	0

char. table

Sol

A	B	Q_n	Q_{n+1}	J	K
0	0	0	$Q_n' = 1$	1	X
0	0	1	0	X	1
0	1	0	1	1	X
0	1	1	1	X	0
1	0	0	0	0	X
1	0	1	1	X	0
1	1	0	0	0	X
1	1	1	0	X	1

Excitation table

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	0	0

$J(A, B, Q_n)$

	$B Q_n$	$\bar{B} Q_n$	$B \bar{Q}_n$	$\bar{B} \bar{Q}_n$
$\bar{A}$	1	X	X	1
A	0	X	X	0

$$J = \bar{A} //$$

X	1	0	X
X	0	1	X

$$K = \bar{A} \bar{B} + A B //$$

CONCEPT OF STATE DIAGRAM, STATE TABLE

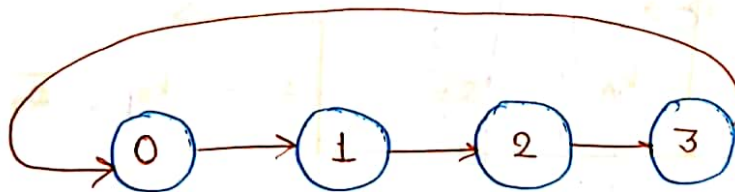
& SYNCHRONOUS COUNTER

Upcounter
→ Must start from 00 state

* Mod 4 Up Counter

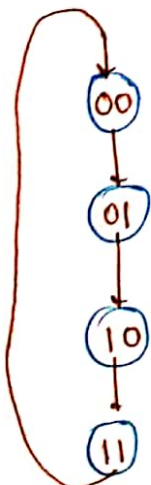
(DESIGN & ANALYSIS)

State diag.



State Table

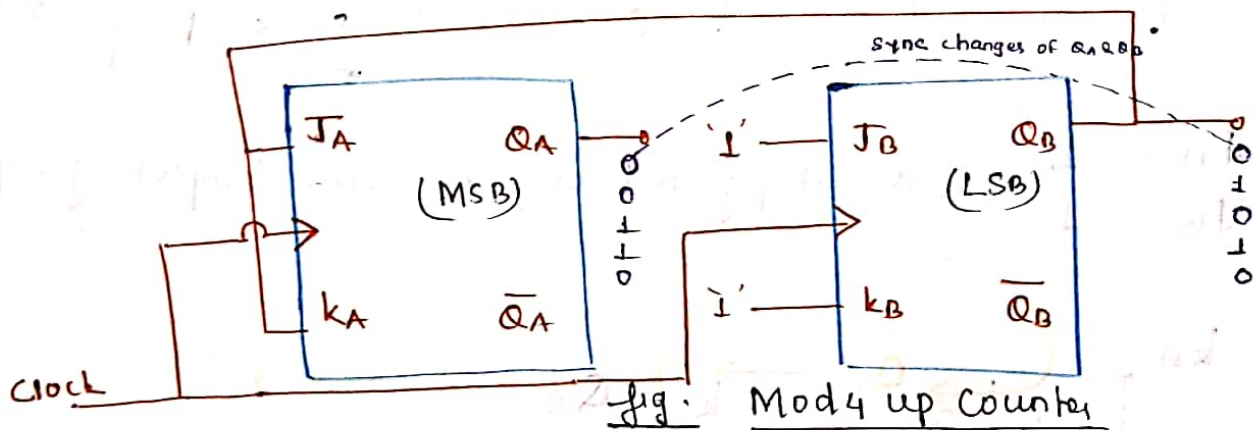
Present state		Next state	
Q_A	Q_B	Q_{A+1}	Q_{B+1}
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0



- Design Mod 4 upcounter using JK flipflop:

Present state		Next state		FF Input			
Q_A	Q_B	Q_{A+1}	Q_{B+1}	J_A	K_A	J_B	K_B
0	0	1	1	0	X	1	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	0	X	1	X	1

We have state and we have to find Input of J, k (designing) {using excitation of JK}



J_A :-

Q_A	Q_B	$\bar{Q}_B$	Q_B
$\bar{Q}_A$	0	1	X
Q_A	X	1	X

K_A :-

Q_A	Q_B	$\bar{Q}_B$	Q_B
$\bar{Q}_A$	X	1	0
Q_A	0	1	1

$$J_A = Q_B$$

$$K_A = Q_B$$

J_B :-

Q_A	Q_B	$\bar{Q}_B$	Q_B
$\bar{Q}_A$	1	X	1
Q_A	1	X	1

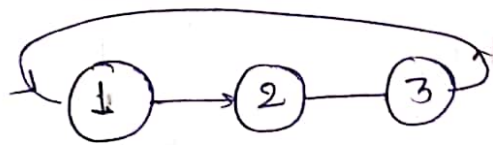
K_B :-

Q_A	Q_B	$\bar{Q}_B$	Q_B
$\bar{Q}_A$	X	1	1
Q_A	X	1	1

$$J_B = 1$$

$$K_B = 1$$

Ques. Design Mod 3 Counter



Present state		Next state		FF I/P			
Q_A	Q_B	Q_{A+1}	Q_{B+1}	J_A	k_A	J_B	k_B
0	0	X	X	X	X	X	X
0	1	1	0	1	X	X	1
1	0	1	1	X	0	1	X
1	1	0	1	X	1	X	0

$J_A = 1$ (directly from table no zero complete group)
 $J_B = 1$

k_A

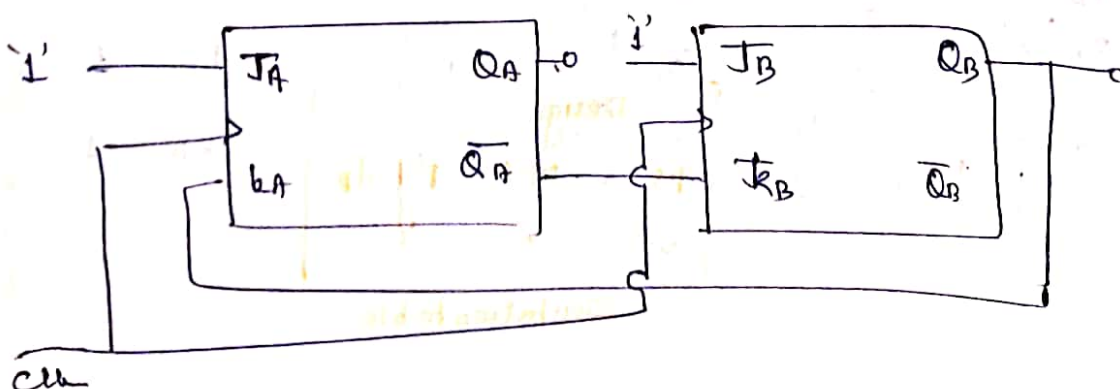
X	X
0	1

$$k = Q_B$$

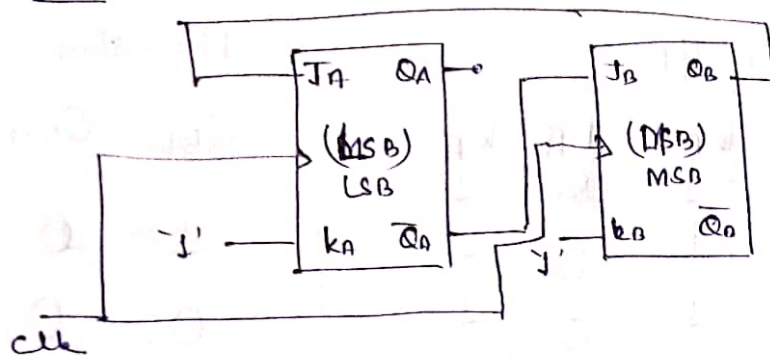
k_B

X	1
X	0

$$k_B = \overline{Q_A}$$



Que Given ~~state~~ Circuit



Note:
Q_B is MSB

Sol $k_A = 1$
 $k_B = 1$

$$J_A = Q_B$$

$$J_B = \overline{Q_A}$$

(Note B is MSB)

Q _B	Q _A	Q _{B+1}	Q _{A+1}
0	0	1	0
0	1	0	0
1	0	0	1
1	1	0	0

Analysis 1

Every flipflop has to follow its next state equation

$$NSE = Q_{n+1} = J\overline{Q_n} + \overline{K}Q_n$$

$$Q_{n+1} = J_A \overline{Q_A} + \overline{k_A} Q_A$$

$$= Q_B \overline{Q_A} + 1 Q_A$$

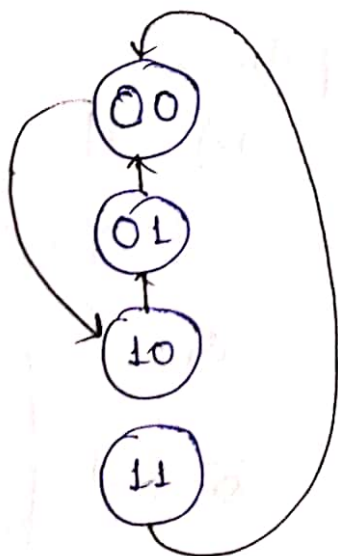
$$Q_{n+1} = Q_B \overline{Q_A}$$

for FF B -

$$Q_{B+1} = J_B \overline{Q_B} + \overline{k_B} Q_B$$

$$= \overline{Q_A} \overline{Q_B} + 0 \cdot Q_B$$

$$Q_{B+1} = \overline{Q_A} \overline{Q_B}$$



Note:- This question was example of Analysis

Circuit $\rightarrow$ Nextstate

Design		
PS	NS	FF/JP
✓	✓	?

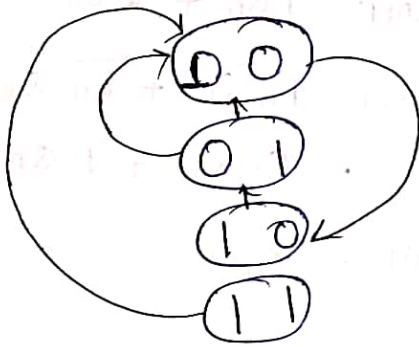
Excitation table

Analysis		
FF/JP	PS	NS
✓	✓	?

Check table

Analysis 2

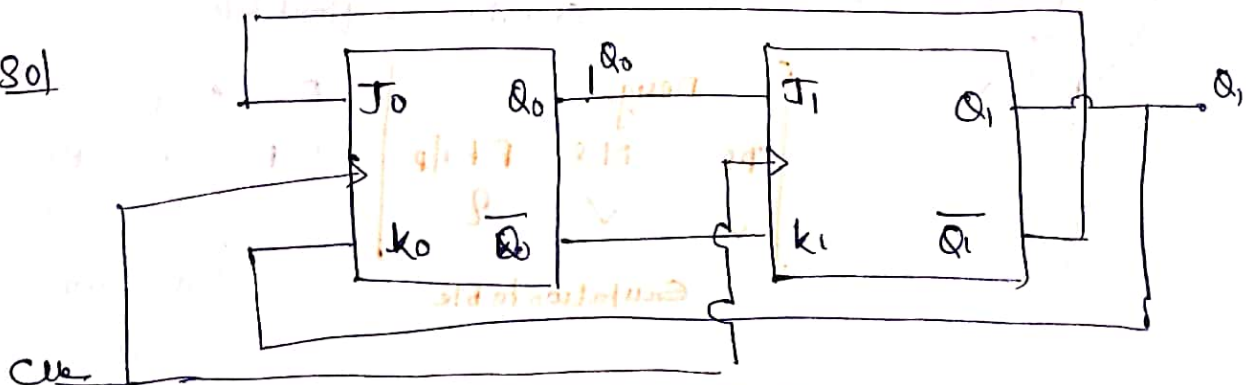
Present state		FF 1/p				Next state	
Q_B	Q_A	$J_B = \overline{Q_A}$	$k_B = 1$	$J_A = Q_B$	$k_A = 1$	Q_{B+1}	Q_{A+1}
0	0	1	1	0	1	1	0
0	1	0	1	0	1	0	0
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	0



Que. In the following sequential ckt, the initial state (before the first clock pulse) of the circuit is $Q_1 Q_0 = 00$. The state (Q_1, Q_0) , immediately after the 333rd clock pulse is.

i) 00 ii) 01 iii) 10 iv) 11

Sol



Sol

$$J_1 = Q_0$$

$$K_1 = \overline{Q_0}$$

$$J_0 = \overline{Q_1}$$

$$K_0 = Q_1$$

Note: If 10 MHz clk applied to Cascade of Mod 5 Mod 2 & two Mod 10 Counter then lowest clk freq is

Total states to be counted = $5 \times 2 \times 10 \times 10 = 4000$

$$f_{out} = \frac{f_{clk}}{4000} = 2.5 \text{ kHz}$$

By Method 1.

FFD

$$NSE \quad Q_{n+1} = J \overline{Q_n} + \overline{K} Q_n$$

$$Q_0^+ = J_0 \overline{Q_0} + \overline{K_0} Q_0$$

$$= \overline{Q_1} \overline{Q_0} + \overline{Q_1} Q_0$$

$$= \overline{Q_1} (Q_0 + \overline{Q_0})$$

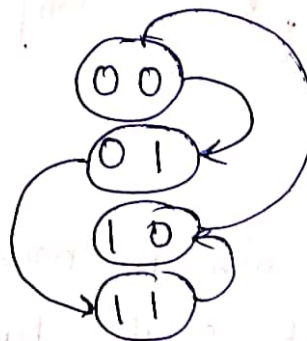
$$Q_0^+ = \overline{Q_1}$$

$$\text{for } Q_1^+ = J_1 \overline{Q_1} + \overline{K_1} Q_1$$

$$= Q_0 \overline{Q_1} + \overline{Q_0} Q_1$$

$$Q_1^+ = Q_0$$

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	1
1	0	0	0
1	1	1	0



By Method 2

Present st.

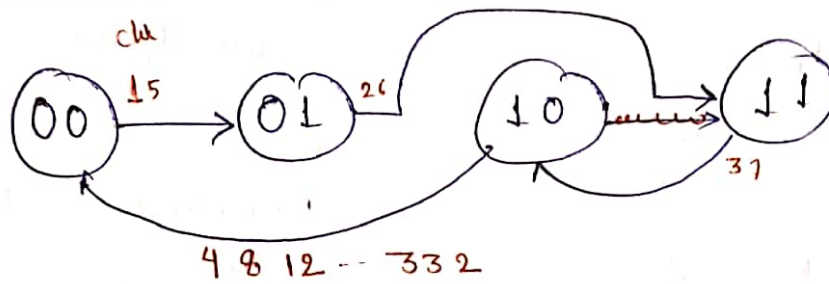
Q_1	Q_0
0	0
0	1
1	0
1	1

ff // p

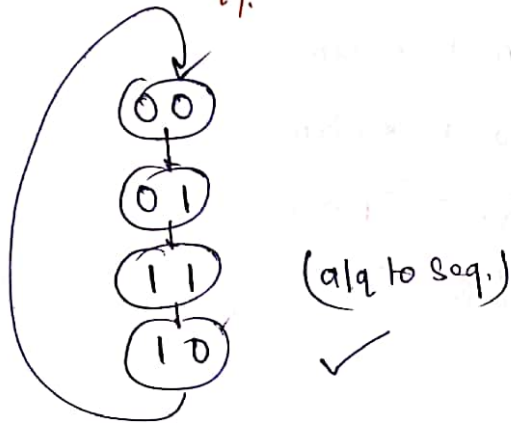
J_0 $= \overline{Q_1}$	K_0 $= Q_1$	J_1 $= Q_0$	K_1 $= \overline{Q_0}$
1	0	0	1
1	0	1	0
0	1	0	1
0	1	1	0

Next stat

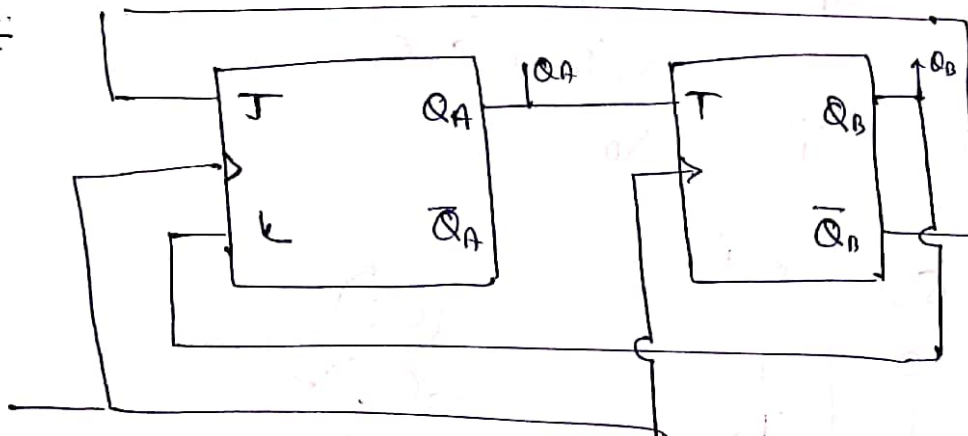
Q_1^+	Q_0^+
0	1
1	1
0	0
1	0



Q. 333 = 01 //



Que.



A 2 bit Counter is shown. If state $Q_A Q_B$ of Counter at clock time t_n is "10" then state $Q_A Q_B$ of Counter at t_{n+3} (after 3 clock cycle) will be

Sol $T_B = Q_A$ $J_A = \overline{Q_B}$
 $K_A = Q_B$

Method 1

NSE:

$$FF1: Q_{A+1} = J \bar{Q}_n + \bar{k} Q_n$$

$$= \bar{Q}_B \bar{Q}_n + Q_B Q_n$$

$$Q_{A+1} = Q_B \otimes Q_n$$

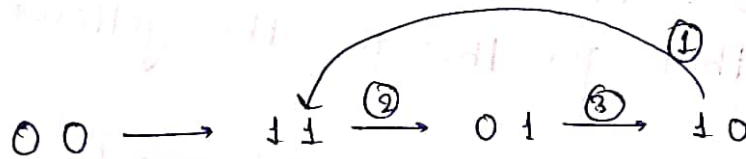
Q_n	Q_B	$Q_{B+1} Q_{A+1}$
0	0	0 1
0	1	1 0
1	0	1 0
1	1	0 1

FF2:

$$Q_{B+1} = Q_B \oplus T$$

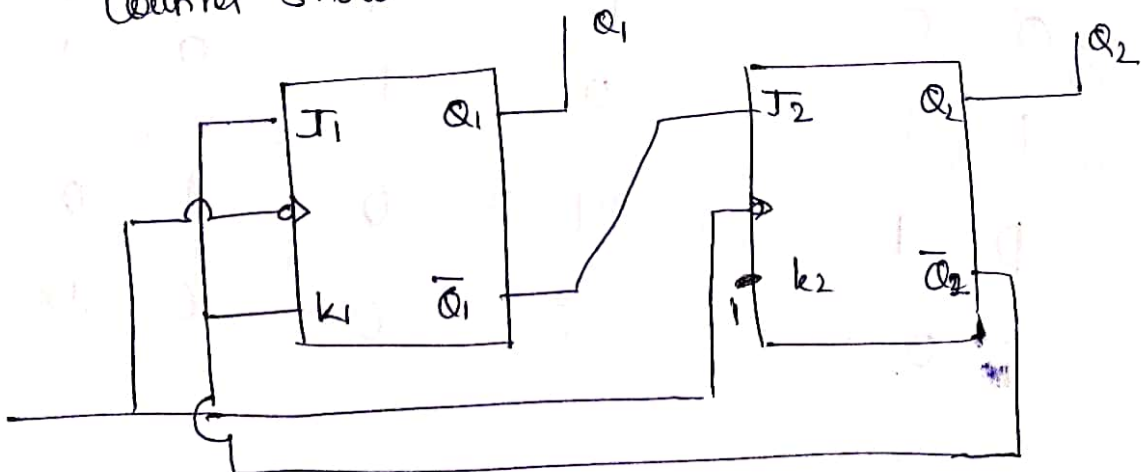
Method 2

Q_n	Q_B	$J_A = \bar{Q}_B$	$k_A = Q_B$	$T_B = Q_A$	Q_{A+1}	Q_{B+1}
0	0	1	0	1	1	1
0	1	0	1	0	0	0
1	0	1	0	1	1	1
1	1	0	1	0	0	0



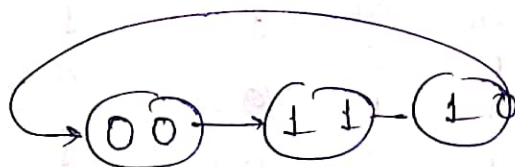
Ans 10

Que. what are the counting states (Q_1, Q_2) for counter shown.



Sol-

Present state		ff i/p				MS
Q_1	Q_2	$J_1 = \overline{Q_2}$	$K_1 = \overline{Q_2}$	$J_2 = \overline{Q_1}$	$K_2 = 1$	$Q_1^+ Q_2^+$
0	0	1	1	1	1	1 1
1	1	0	0	0	1	1 0
1	0	1	1	0	1	0 0



Que 2 Two D ff are Connected as a Synchronous Counter that goes through the following $Q_B Q_A$ Sequence

$00 \rightarrow 11 \rightarrow 01 \rightarrow 10 \rightarrow 00$

The Connection to i/p D_A & D_B are

<u>Sol</u>	Q_A	Q_B	Q_{A+1}	Q_{B+1}	D_A	D_B
	0	0	1	1	1	1
	1	1	0	1	0	1
	1	0	1	0	1	0
	0	1	0	0	0	0

Designing question

Excitation table of D FF

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

note:
Adjust truth table to before using kmap

$$D_A = \overline{Q_A} \overline{Q_B} + \overline{Q_A} Q_B = \overline{Q_A}$$

$$D_B = \overline{Q_A} \overline{Q_B} + Q_A Q_B = \oplus$$

But Question asked Q_B, Q_A

Hence $D_B = \overline{Q_B}$

	Q_B	Q_A	Q_{B+1}	Q_{A+1}	D_B	D_A
0	0	0	1	1	1	1
3	1	1	0	1	0	1
1	0	1	1	0	1	0
2	1	0	0	0	0	0

$D_B = \overline{Q_B}$

$Q_B \backslash Q_A$	0	1
0	1	1
1	0	0

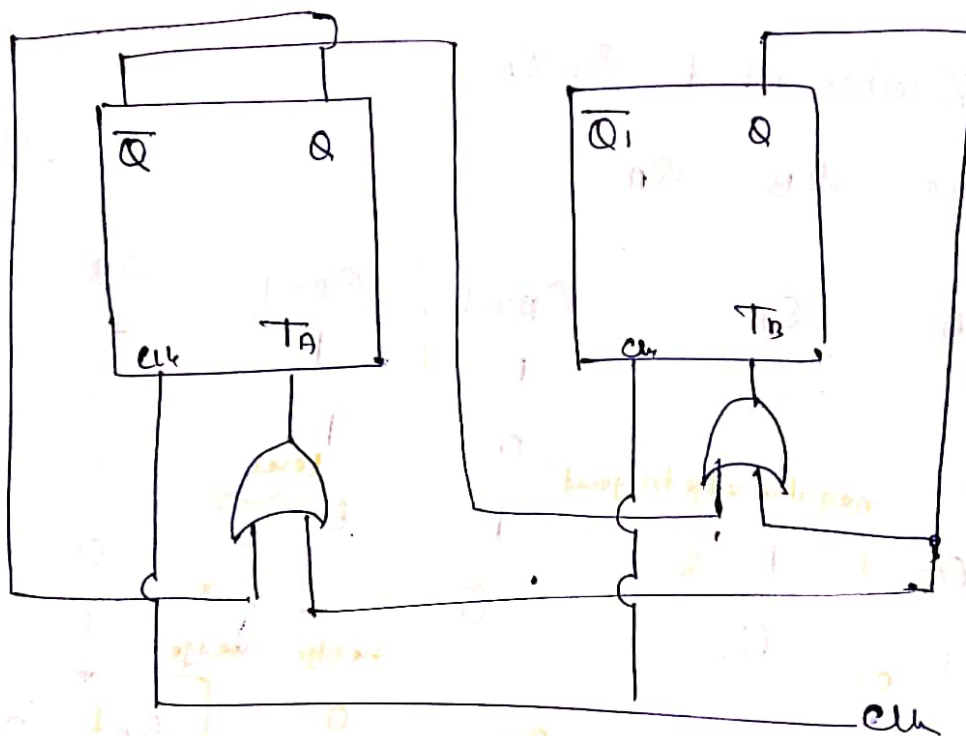
$$D_B = \overline{Q_B}$$

$$D_B = \overline{Q_B}$$

$$D_A = Q_A \odot Q_B$$

Que. A Sequential ckt is shown in fig. Let the state of ckt be encoded as $Q_1 Q_0$. The notation $X \rightarrow Y$ implies that state Y is reachable from X in a finite no. of clock X state. Identify incorrect statement

- A) $00 \rightarrow 00$
 B) $11 \rightarrow 01$
 ✓ C) $01 \rightarrow 11$
 d) $01 \rightarrow 10$



Sol:

$$T_A = Q_A + Q_B$$

$$T_B = \overline{Q_A} + Q_B$$

Present state

FF i/p

Q_A Q_B

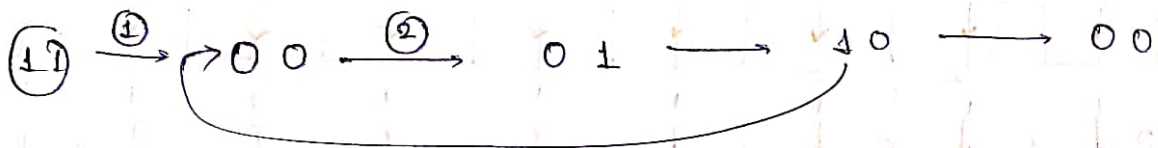
$T_A = Q_A + Q_B$

$T_B = \overline{Q_A} + Q_B$

next state

Q_{A+1} Q_{B+1}

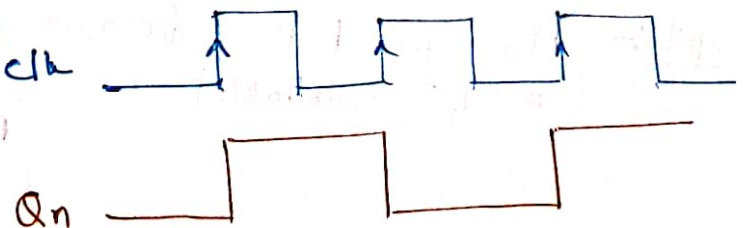
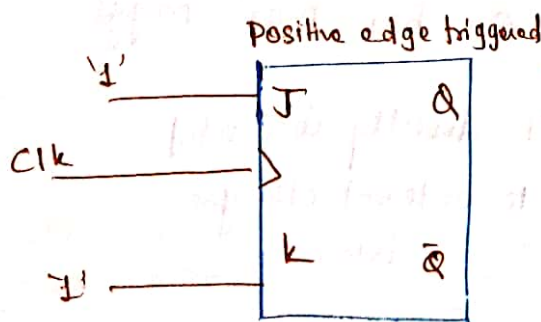
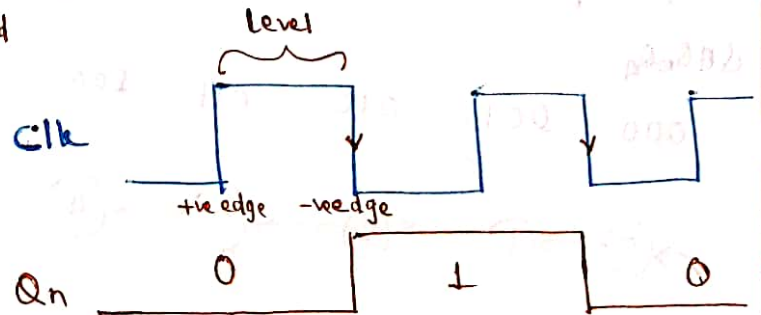
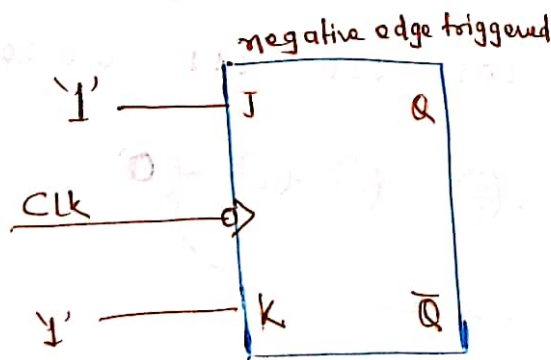
0	0	0	1	0	1
0	1	1	1	1	0
1	0	1	0	0	0
1	1	1	1	0	0



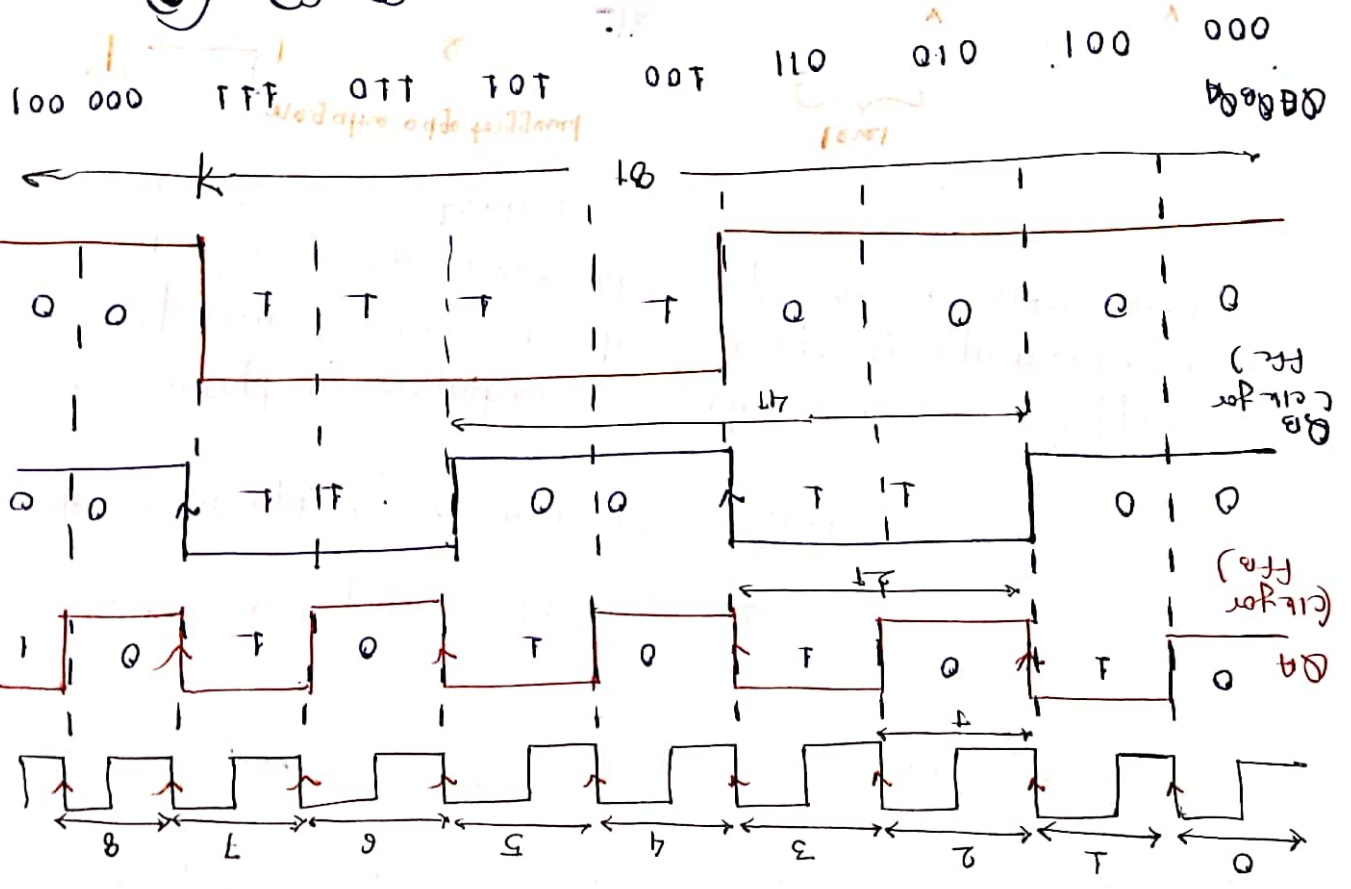
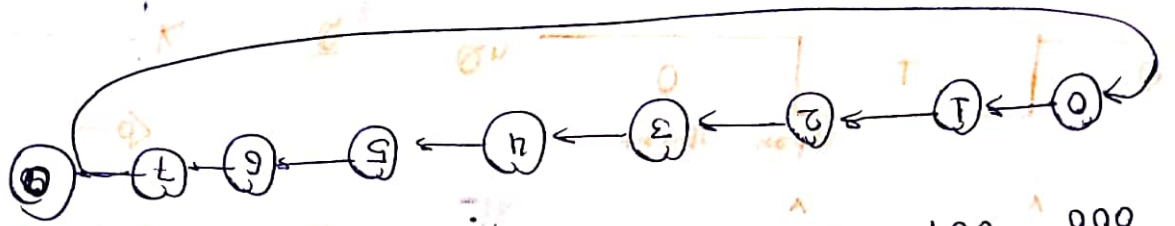
After 2 clock pulse 11 → 01

* Concept of Asynchronous Counters

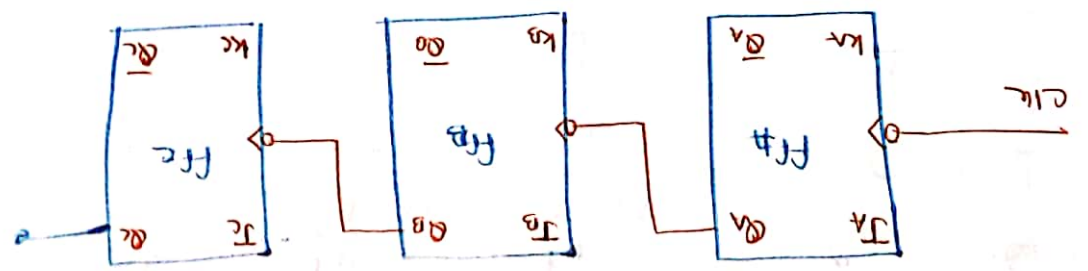
- Concept of Asynchronous Counter is based on Toggling operation where as the Concept of Synchronous Counter is based on next state eqn (or characteristic table or Excitation table)



Note: FFs give LSB because 1st has Max no going to external clk give
 (as LSB of truth table)



All $T = K = 1$



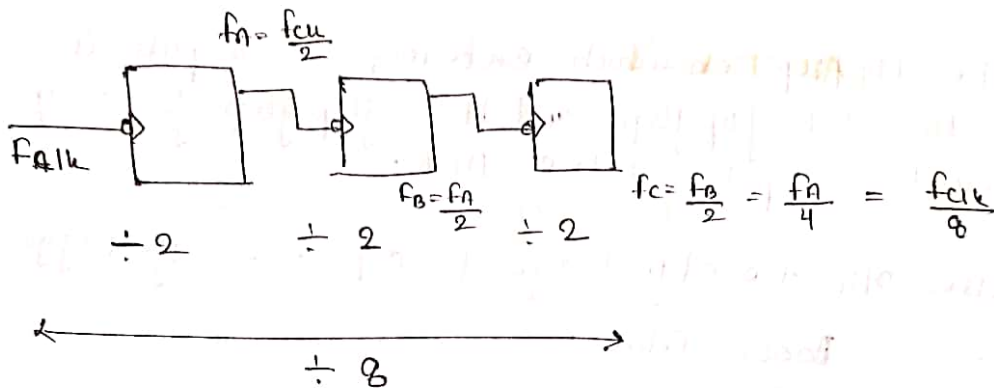
• Mod 8 Counter up Asynchronous Counter
 3bit up Counter

Divide by 8 up asyn. Counter

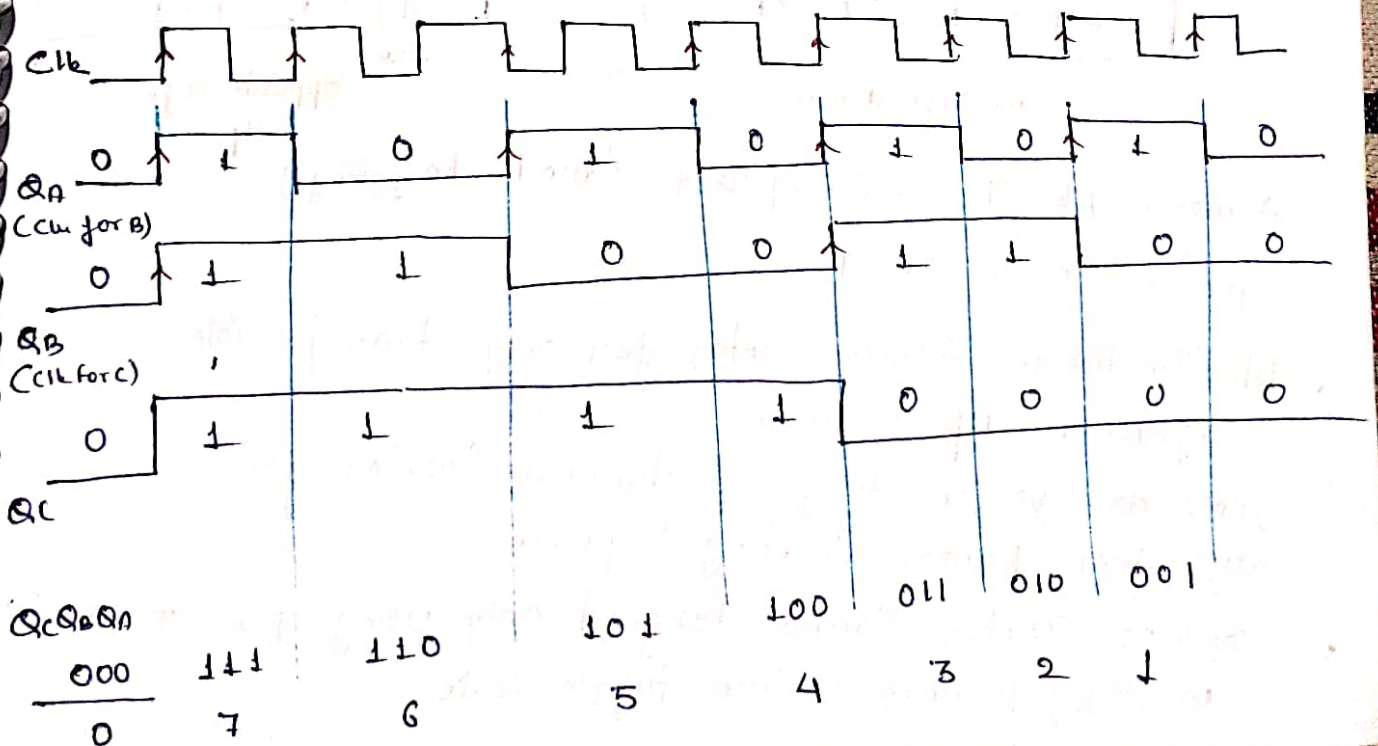
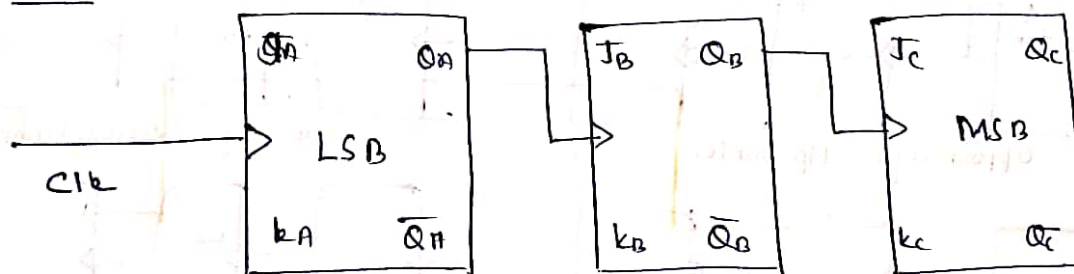
$$f_c = \frac{f_{clk}}{8}$$

{ frequency }

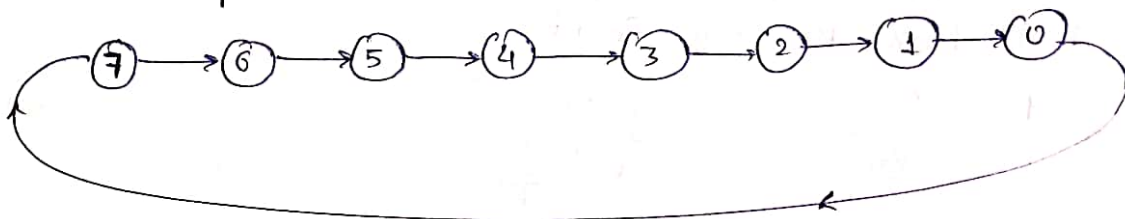
$$f = \frac{1}{T}$$



Que.



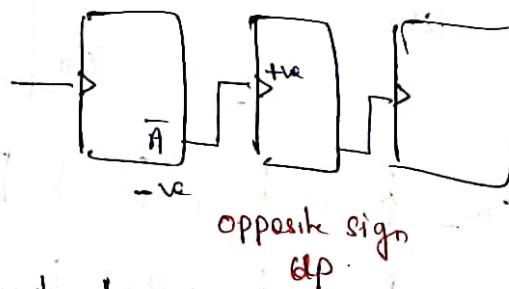
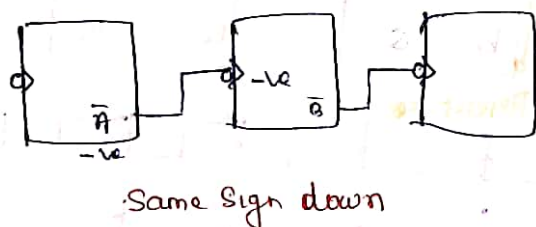
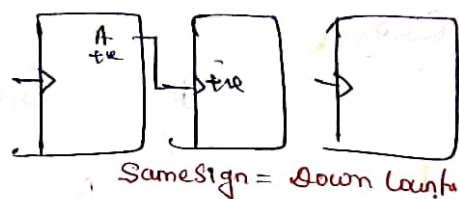
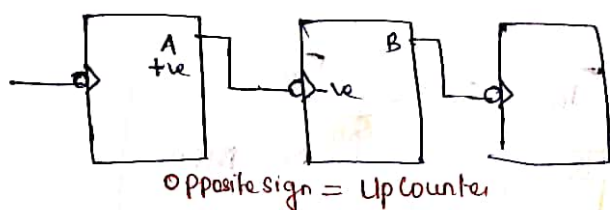
It is Mod 8 Down asyn. Counter / 3 bit down Asyn Counter.
Divide by 8 down asyn Counter.



Note:- The Flipflop on which external clock pulse is applied is the LSB flipflop and the flipflop far away from external clock pulse is the MSB.

Note:- Use All the edge triggered or $\neg$ edge triggered clock. Do not Mix.

* Identification of Asyn. Counter



Asyn $\rightarrow$ JK, T flip only used (due to toggling)

Sync $\rightarrow$ SR, D, JK, T

- We can design Asyn Counter for only two possible Sequence; Up & down

Where as We can design Synchronous Counter for any given Random Counting Sequence.

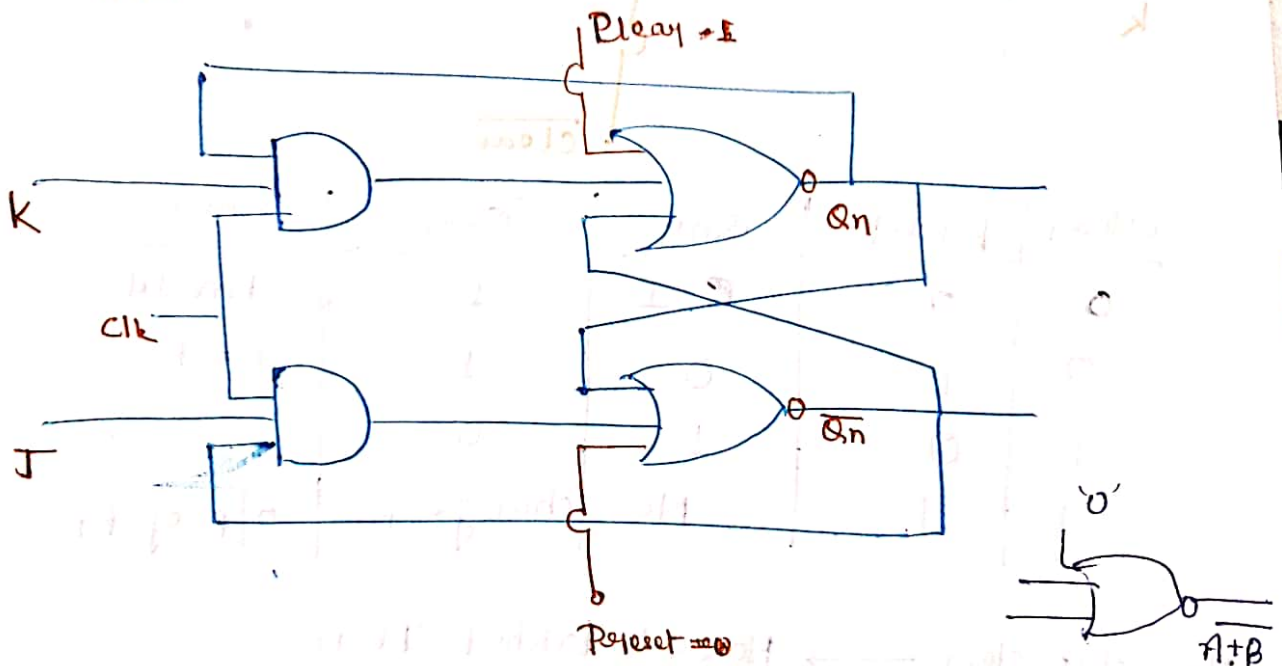
- Asyn. Counter can be designed only using JK & T flip or flipflop which contain toggle state.

- Under the Input Condition $J = K = 1$ or $T = 1$
- Sync. Counter can be designed using SR flip flop, D flip flop, JK FF, T, FF or any other flip flop whose next state eqⁿ is given.

* Asynchronous Inputs (Clear & preset)

OR

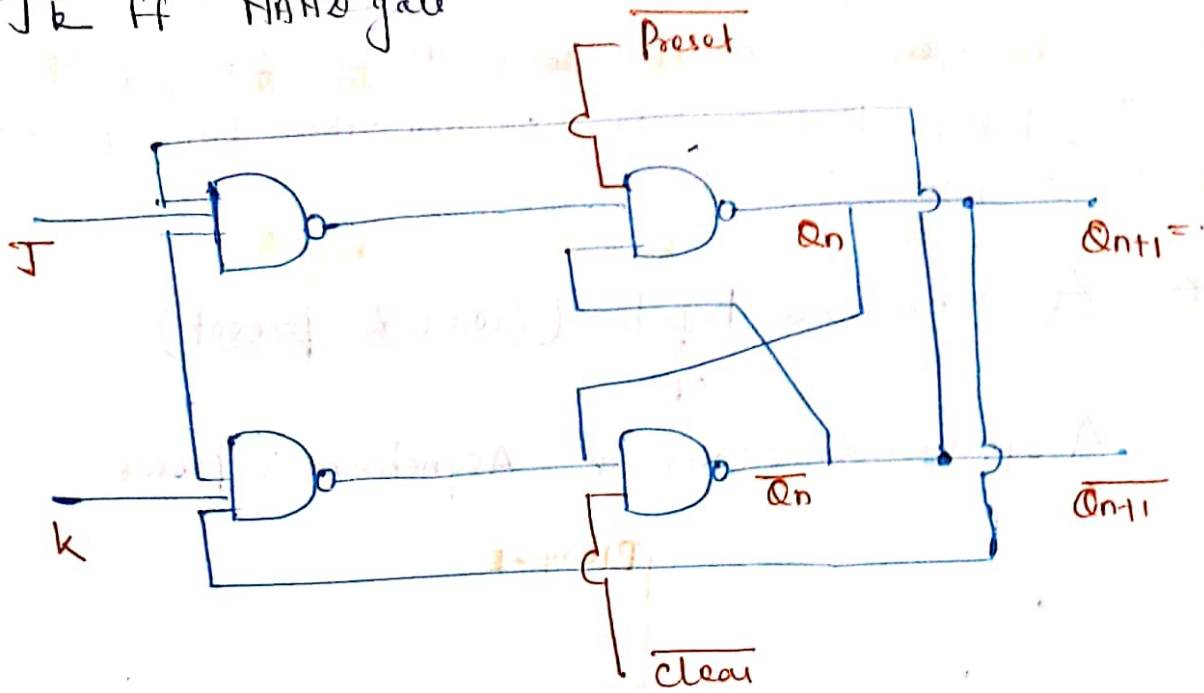
A Synchronous clear & Asynchronous preset



Clear	Preset	Q_{n+1}	$\overline{Q_{n+1}}$	State
0	0	No changes in o/p of ff		
0	1	1	0	Set
1	0	0	1	Reset (Clear)
1	1	0	0	Invalid

↔ we never use this i/p combination.

Jk ff NAND gate

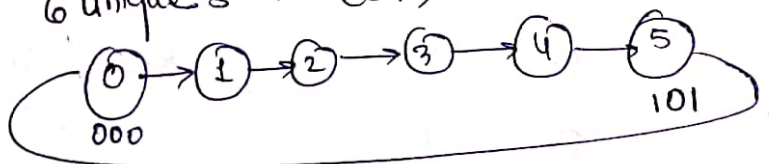


$\overline{\text{Clear}}$	$\overline{\text{Preset}}$	Q_{n+1}	$\overline{Q}_{n+1}$	State
0	0	0	1	Invalid
0	1	0	1	Reset
1	0	1	0	Set
1	1	No	change in	o/p of ff

Active High clear $\rightarrow$ NOR gate latched Jk ff
 Active low clear $\rightarrow$ NAND gate latched Jk ff

* Mod 6 Asy. Up Counter

6 unique states (WP)



All J & K = 1

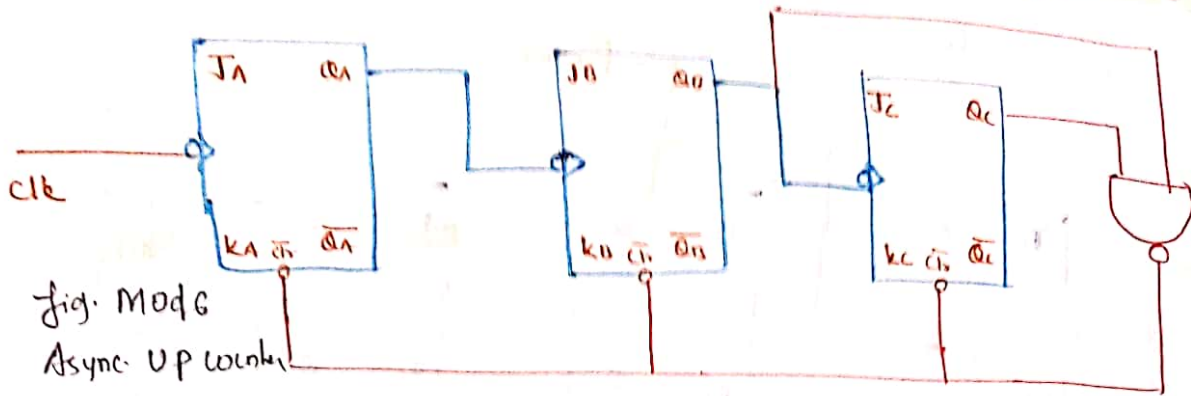


Fig. Mod 6
Async. Up counter

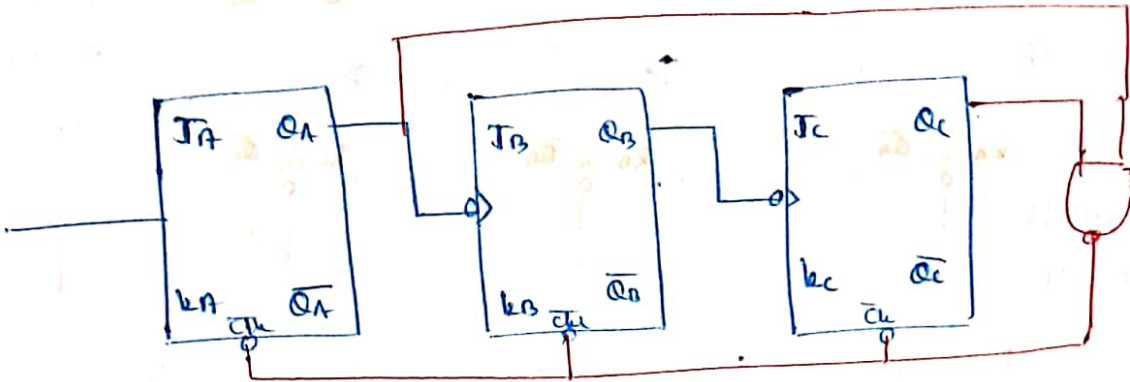
Q_C	Q_B	Q_A (LSB, clk input)	$\overline{Clr}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Q_C	Q_B	Q_A	$Q_B Q_A$	$Q_B \overline{Q_A}$	$\overline{Q_B} Q_A$	$\overline{Q_B} \overline{Q_A}$
0	0	0	1	1	1	1
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	1	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

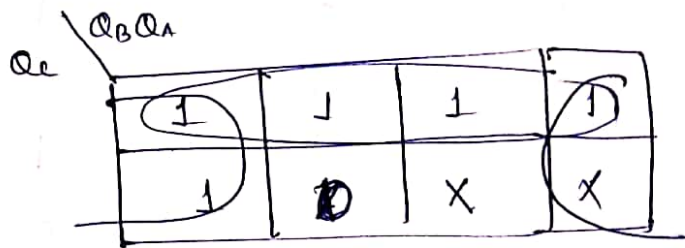
$\overline{Q_C} + \overline{Q_B}$
 $\Rightarrow \overline{Q_C Q_B}$

Mod 6
Counter
So up to 5
it will not
use clr

Mod 5 Counter:

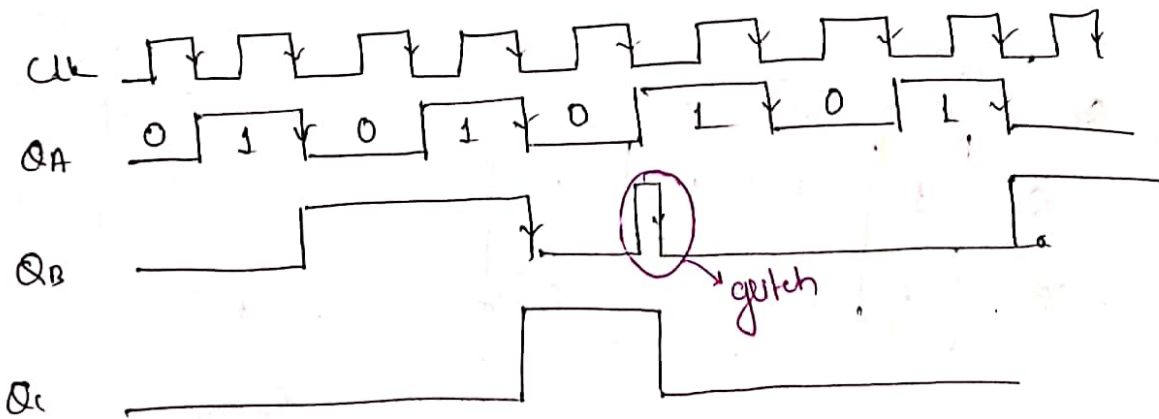


Q_C	Q_B	Q_A	$\overline{Clr}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0



$$\overline{Q_A} + \overline{Q_C}$$

$$\overline{Clr} = \overline{Q_A} \cdot \overline{Q_C}$$

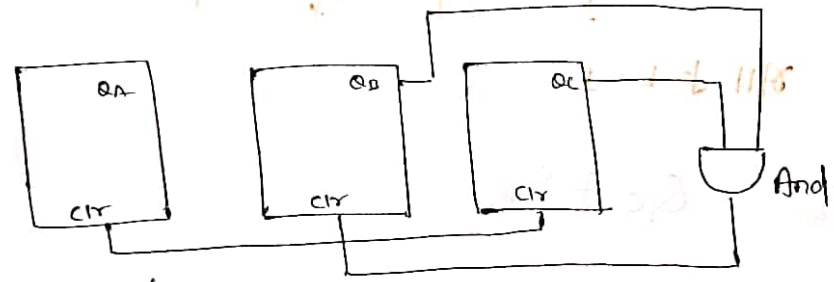


Note: CTR Input for all Mod n Counter

Q _c	Q _B	Q _A	CTR	for High clear
0	0	0		
0	0	1	$\overline{Q_A}$	Mod 1
0	1	0	$\overline{Q_B}$	Mod 2
0	1	1	$\overline{Q_B Q_A}$	Mod 3
1	0	0	$\overline{Q_C}$	Mod 4
1	0	1	$\overline{Q_C Q_A}$	for Mod 5
1	1	0	$\overline{Q_C Q_B}$	for Mod 6
1	1	1	$\overline{Q_C Q_B Q_A}$	for Mod 7

Diagram of a 3-input AND gate with inputs $\overline{Q_A}$, $\overline{Q_B}$, and $\overline{Q_C}$. The output is labeled '0'.

* Mod 6 Active High clear



Q_B Q_A

0	0	0	0
0	0	X	1

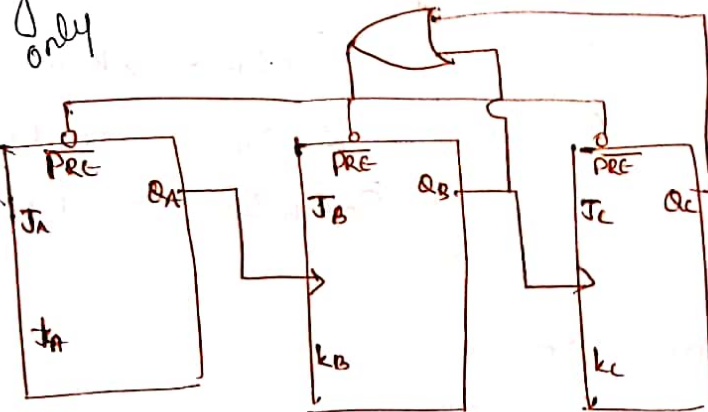
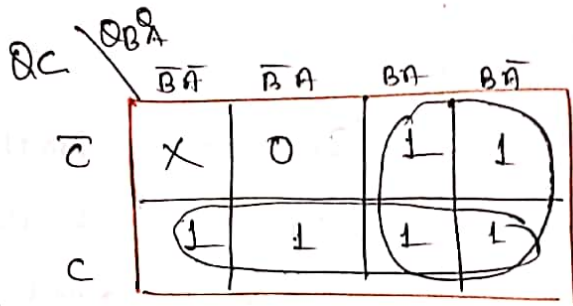
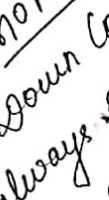
Q_C

Q_C Q_B

Q _C	Q _B	Q _A	Clr
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

Mod 6 Asynchronous Down Counter

note:
down counter
always start from
111 state only
and up counter
start from
000 only


$$2f_{11} \quad J=L=1$$

$$\overline{P_{RE}} = Q_C + Q_D$$

Q_C	Q_B	Q_A	$\overline{PRE}$
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	1
0	0	1	0
0	0	0	X

Q_C	Q_B	Q_A	$\overline{PRE}$	PRE
1	1	1	↓	
1	1	0	Mod 1	$\overline{Q_A}$
1	0	1	Mod 2	$\overline{Q_B}$
1	0	0	Mod 3	$\overline{Q_B + Q_A}$
0	1	1	Mod 4	$\overline{Q_C}$
0	1	0	Mod 5	$\overline{Q_C + Q_A}$
0	0	1	Mod 6	$\overline{Q_C + Q_B}$
0	0	0	Mod 7	$\overline{Q_C + Q_B + Q_A}$

Que 1 find $\overline{CTR}$ input Mod 21 Asyn. up Counter
 $(21)_{10} \rightarrow (10101)_2$

$Q_E \quad Q_D \quad Q_C \quad Q_B \quad Q_A$
 $1 \quad 0 \quad 1 \quad 0 \quad 1$

$$\overline{CTR} = \overline{Q_E Q_C Q_A}$$

Que 2 Mod 37 asyn. Down with pre input

Sol $(37)_{10} \rightarrow (100101)_2$

$Q_F \quad Q_E \quad Q_D \quad Q_C \quad Q_B \quad Q_A$
 $1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1$

Method 1

Targeting '0'

$100101 \xrightarrow{\text{down } Q_F} 011010$

$$\overline{Q_F + Q_C + Q_A}$$

Method 2

Targeting '1'

$$\overline{Q_F + Q_C + Q_A}$$

If Target '1'
down Count
Then Not gate

Que 3 Mod 14 asyn. down using pre

Sol

$(14)_{10} \rightarrow (1110)_2$ 0001

$Q_D \quad Q_C \quad Q_B \quad Q_A$
 $1 \quad 1 \quad 1 \quad 0$

targeting '0'

$$\overline{Q_A}$$

target '1' down

0001

$$\text{pre} = \overline{Q_A}$$

note Always target '1'

② Mod 25 up using clr

sol $(25)_{10} \longrightarrow 11001$

(11001)

$Q_E Q_D Q_C Q_B Q_A$

$\downarrow \quad \downarrow \quad \downarrow$

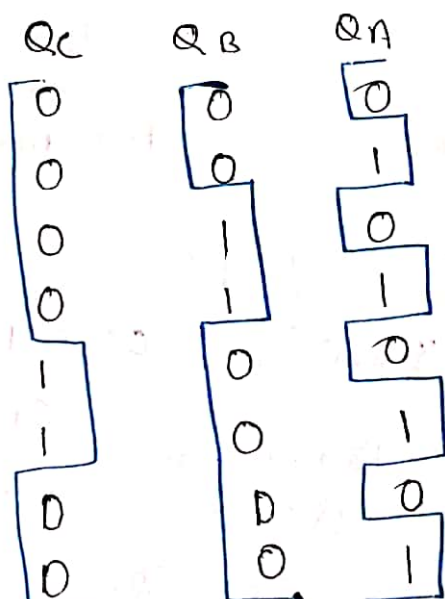
$Q_E Q_D Q_A$

$clr = Q_E Q_D Q_A //$

① Mod 14 down Counter ^{pre}
 $(14)_{10} \longrightarrow (1110)_2$

Down to 0001
 Target zero:- $Q_D + Q_C + Q_B$

* Mod 6 Clock frequency ÷

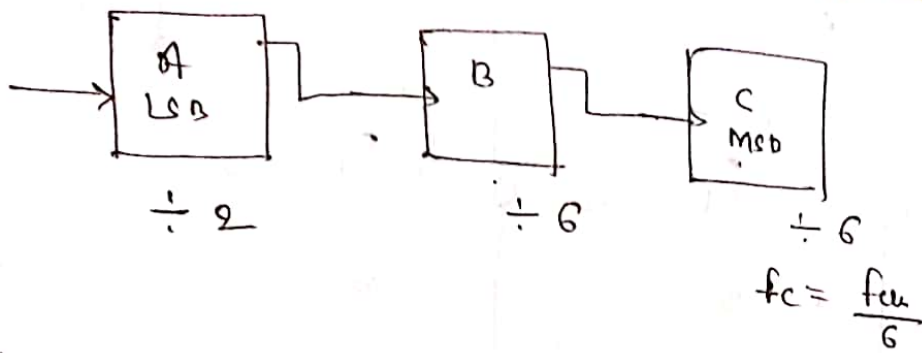


$f_A = \frac{f_{ck}}{2}$

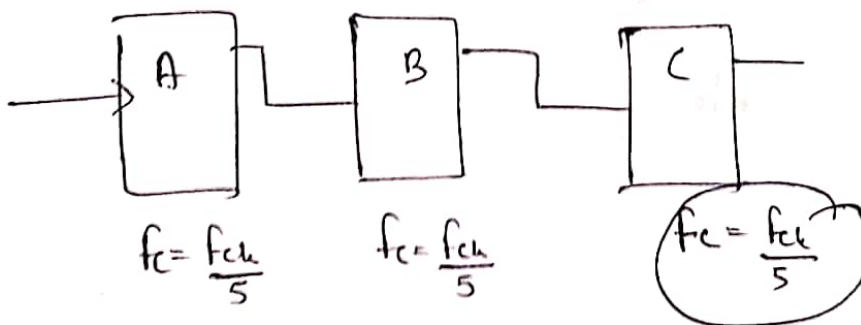
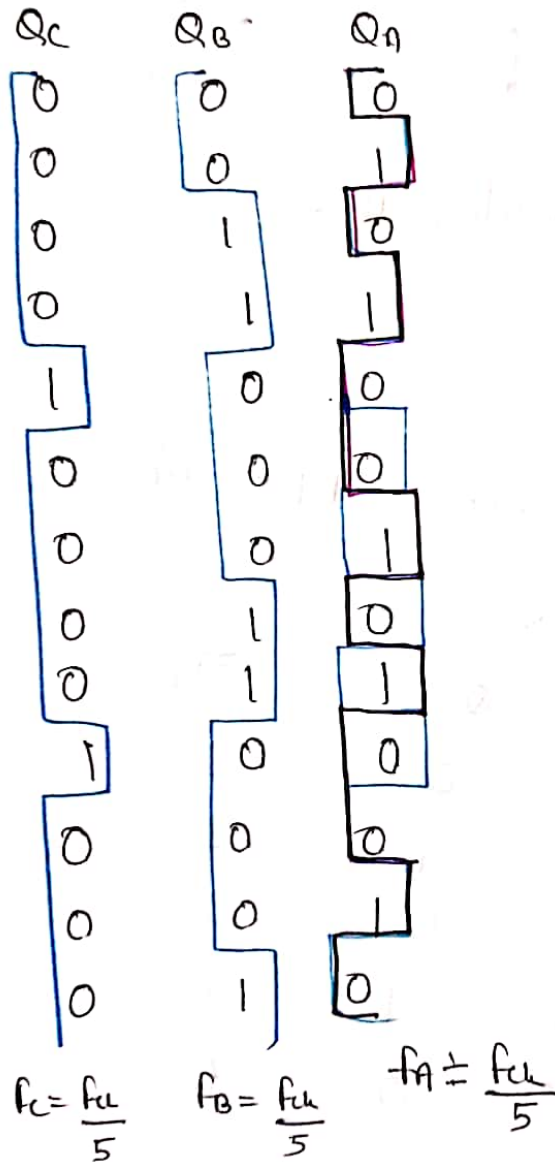
MSB has
 $\div 6$
 Hence
 mod 6

$f_c = f_k / 6$

$f_B = \frac{f_{ck}}{6}$



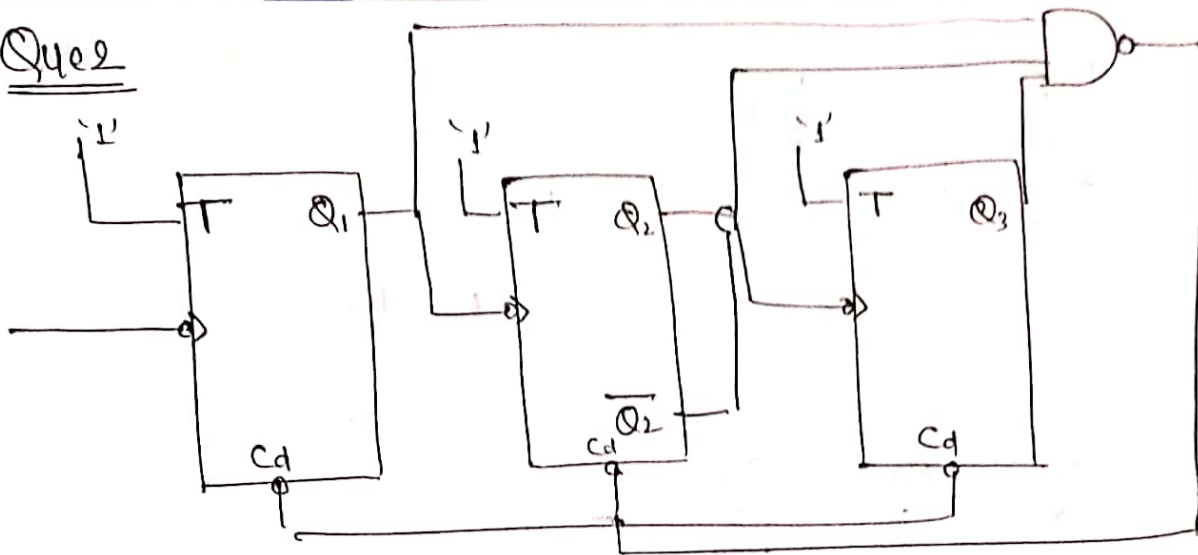
Mod 5



Note: check MSB and Comment on Mod 14 Counter

That's Why $\div 5$ Counter MSB*

Que 2



The Mod Value of Counter is —

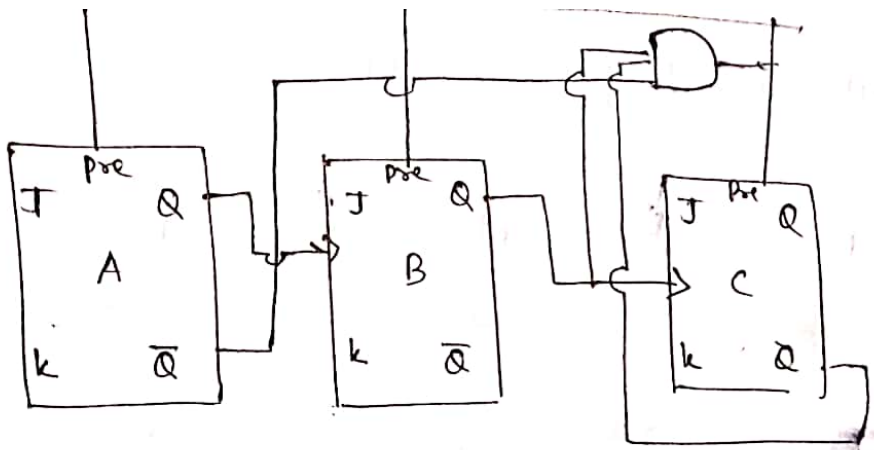
Sol Observation:-

- i) $\overline{CLR}$ is active low
- ii) +ve -ve connection UP counter
- iii) $\overline{CLR} = Q_1 \overline{Q_2} Q_3$

Q_1	Q_2	Q_3	$\overline{CLR}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0

Mod 5 Counter.

Que.



All $J = K = 1$

sol which Counter?

+ve +ve $\rightarrow$ down Counter

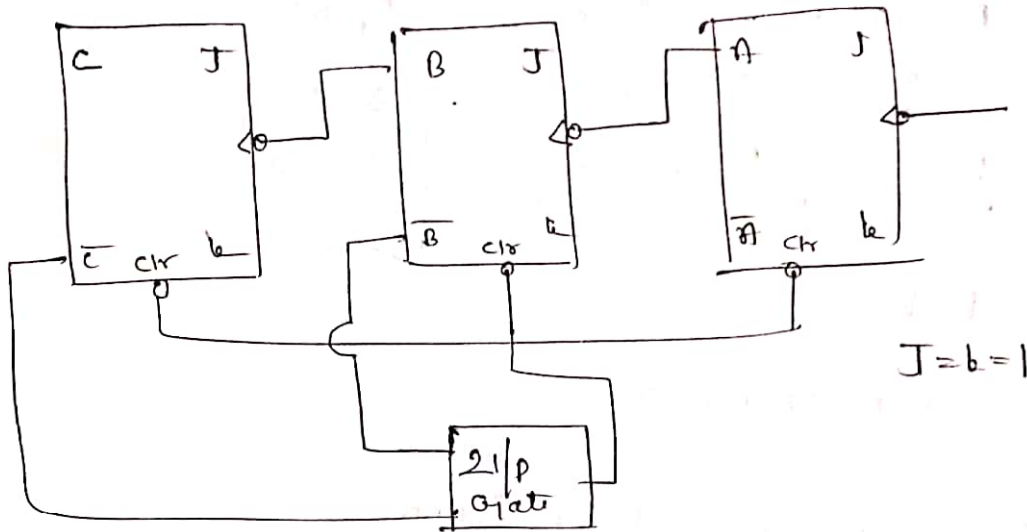
$$Pre = \overline{Q_C} * Q_B * \overline{Q_A} \quad \overline{Q_C} \quad Q_B \quad \overline{Q_A}$$

Q_C	Q_B	Q_A	Pre
1	1	1	0
1	1	0	0
1	0	1	0
1	0	0	0
1	0	1	0
0	1	1	0
0	1	0	1
0	0	1	

Mod 5 down Counter

preset 1 i.e again start
1 1 1

Que. Modulo-6 J-K Asynchronous Counter



Sol A is B (clk applied)
C MSB

+ - Up Counter

$$\text{Clr} = \overline{C} \cdot \overline{B}$$

up Counter i.e	C	B	A	Modulo 6 $\overline{\text{Clr}}$
	0	0	0	1
	0	0	1	1
	0	1	0	1
	0	1	1	1
	0	0	0	1
	1	0	1	1
	1	1	0	0

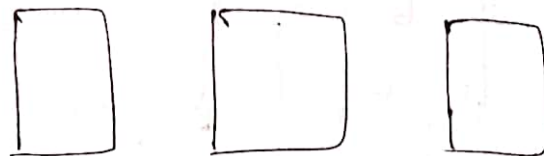
$\overline{\text{Clr}} = 1$
for not
activate

$$\overline{C} + \overline{B} = 0$$

Hence OR

Que. A 0 to 6 Counter consist of 3 FF and a Combinational Circuit of 2 input gate(s). The Combination Circuit consist of —

Sol



~~$Q_c Q_b Q_a$~~
+ + +
understand

0 to 6 means Mod 7 up Counter

not req. $\downarrow$ $\downarrow$ $\downarrow$
C B A

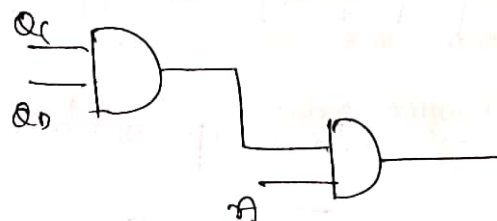
i) $\overline{clr}$

$Q_c Q_b Q_a$

not possible (two 1/p gate)

ii) clr

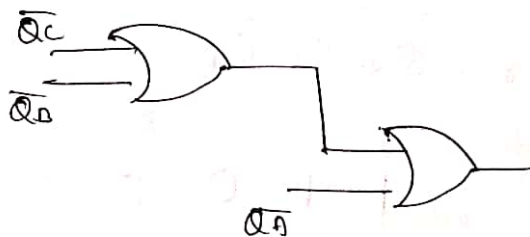
$Q_c Q_b Q_a$



also

~~$Q_c Q_b Q_a$~~ $\rightarrow \overline{Q_c} + \overline{Q_b}$

$\overline{Q_c} + \overline{Q_b} + \overline{Q_a}$



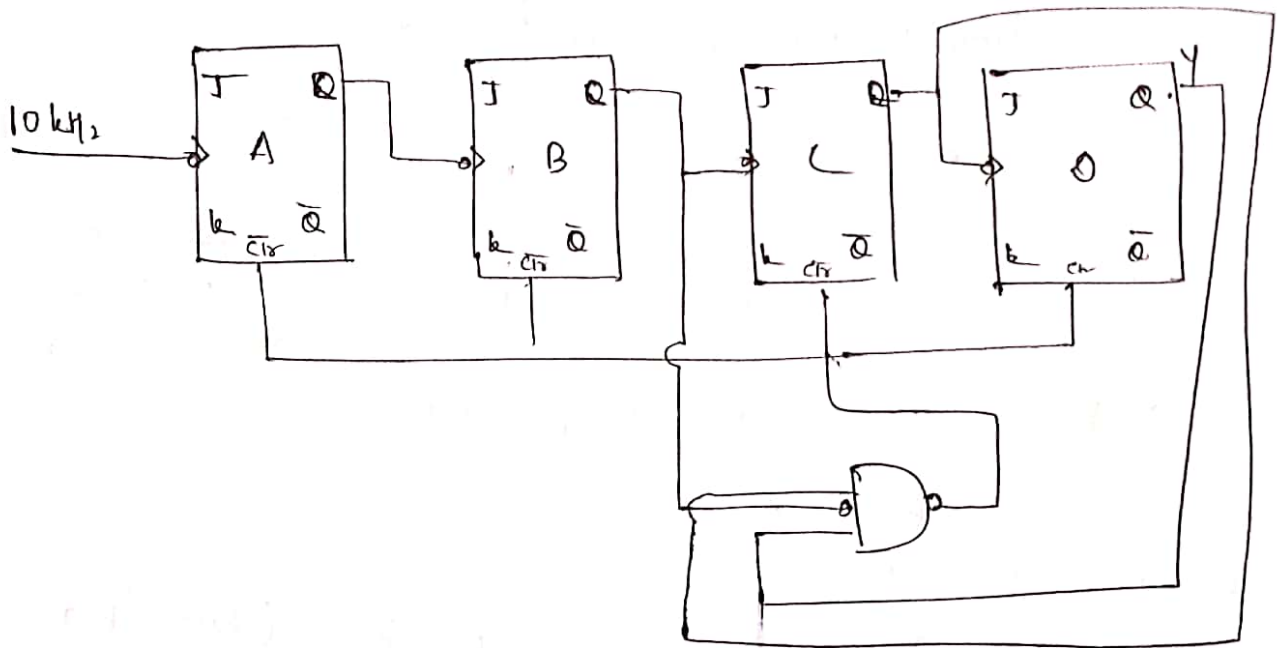
Ans. Two And or Two OR gates

Que

P.T.O

Note: Up counter uses clr or $\overline{clr}$

Que.



The freq of sig at o/p y is

Sol) $\therefore$ i.e up counter

$$\overline{\text{CLR}} = \overline{Q_B Q_C Q_D}$$

$$\overline{\text{CLR}} = Q_D Q_C \overline{Q_B}$$

$\overline{\text{CLR}}$ i.e 1/p 0 to set

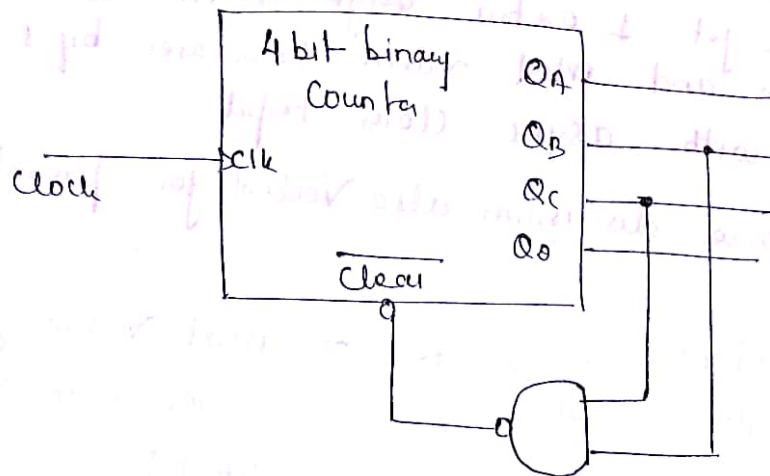
1 1 0 0 $\rightarrow$ it will set clr to 1

i.e Mod. 12

$$f_{out} = \frac{10}{12} = 5/6 \text{ kHz}$$

* Synchronous Clear & Synchronous 1/p
i.e Synchronized with the clock pulse.

Que. A Modn Counter using a Sync. binary up Counter with Synchronous clear input is shown in the figure. The value of n is —



Sol. - Synchronous clear $\rightarrow$ Work same as Async clear but activate on -ve or +ve edge of clock pulse depending on active High or Active Low Clk.

$$\overline{\text{Clear}} = \overline{Q_C Q_D}$$

$$\overline{\text{Clear}} = \overline{Q_C Q_D} = (0)$$

Q_D	Q_C	Q_B	Q_A
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
1	1	0	0
1	1	0	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1

In this que. it doesn't matter Q_A or Q_B be MSB or LSB because Q_C & Q_D are used for clear

$$\overline{Q_C Q_D} = 0 \Rightarrow \overline{\text{Clear}} = 1$$

$$N = 8$$

but due to Sync clear it wait for next cycle $N = 8 + 1 = 9$

In case of Sync. clear input, the clear i/p cannot be activated between the two active edges.

It has to wait for next active edge.

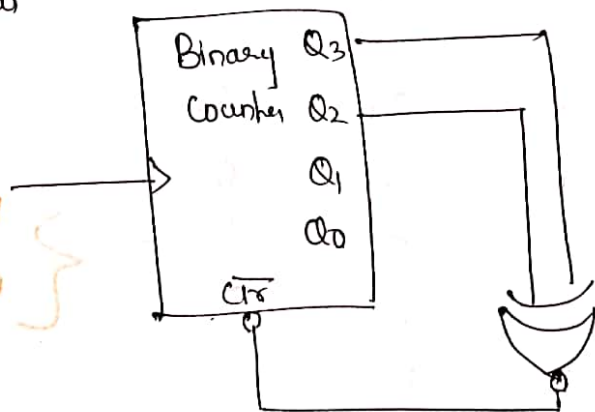
Hence we get 1 extra ~~edge~~ ^{state} in the counting sequence and Mod value increases by 1

in Comp. with async. clear input.

Note: Above discussion also Valid for preset i/p

$$\text{Mod Value of Counter (Sync or Async)} = \text{Mod Value of Same Counter with async. clear i/p} + 1$$

Que. Figure shows a binary Counter with Sync. clear i/p with decoding logic shown, the counter work as



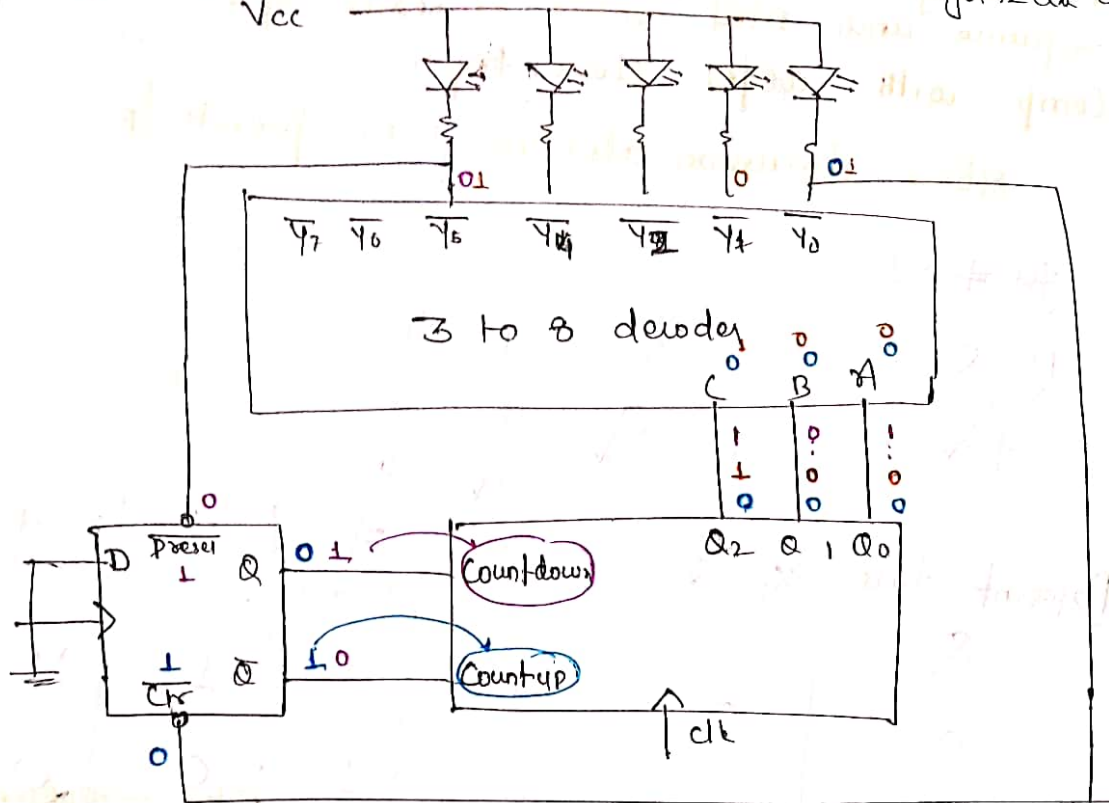
Clr $\rightarrow$ up/down
preset $\rightarrow$ down

Sol $\overline{\text{Clr}} = Q_3 \oplus Q_2$

Q_3	Q_2	Q_1	Q_0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1

$Q_3 Q_2 = 0 \Rightarrow \text{Mod } 4$
but Sync clear then $\text{Mod } 4 + 1 = \text{Mod } 5$

Q.4e. Circuit shows up/down Counter working with decoder & FF. preset & clear of FF are active low input. Assuming that initial value of Counter o/p ($Q_2 Q_1 Q_0$) as zero. The Counter o/p in decimal for 12 clock cycles as



Sol: $Q_2 \quad Q_1 \quad Q_0 \rightarrow \begin{matrix} C & B & A \\ Q_2 & Q_1 & Q_0 \\ 0 & 0 & 0 \end{matrix}$

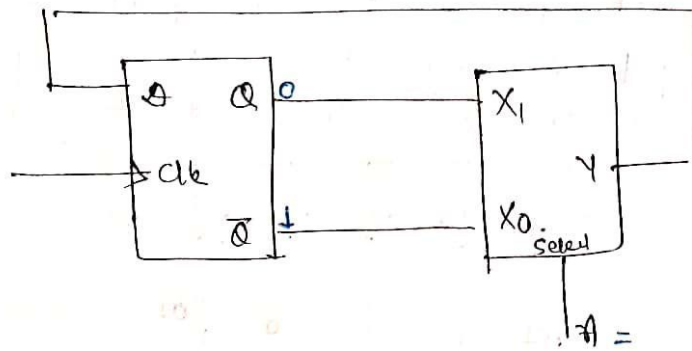
$$\overline{clr} = \overline{Y_0}$$

$$\overline{pre} = \overline{Y_5}$$

Sequence $\rightarrow 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 4 \ 3 \ 2 \ 1 \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 //$

Note: clock is grounded So 0 i/p will never pass to Q

Que. The state transition diagram for logic circuit shown is

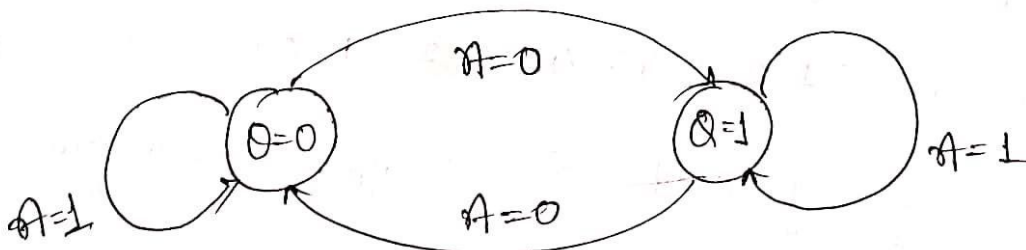


Sol for $A = 1$

$Q = 0$
 $A = 0$

$Q = 1$
 $A = 0$

Present state	$X_1 = Q$	$X_0 = \bar{Q}$	A assumption	Y	Next state
Q				$Y = 0$	
0	0	1	0	1	1
1	1	0	1	0	0
			$A = 1$		
1	1	0	0	0	0
			1	1	1



* Programmable Logic Devices (PLDs)

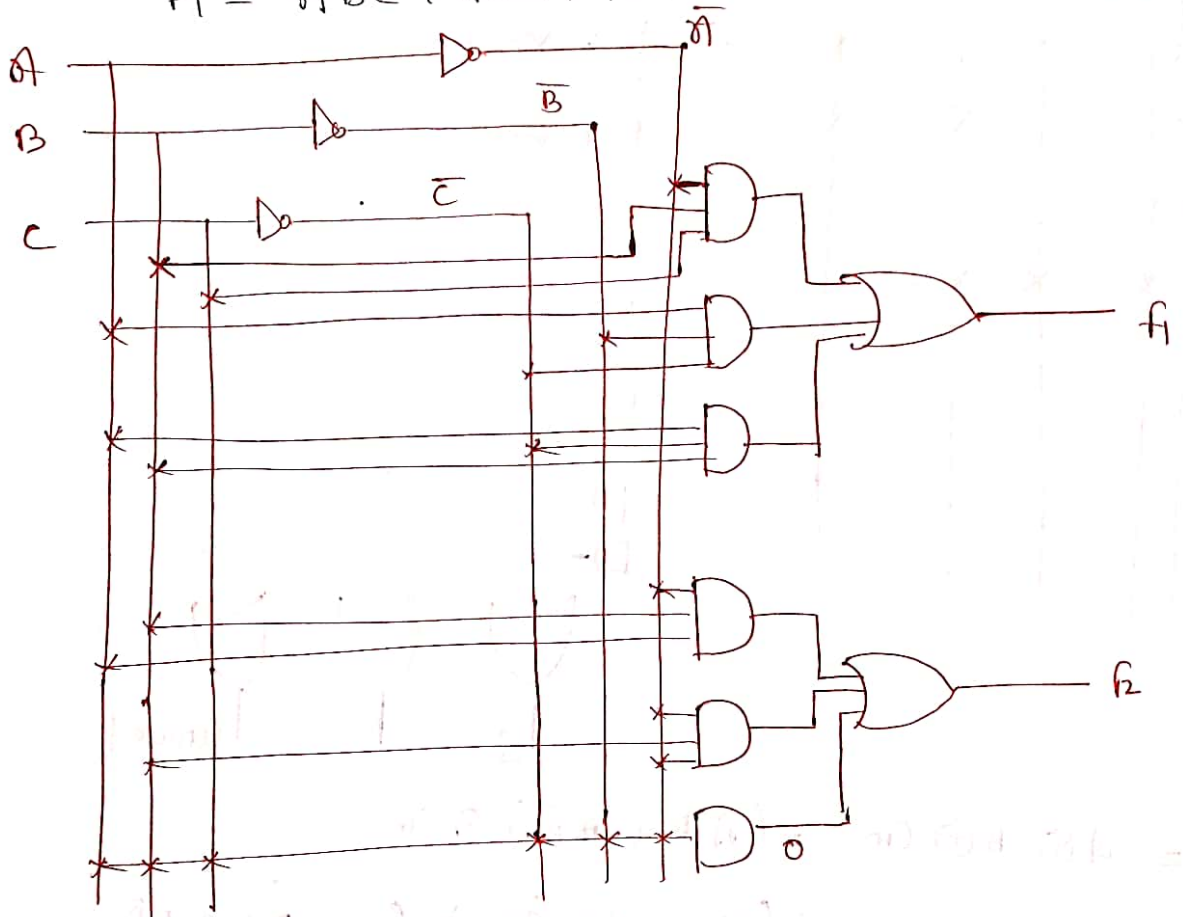
I. Programmable Array Logic (PAL)

* Programmable AND gates & fixed OR gate

eg. $f_1 = \overline{A}BC + A\overline{B}C$ & $f_2 = \overline{A}B$

$$f_1 = \overline{A}BC + A\overline{B}C + AB\overline{C}$$

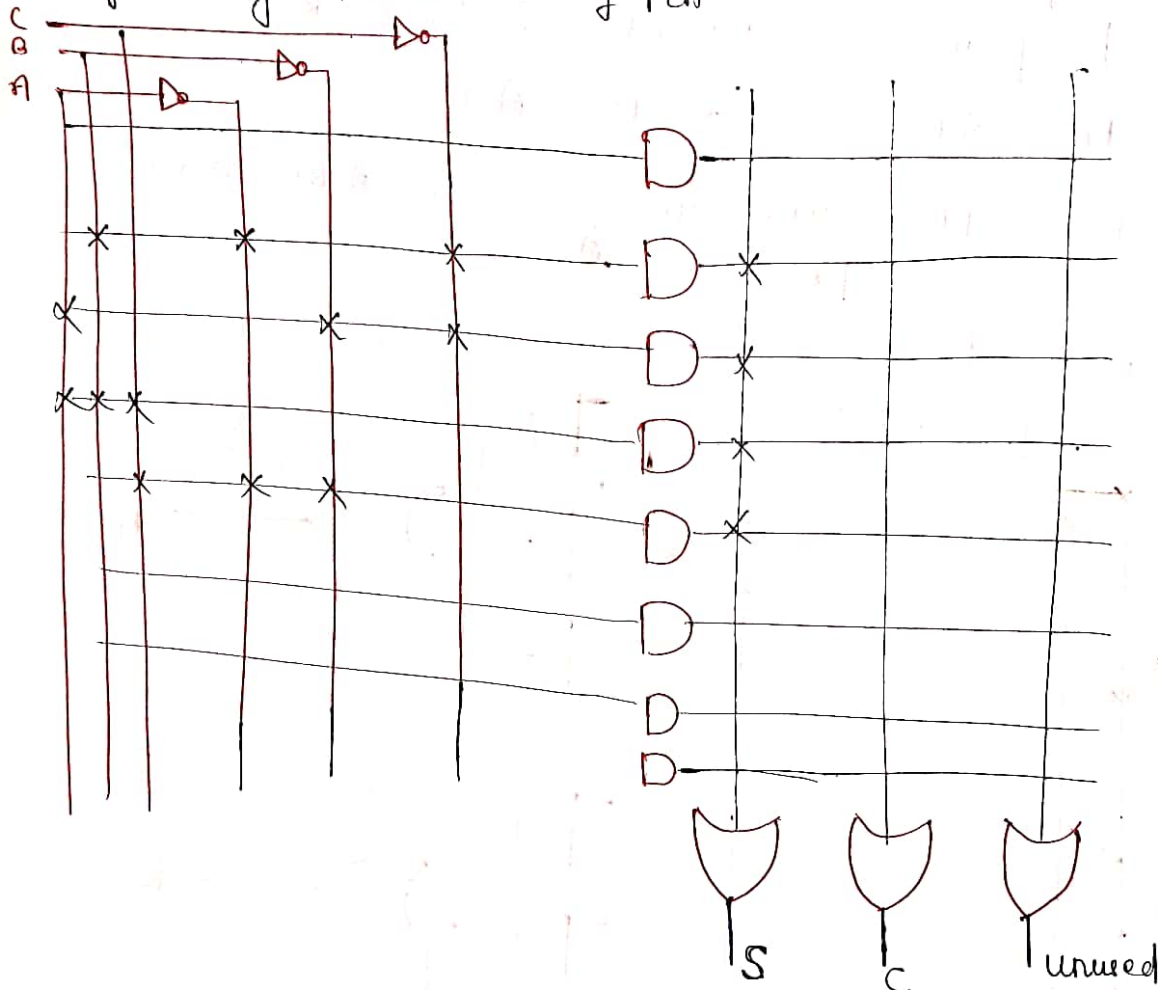
$$f_2 = \overline{A}BC + \overline{A}B\overline{C}$$



* Programmable Logic Array (PLA)

* Programmable AND gate and ^{Prog.} OR gate.

eg. Design Fulladder using PLA



$$S = A \oplus B \oplus C_{in} = (\overline{A}B + A\overline{B}) \oplus C_{in}$$

$$= (\overline{A}B + A\overline{B}) \overline{C_{in}} + C_{in} (\overline{A}B + A\overline{B})$$

$$= \overline{A}B \overline{C_{in}} + A\overline{B} \overline{C_{in}} + C_{in} (\overline{A}B + A\overline{B})$$

$$\Rightarrow \overline{A}B \overline{C_{in}} + A\overline{B} \overline{C_{in}} + C_{in} A + C_{in} \overline{A}$$

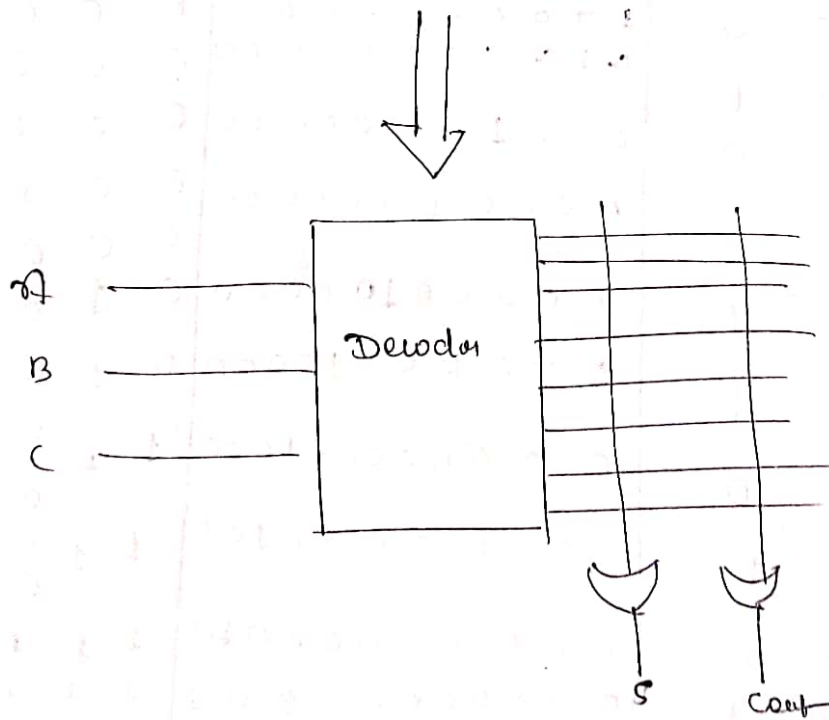
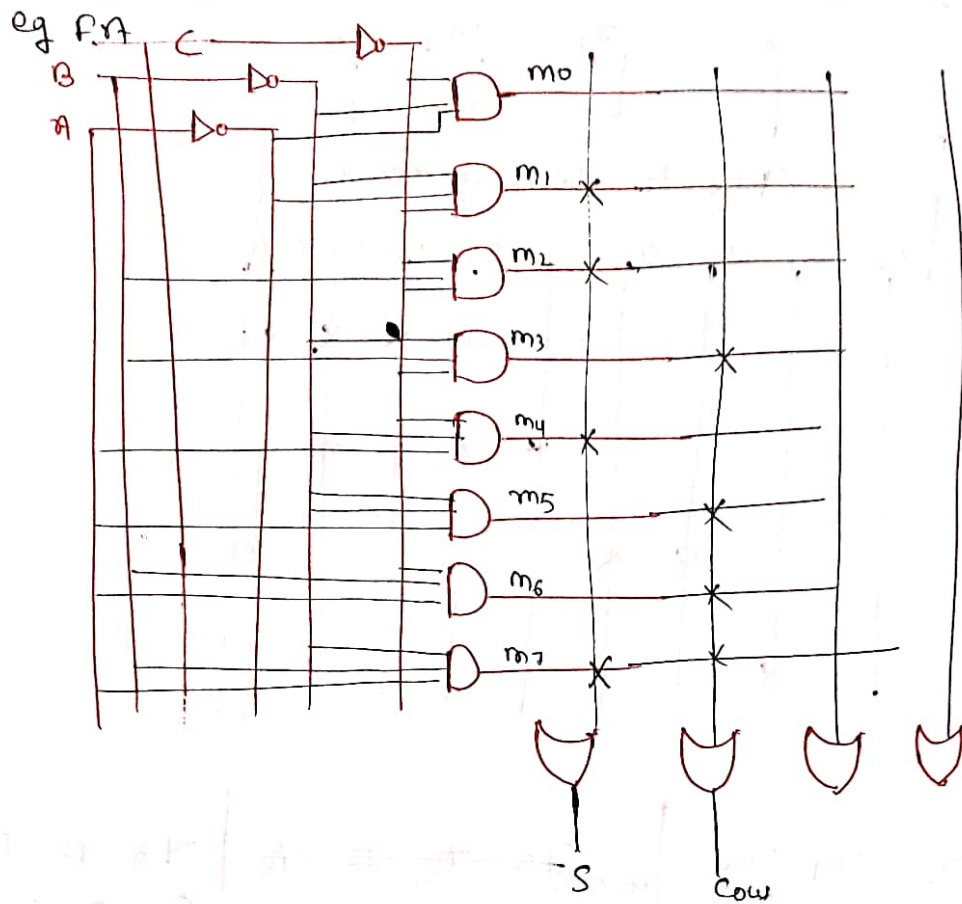
*

$$\text{Shortcut } S \oplus m = m(1, 2, 4, 7)$$

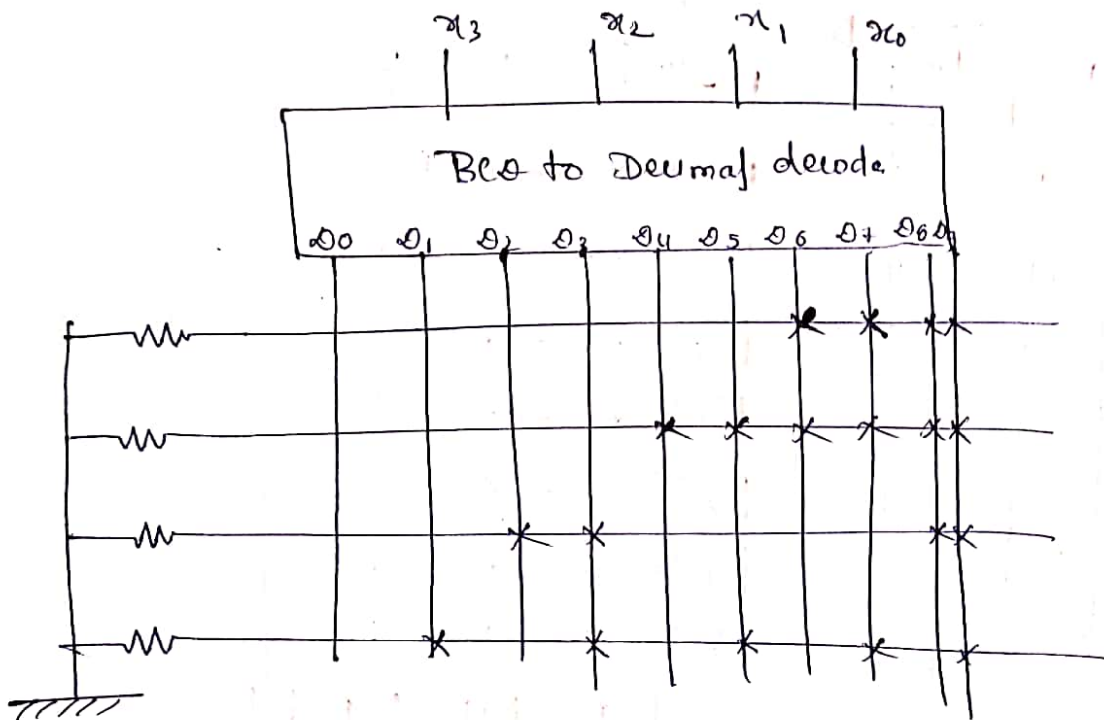
$$C = m(3, 5, 6, 7)$$

note

* PROM:
(fixed)



Que. If input $x_3 x_2 x_1 x_0$ to ROM in fig are 8 4 2 1 BCD number then the o/p $y_3 y_2 y_1 y_0$



<u>Sol</u>	x_3	x_2	x_1	x_0	00	01	02	03	04	05	06	07	08	09	y_3	y_2	y_1	y_0
	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	1	0
	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0
	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	1	1
	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0
	0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	1
	0	1	1	0	0	0	0	0	0	0	1	0	0	0	1	0	1	0
	0	1	1	1	0	0	0	0	0	0	0	1	0	0	1	1	0	1
	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
	1	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0	1

2 4 2 1

Q43.

Sol:

Q ₁	Q ₂	A	S	A	S	Q ₂
0	0	0		0	0	1
0	1	0		0	1	0
1	0	1		1	0	0
1	1	1		1	1	1

- A) $A \rightarrow \overline{Q_2}$ & $S \rightarrow \overline{S} \Rightarrow Q_2 = \overline{Q_2} \odot S = Q_2 \odot S \neq$
- B) $A \rightarrow \overline{Q_2} \Rightarrow Q_2 = \overline{Q_2} \odot S = Q_2 \oplus S \neq Q_1 \oplus S$
- C) $A \rightarrow \overline{Q_1}$ & $S \rightarrow \overline{S} \Rightarrow Q_2 = \overline{Q_1} \odot \overline{S} = Q_1 \odot S \neq$
- D) $A \rightarrow \overline{Q_1} \Rightarrow Q_2 = \overline{Q_1} \odot S = Q_1 \oplus S \checkmark$
desired

$\overline{A} \oplus B = \overline{A} \odot B$	$\overline{A} \odot \overline{B} = \overline{A \odot B}$
$\overline{A} \odot B = A \oplus B$	$\overline{B} \oplus \overline{A} = A \oplus B$

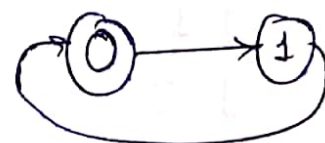
3-9

Q48

Sol

$T=1$ { always o/p of NAND with one i/p 0 zero }

Q	FF	HS
P.S	i/p	o/p
0	1	1
1	1	0

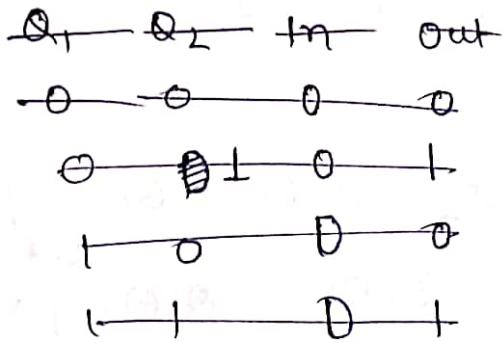
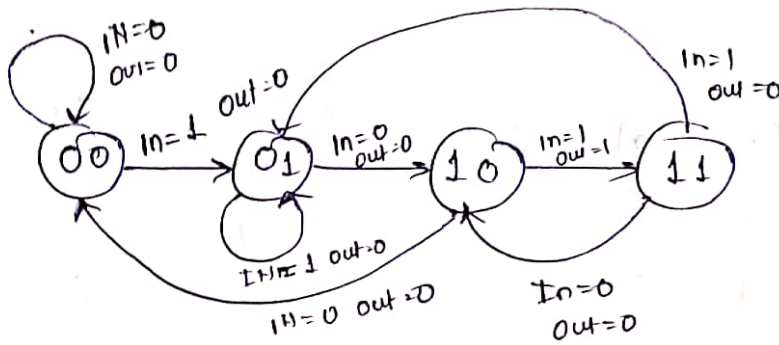


Mod 2 Counter
 $\div 2$

$$f = \frac{f_{clk}}{2}$$

$$= \frac{1}{2} = 0.5 \text{ kHz} //$$

Q80.



first copy input
in table
then fill Next state
and o/p a/q to
FSM

Present state	Input	Next state	O/p
0 0	1	0 1	0
0 1	0	1 0	0
1 0	1	1 1	1
1 1	0	1 0	0
1 0	1	1 1	1
1 1	1	0 1	0
0 1	0	1 0	0
1 0	1	1 1	1
1 1	0	1 0	0
1 0	0	0 0	0
0 0	1	0 1	0
0 1	1	0 1	0
0 1	0	1 0	0
1 0	1	1 1	1

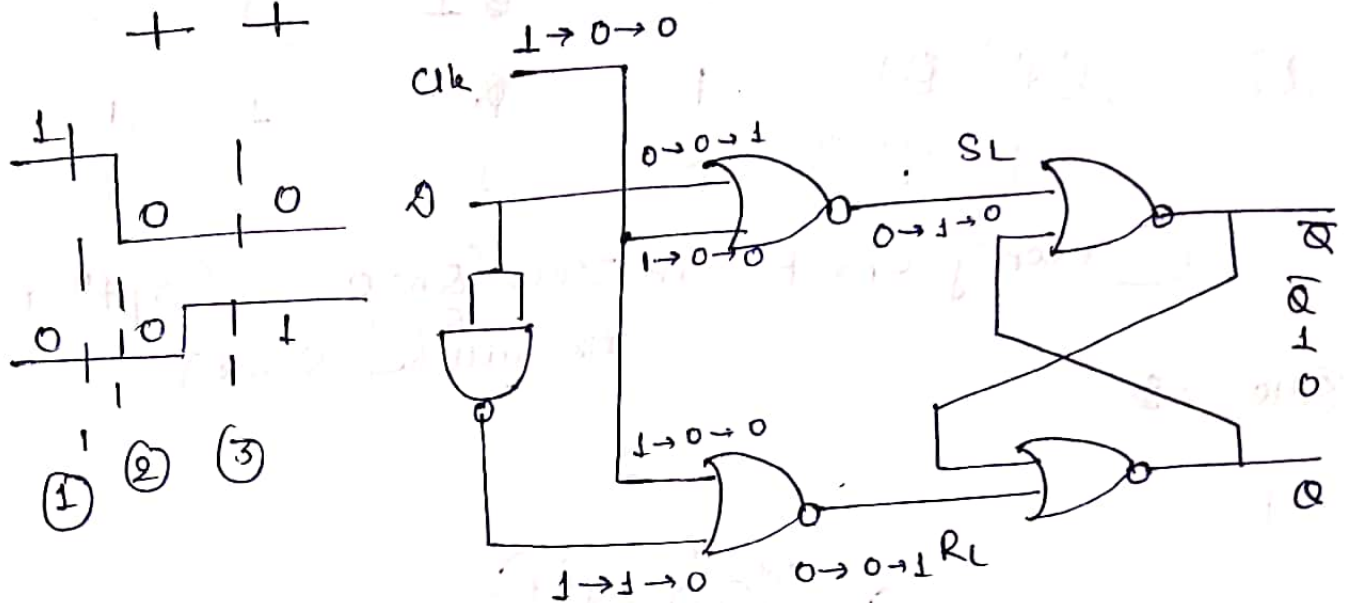
4 1's

Q19.

Sol)

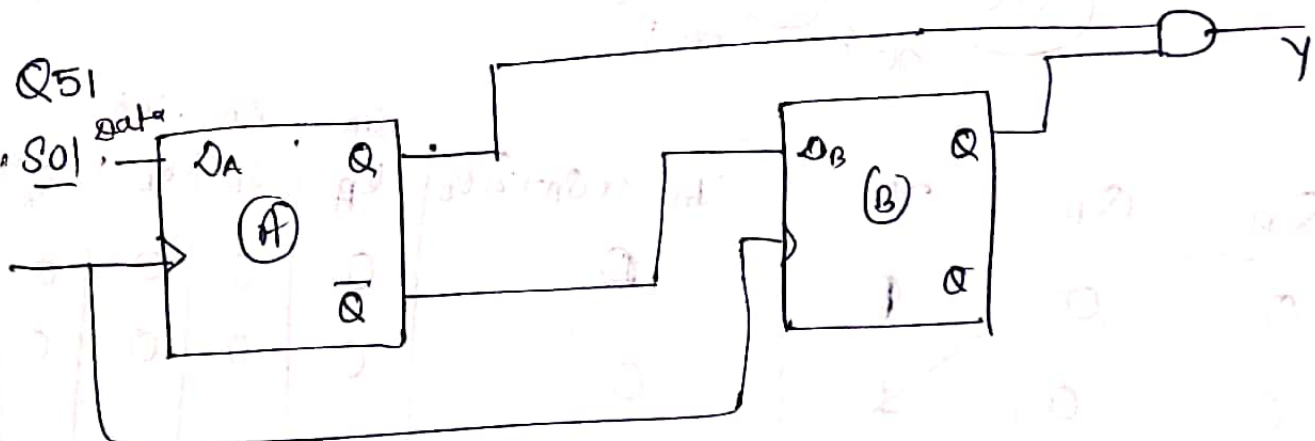
	clk	S	Q
①	0	0	0
②	0	1	1
③	1	0	0

optⁿ c)



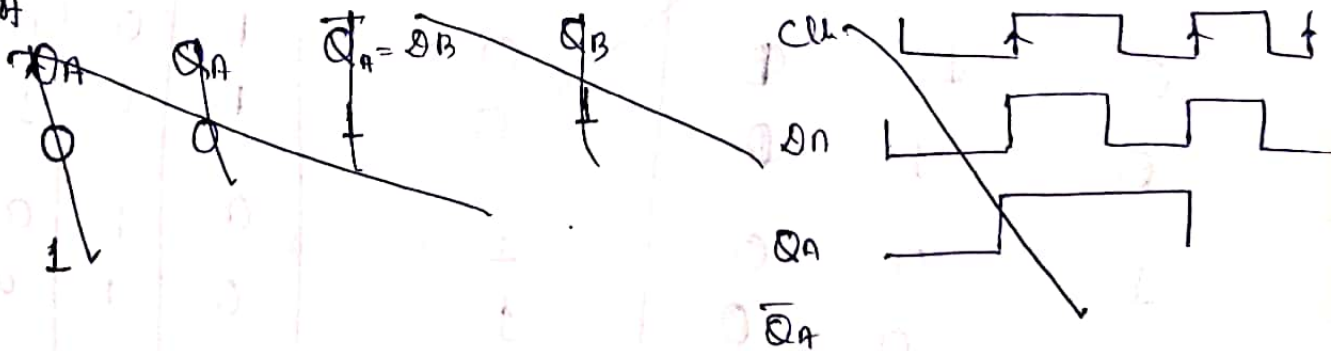
Q51

Sol)



optⁿ A

①



Sol

Present State

Flipflop Input

Next-state

Q_A Q_B

D_A
Data

$D_B = \bar{Q}_A$

Q_{A+1}

Q_{B+1}

optⁿ

A)

no info

(NI) (NI)

0

(NI)

0

(NI)

0

(NI)

1

1

1

1

$\Rightarrow \gamma = 1$

B)

(NI) (NI)

1

(NI)

1

(NI)

$\Rightarrow \gamma = 0$

1

(NI)

0

1

0

0

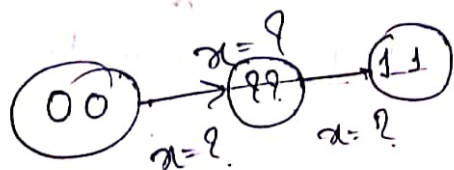
α

C even if start with "zero" $Q_A=0$
 $Q_B=0$
Ans will be same)

optⁿ a)

Que. 58

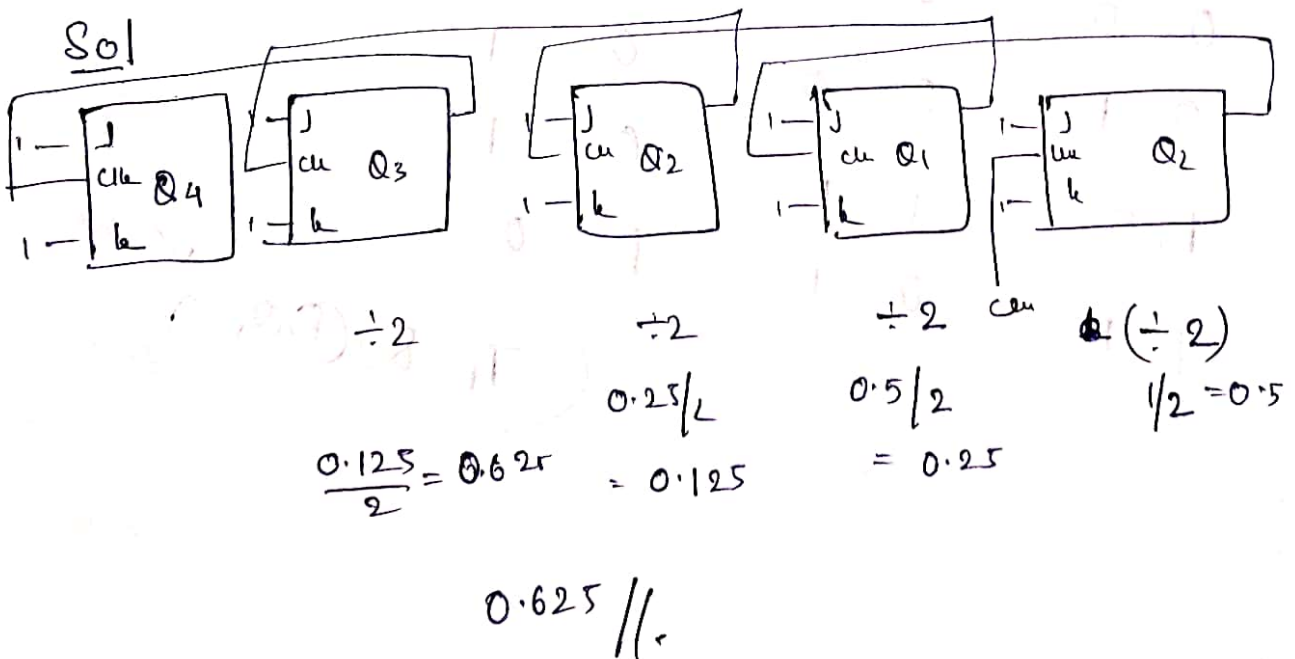
Sol



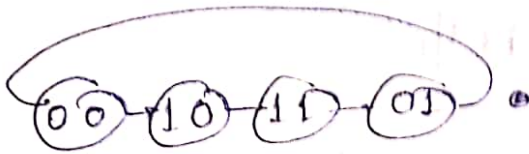
Q_A	Q_B	x	$J_A = xQ_B + \bar{x}\bar{Q}_B$	Q_B k_A	$D_A = \bar{x}Q_A$ J_B	k_B	Q_A^{+}	Q_B^{+}
0	0	1	1	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	1	1	0	1	0	0	1	0
0	1	0	0	1	0	1	1	0
1	0	0	1	0	1	0	1	0
1	0	1	1	0	1	0	1	0
1	1	0	0	0	0	0	0	1
1	1	1	0	0	0	0	0	1

Present State		Input	ff i/p		Next ch.		
Q_A	Q_B	x	$J_A = xQ_A + \bar{x}\bar{Q}_0$	$k_A = Q_0$	$J_B = Q_A$	$k_B = \bar{x}\bar{Q}_A$	$Q_A^+ Q_B^+$
A)	0	1	0	0	0	0	0 0
	0	1	0	0	0	0	0 0 α
B)	0	1	0	0	0	0	0 0 α
	0	0	1	0	0	1	1 0 α
C)	0	0 0	1	0	0	1	1 0
	1		1	0	1	0	1 1 $\checkmark$

Que 74 { changes $\rightarrow Q_0 \rightarrow Q_1 Q_2 Q_3 Q_4$
all JK Connected to 1 }



Q67



$$T_2 = Q_1 \oplus Q_2$$

$$T_1 = X$$

Q_2	Q_1	$X = T_1$	$T_2 = Q_1 \oplus Q_2$	Q_2^+	Q_1^+
0	0	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	0
1	1	0	0	1	1

$X = (Q_2 \oplus Q_1)$

P.S		N.S		F.F // P	
Q_2	Q_1	Q_2^+	Q_1^+	T_2	T_1
0	0	1	0	1	0
0	1	0	0	0	1
1	0	1	1	0	1
1	1	0	1	1	0

$$T_1 = Q_2 \oplus Q_1$$

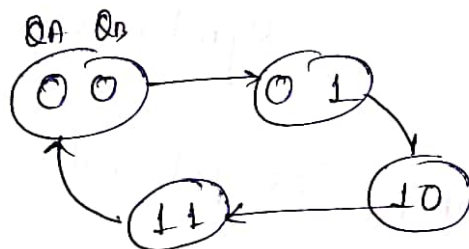
Que. 72

Q_2	Q_1	Q_0	$\overline{Q_2 \cdot Q_0}$
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

{ clock applied LSB
i.e. Q_0
Connected to +ve
i.e. upcount.

Module-5 binary
upcounter.

Que 73.



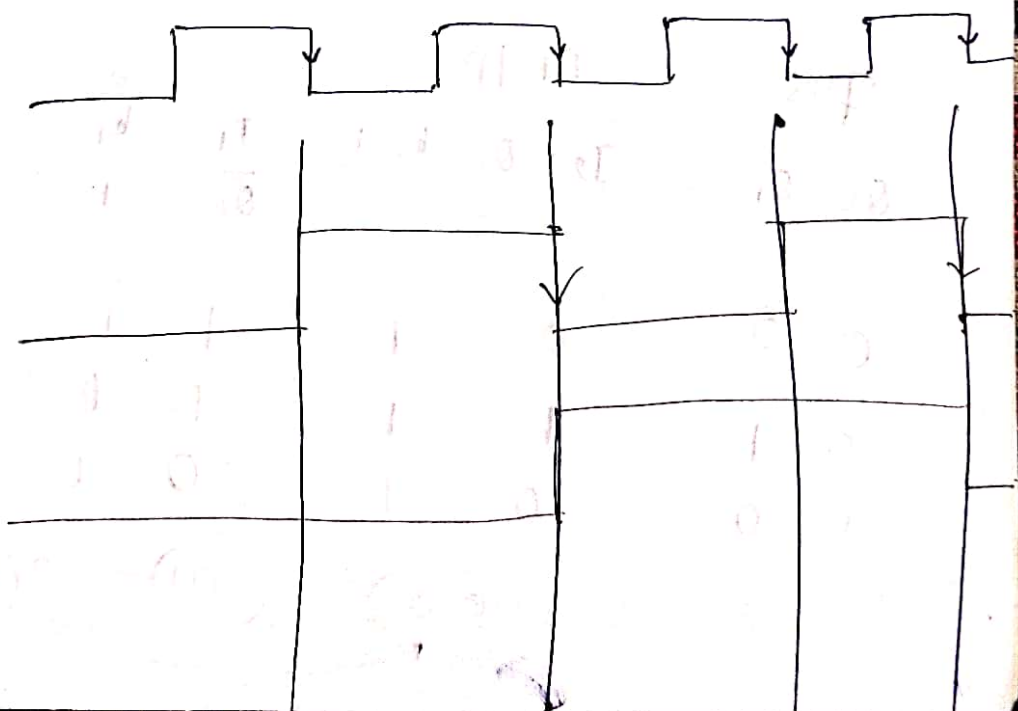
$J = K = 1$
(Toggle)
clk

~~QA~~

~~QB~~

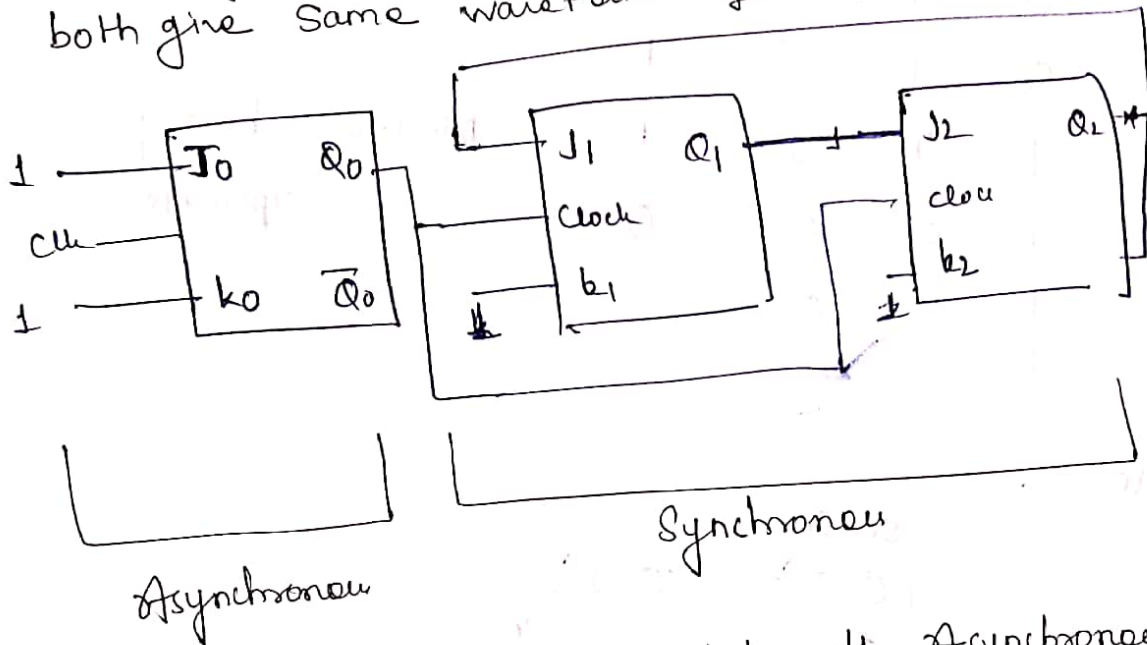
~~QA~~

QA



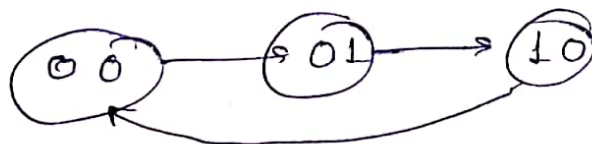
Note for async. LSB change with respect to clock (Q_0 here) and MSB change with no edge of LSB (Q_2 here)

* for Sync. Counter any bit (MSB, LSB) can change with clock pulse. both give same waveform if they have same state delay.

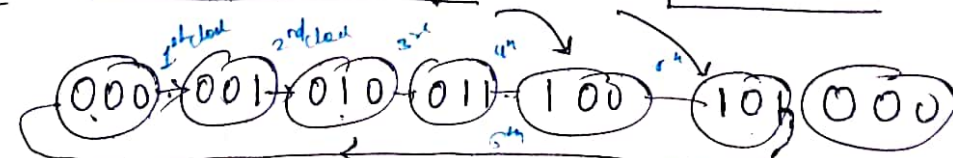
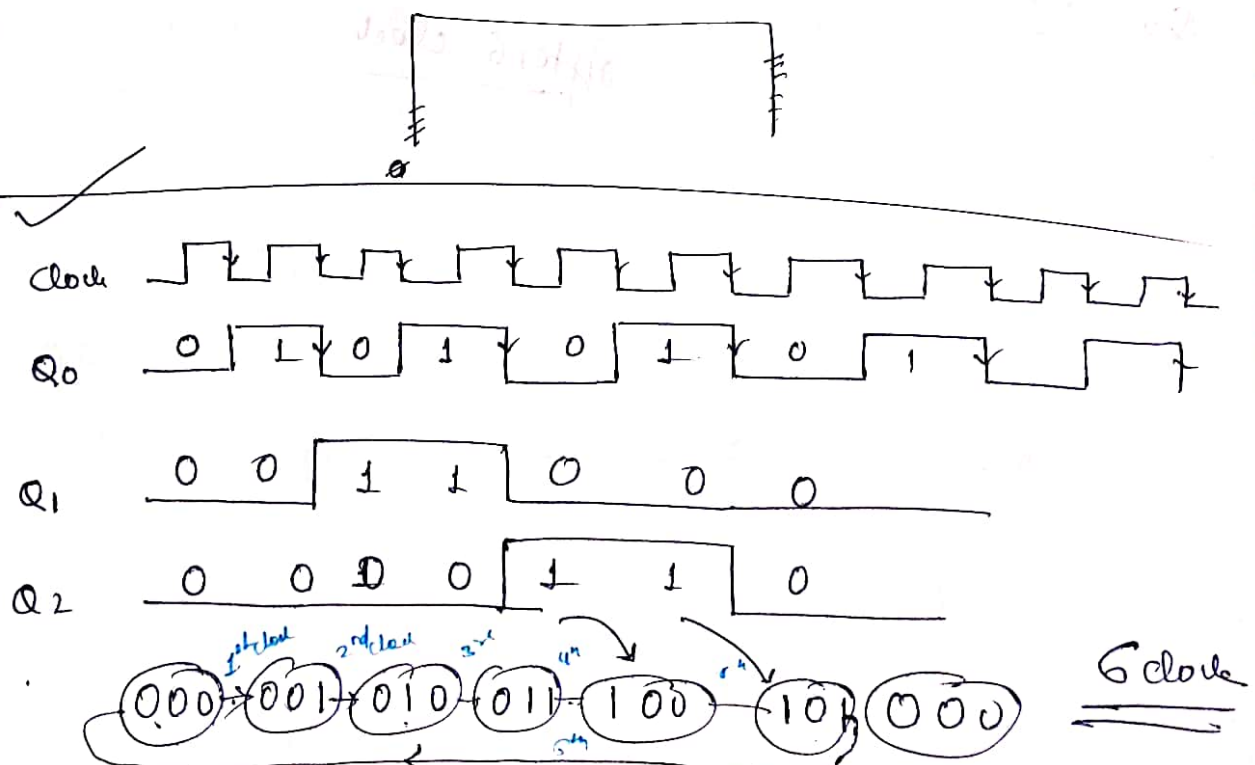
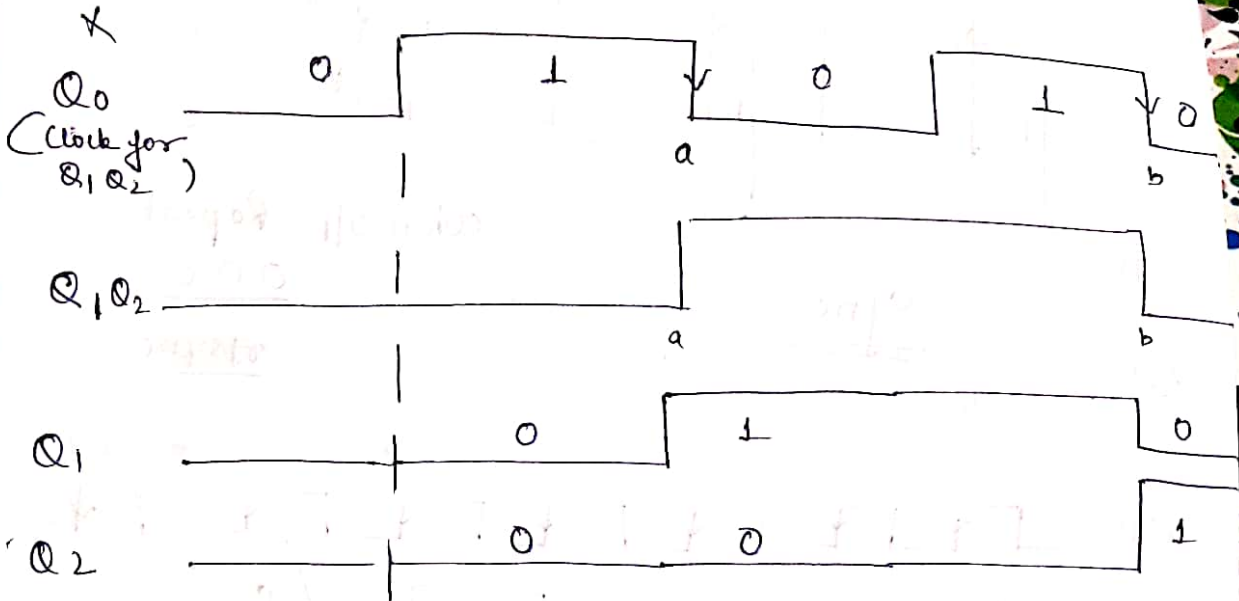


Here Sync. Counter Cascaded with Asynchronous Counter

P.S.		FF / P				N.S	
Q_2	Q_1	$J_2 = Q_1$	$K_2 = 1$	$J_1 = \overline{Q_2}$	$K_1 = 1$	Q_2^+	Q_1^+
0	0	0	1	1	1	0	1
0	1	1	1	1	1	1	0
1	0	0	1	0	1	0	0



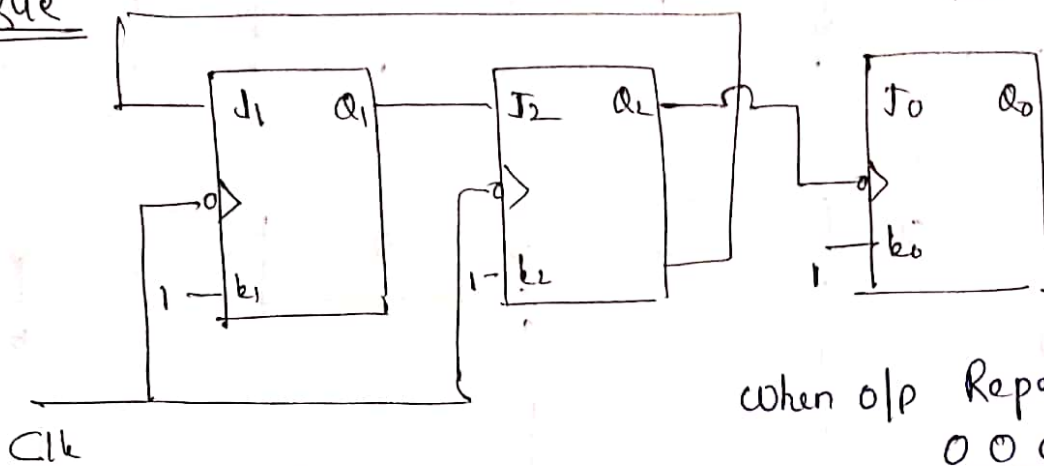
Q₀
Clock
Circuit



6 clock

Que

MSB (fastest from clock)



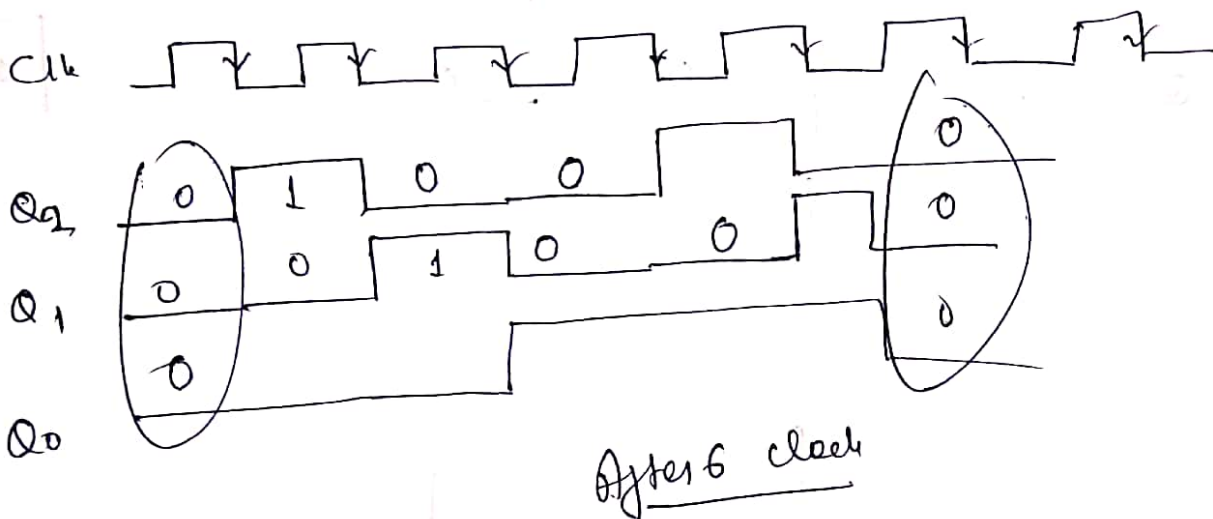
When o/p Repeat

000

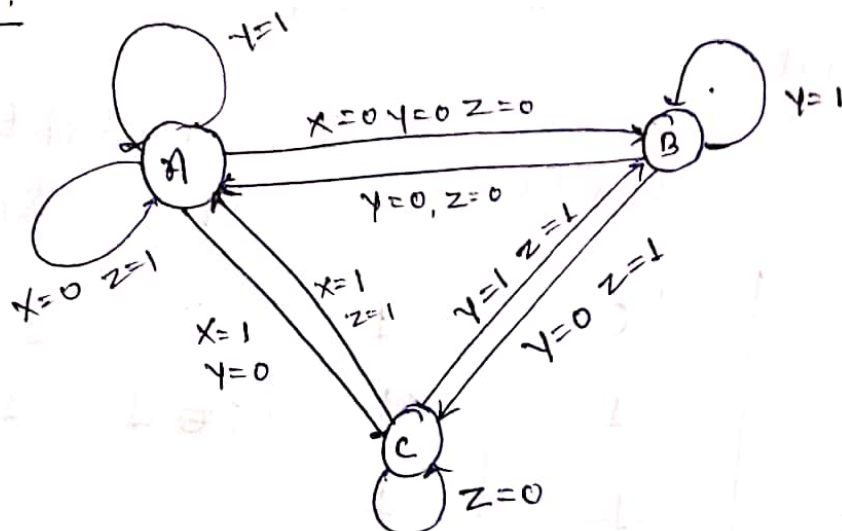
Asy

Sync

Sol



Q34.



State A

Inputs

X Y Z

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

Next state

B

A

A

A

C

A

A

A

State B

Inputs

X Y Z

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

Next

A

C

B

B

A

C

B

B

State C

Input

X Y Z

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

Next

C

A

C

B

C

A

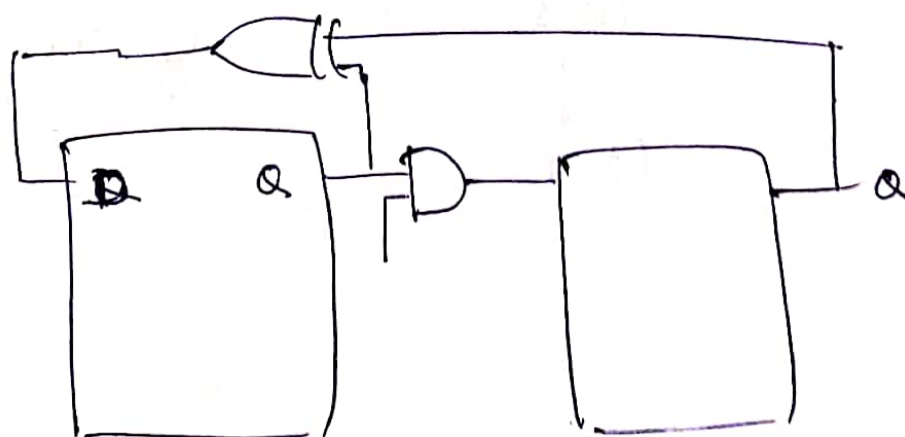
C

B/A

option c

Q3.

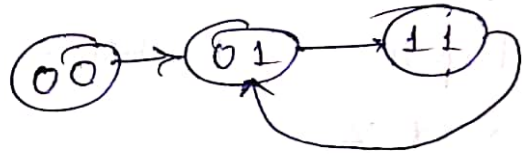
Sol



1) for $x_{in} = 0$ State Diagram (synchronous)

Present state

Q_A	Q_B	$Q_A = Q_A \oplus Q_B$	$Q_B = 1$	Next state Q_{A+1}	Q_{B+1}
0	0	0	1	0	1
0	1	1	1	1	1
1	0	1	1	0	1
1	1	0	1	1	1



ii) for $x_{in} = 1$

PS

Q_A	Q_B
0	0
0	1
1	1

ff I/P

$$Q_A = Q_A \oplus Q_B$$

0
1
0

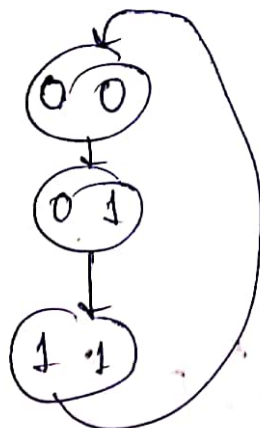
$$Q_B = \overline{Q_A}$$

1
1
0

NS

Q_{A+1} Q_{B+1}

0	1
1	1
0	0



option 0

(Repeat)

States = Cycles

for $x_{in} = 0$ it has 2 states

* SHIFT REGISTER

- 1) Serial In Serial out (SISO)
- 2) Serial In parallel out
- 3) Parallel In Serial out
- 4) Parallel In parallel out

→ Register is a Collection of Multiple flipflop to store Multiple bits of data.

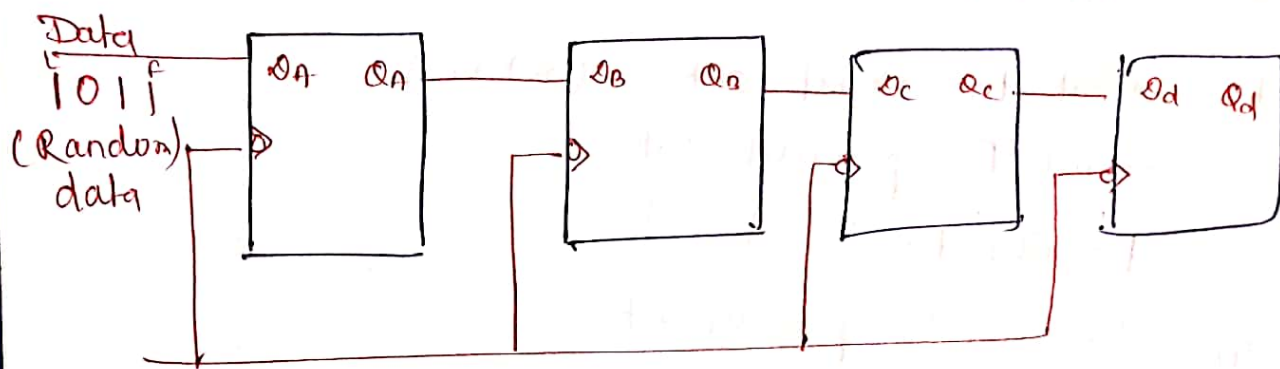
(i) SISO shift Register:- In SISO shift Register 1 bit of data gets into the shift Register with each clock pulse and one bit of data can be checked (output) with each clock pulse.

(ii) SIPO shift Register:- In SIPO shift Register 1 bit of data gets into the shift Reg. with each clock pulse, and all the bits can be checked (output) simultaneously on a single clock pulse.

(iii) PISO:- In PISO shift Register all the bits gets into the shift Reg. simultaneously on a single clock pulse, and one bit can be checked (output) with each clock pulse.

(iv) PIPO:- In PIPO shift Register all the bits gets into the sh. reg. simultaneously on a single clock pulse, and all the bits can be checked (output) simultaneously on a single clock pulse.

SISO/SIPO



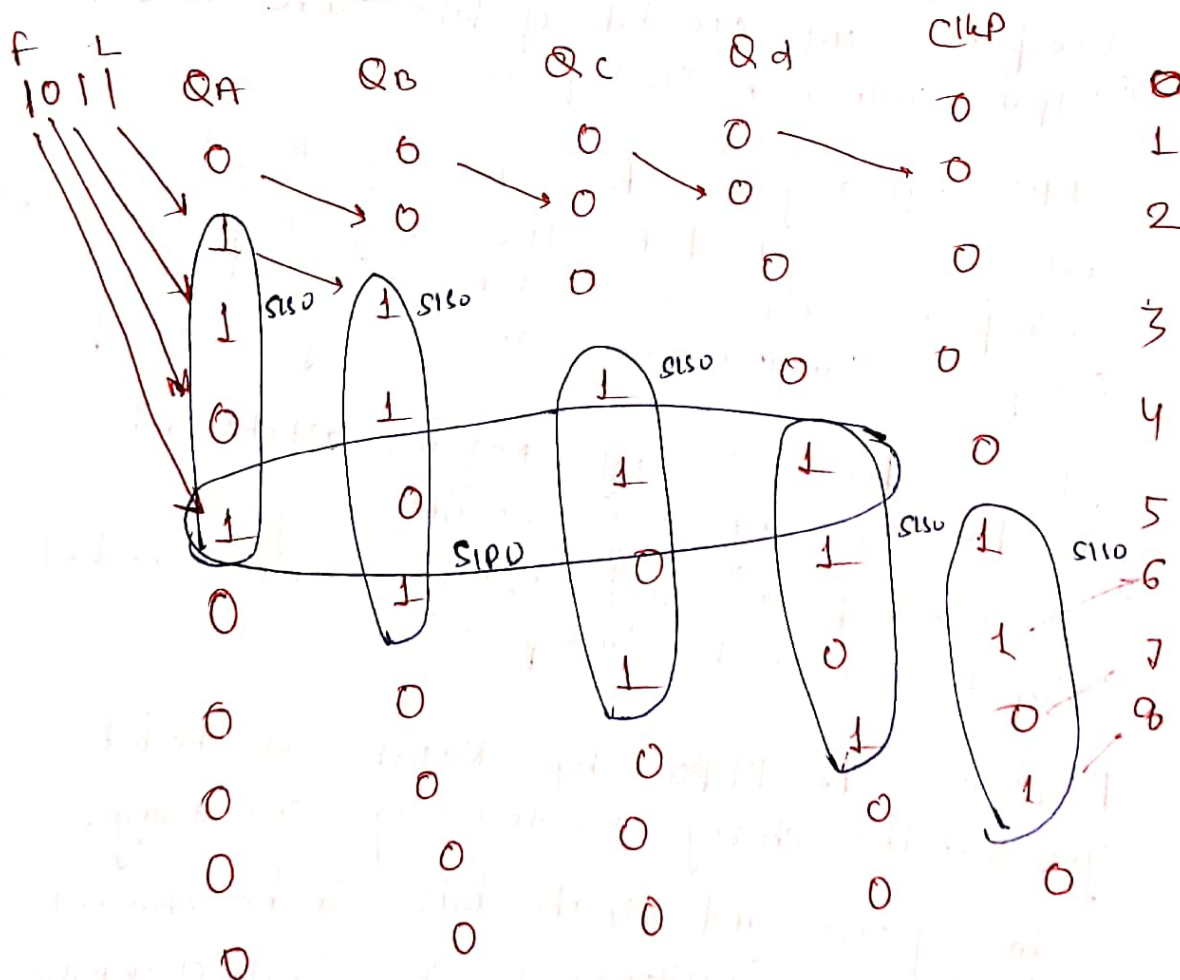
Next state eqⁿ-

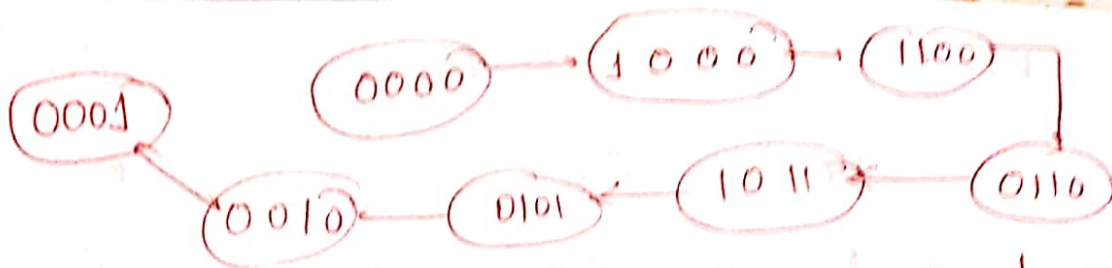
$$Q_{B+1} = Q_A$$

$$Q_{C+1} = Q_B$$

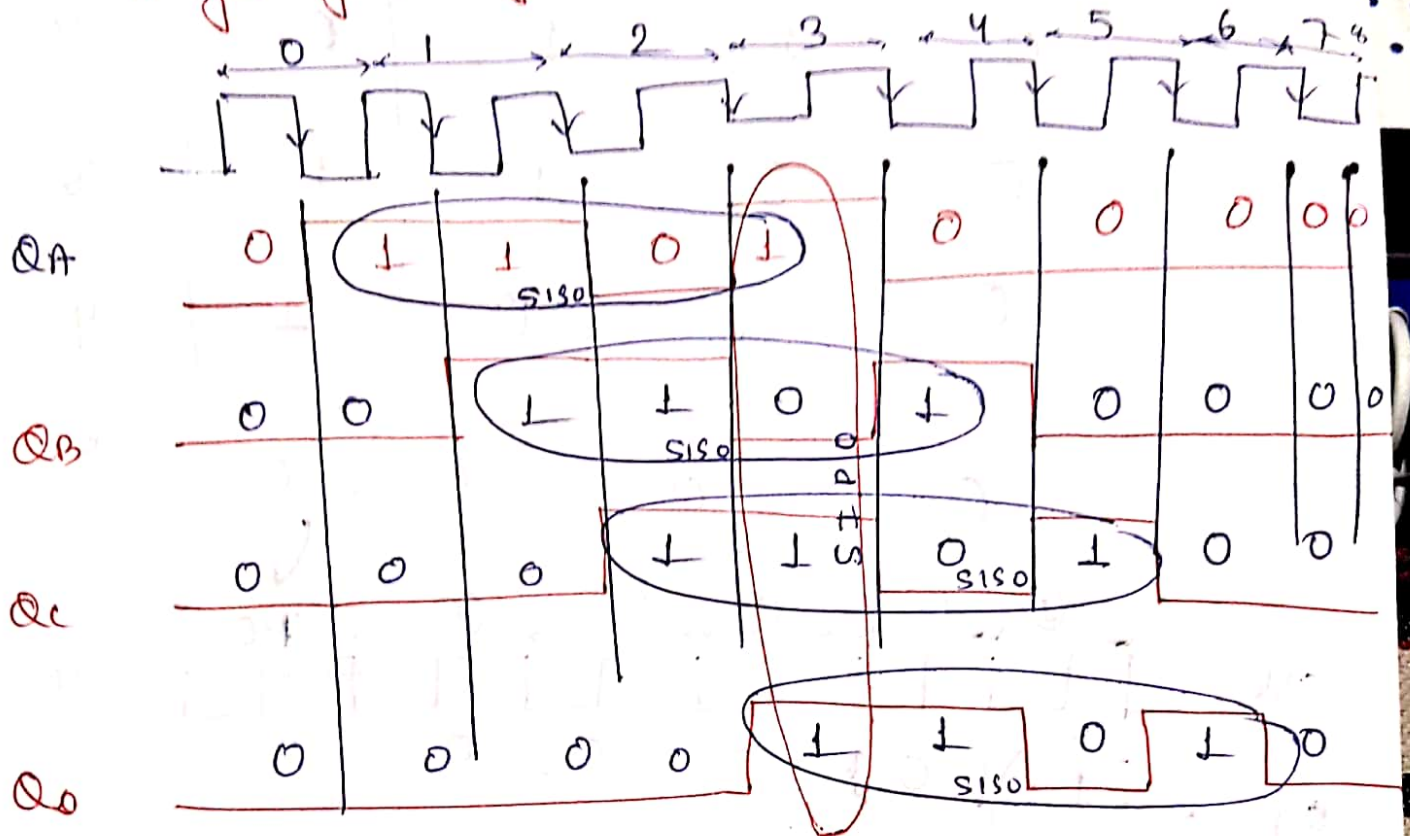
$$Q_{D+1} = Q_C$$

$$Q_{A+1} = \text{data input}$$





We don't discuss state diagram in shift register like 'Counter' because it does not repeat ∞ times it get just shift and fixed to a particular state

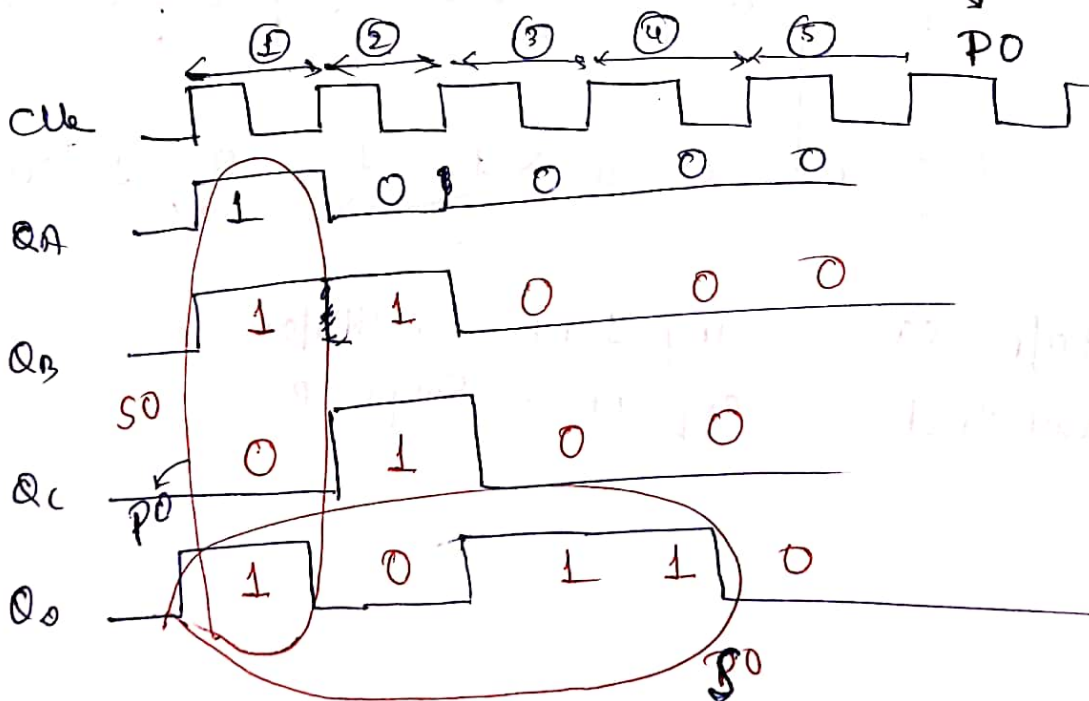
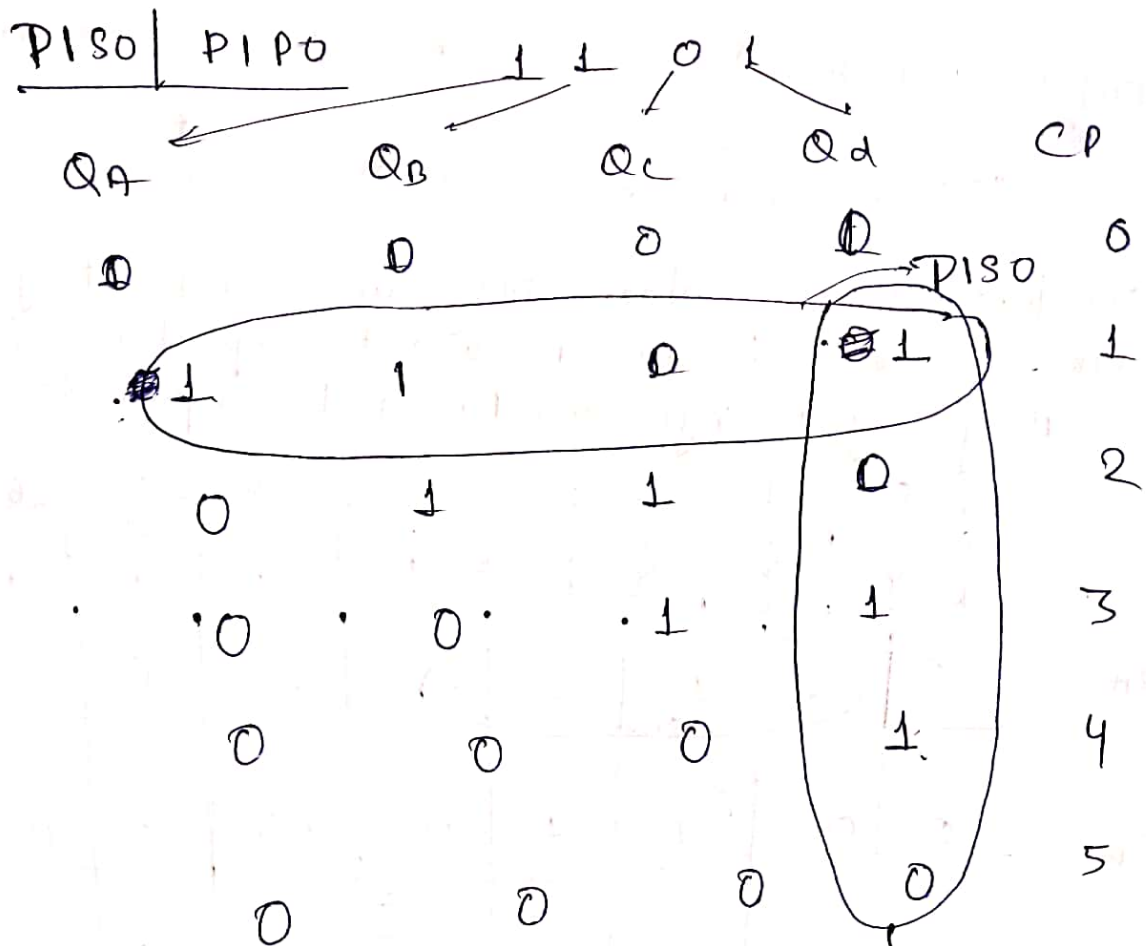


Serial o/p (SO)

Parallel o/p (PO)

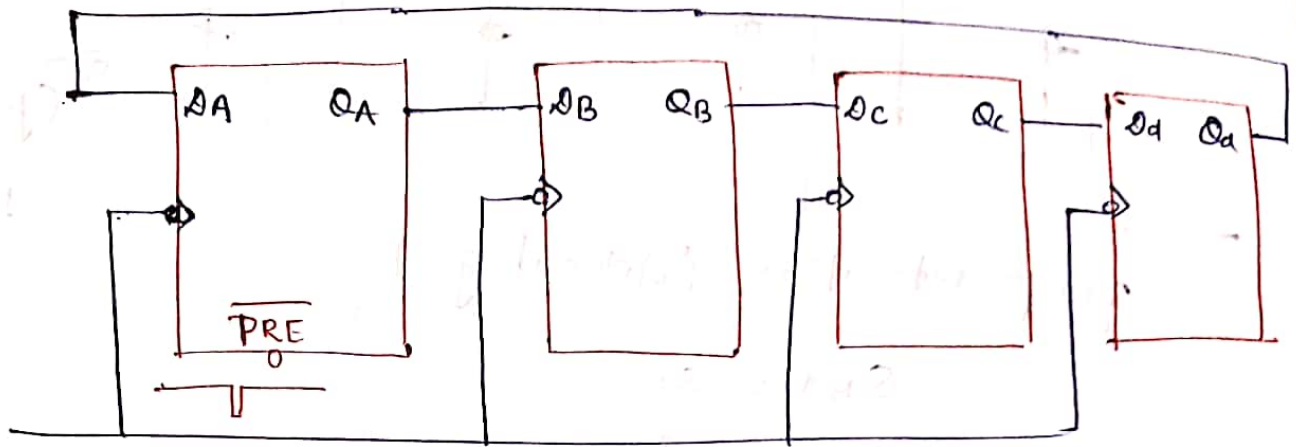
any 1 FF & 4 CPulse

any FF & Single CP



Applications of SR

1) Ring Counter :-



State eqⁿ:

$$Q_{B+1} = Q_A$$

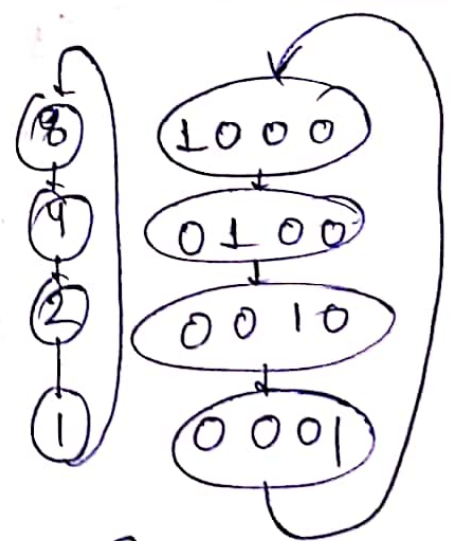
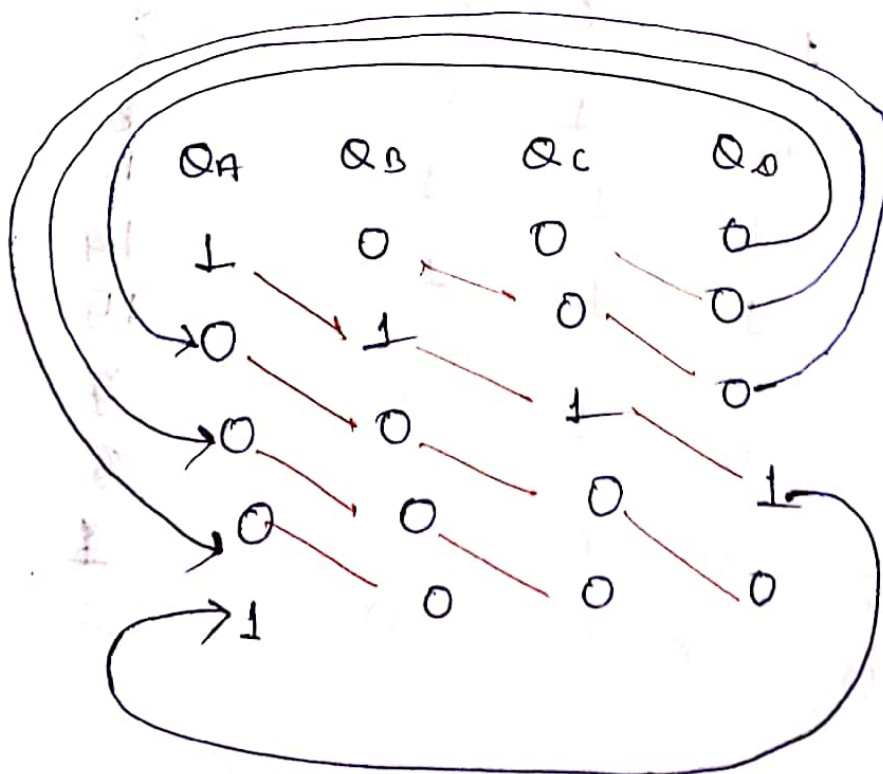
$$Q_{D+1} = Q_C$$

$$Q_{A+1} = Q_D$$

$$Q_{C+1} = Q_B$$

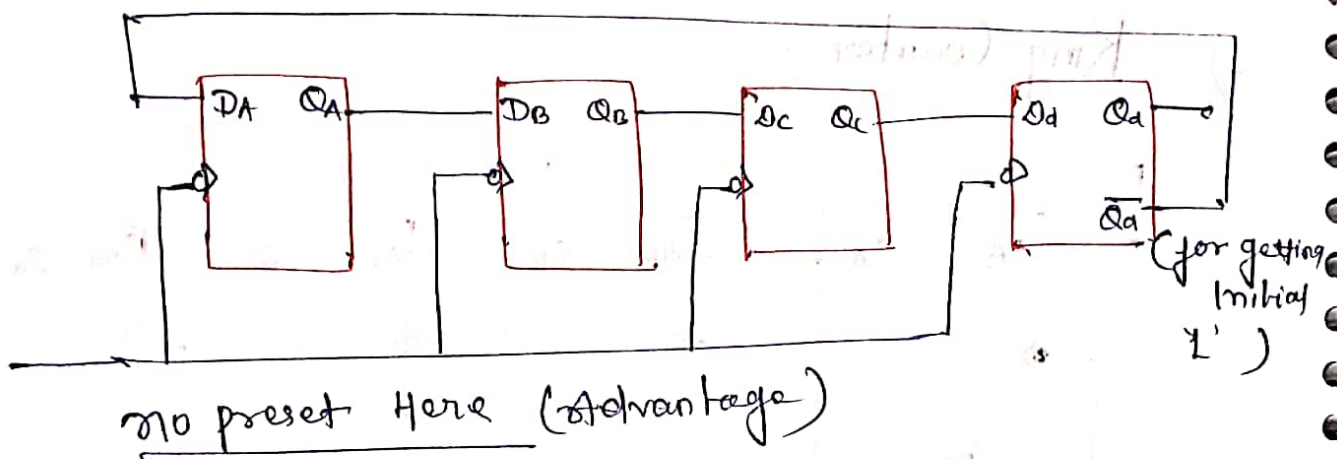
If counting start at 0 0 0 0
next state 0 0 0 0

$\div 4$ RC
Mod-4 RC



{ Sequence does not matter }

11) Johnson's Counter (Twisted Ring Counter)



$$Q_{B+1} = Q_A$$

$$Q_{C+1} = Q_B$$

$$Q_{D+1} = Q_C$$

$$Q_{A+1} = \overline{Q_D}$$

Q_A	Q_B	Q_C	Q_D
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1
0	0	0	0

Mod 8
John-C

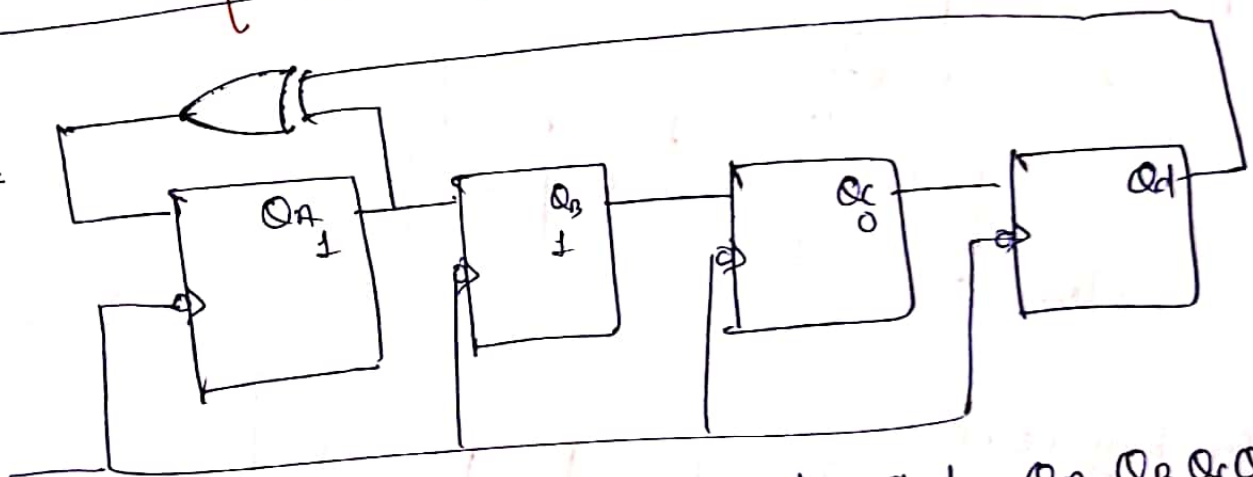
0
1
2
3
4
5
6
7

4 bit Ring $\Rightarrow$ Mod 4 Counter
 n bit Ring $\Rightarrow$ Mod n Counter

4 bit Johnson $\Rightarrow$ Mod 8
 n bit Johnson $\Rightarrow$ Mod $2n$

	Ring	Johnson
used state	n	$2n$
unused state	$2^n - n$	$2^n - 2n$

Que.

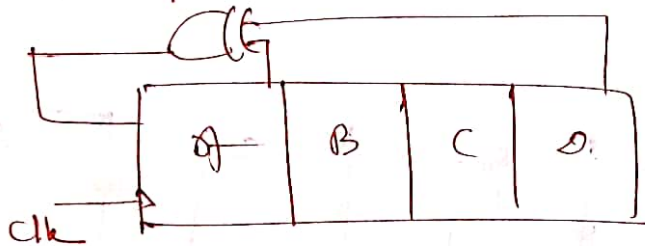


How many clock pulse Req. to get QA QB QC QD = 1111

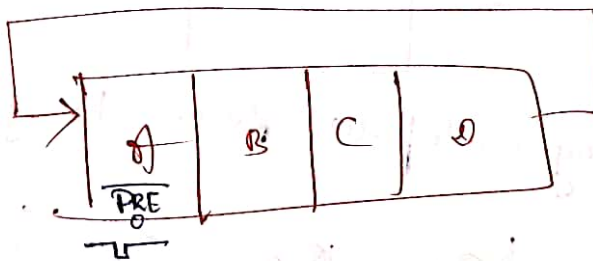
Sol	CP	QA	QB	QC	QD	CP ⁿ
	0	1	1	0	1	1
	1	0	1	1	0	2
	2	0	0	1	1	3
	3	1	0	0	1	4
	4	0	1	0	0	5

Cp	Q _A	Q _B	Q _C	Q _D	Cp
4	0	0	1	0	5
5	0	0	0	1	6
6	1	0	0	0	7
7	1	1	0	0	8
8	1	1	1	0	9
9	1	1	1	1	10
					clk

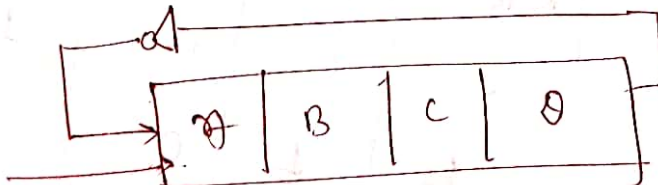
Equivalent Representation



for Ring Counter



Johnson



Q99, 10

Q114. A B C D A_{n+1} B_{n+1} C_{n+1} D_{n+1} = C
 0 1 0 1 0 0 0 0 = A = B

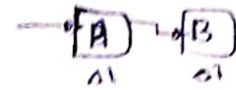
1	1	0	1	0	1	①
2	0	1	1	0	0	②
3	0	0	1	1	0	③
4	0	0	0	1	0	④
5	1	0	0	0	0	⑤
6	0	1	0	0	0	⑥
7	1	0	1	0	0	⑦

Shift Reg. Questⁿ Can also be solved by synchronous Counter approach. (Shift Reg. is Sync. Counter)

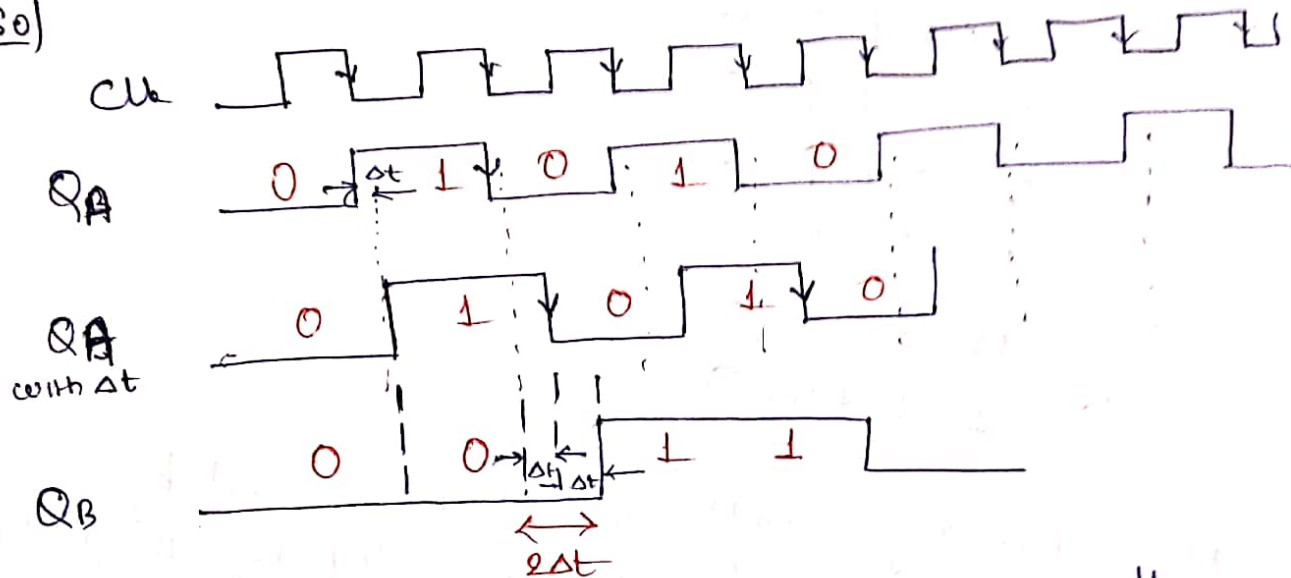
PS				FF Input			NS
Q _A	Q _B	Q _C	Q _D	A =	B =	C =	Q _{A+1} Q _{B+1} Q _{C+1} Q _{D+1}
				$Q_A \oplus Q_D$	Q_A	Q_D Q_C	

Que1 Draw waveform for a Async. Counter with state diagram

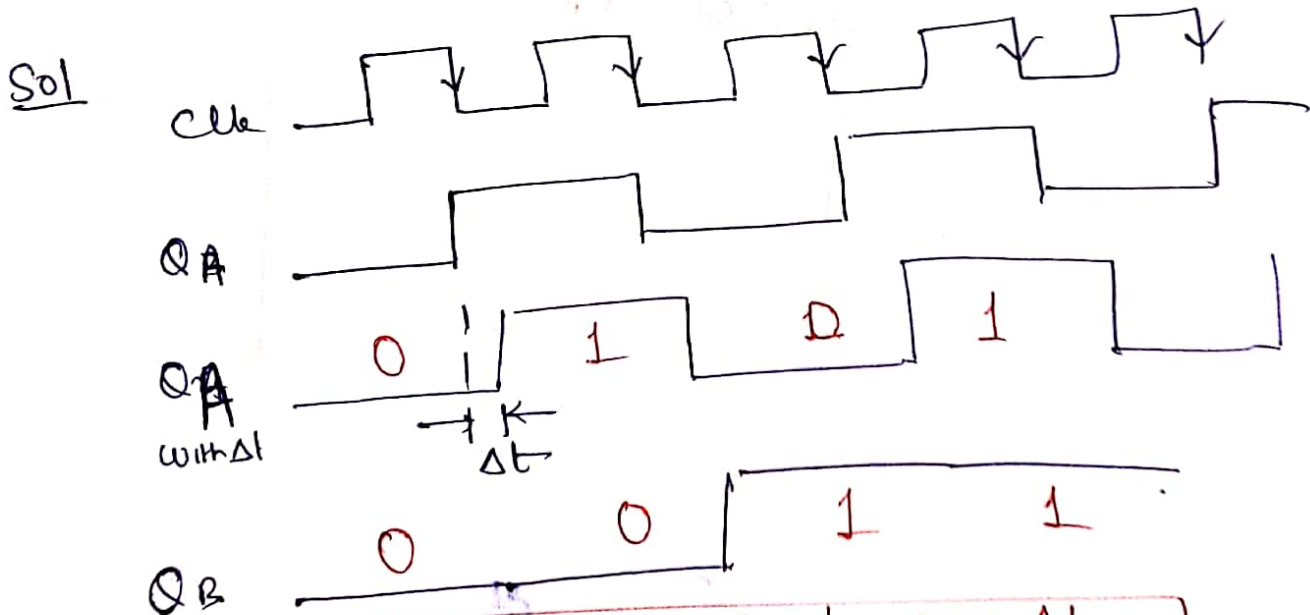
Assume delay of Δt in each FF



Sol



Que2 Draw waveform of Synchronous Counter with state diagram



Conclusion → Delay of Async. Counter = $n \Delta t$
 where n = no. of FF in Async. Counter,
 and Δt is delay in each FF.
 Delay of Syrc. Counter (n FF) = Δt